

MARBLE

Hugh's grade learning binder

Age 4 · 41 topics · 2 subjects

- English
- Mathematics



ENGLISH

20 topics

Sitting and holding a pencil

1 FOR THE GROWN-UP · TEACHING GUIDE

Why it matters. Comfortable posture and a correct pencil grip are the foundation of handwriting. When a child sits correctly at a table and holds the pencil in a comfortable tripod or near-tripod grip, they can form letters more easily and write for longer without tiring.

HOW TO TEACH IT

Set up a good seat first: back straight, feet supported, paper in front. Then show the tripod grip \u2014 thumb and first finger pinching the pencil, resting on the middle finger. Practice one letter at a time, always naming the starting point and the direction the pencil travels, and use lined paper so letters stay a consistent size.

WATCH FOR THESE MIX-UPS

- A tight, white-knuckle grip \u2014 remind your child to hold the pencil gently and comfortably
- Starting letters in the wrong place or drawing them in the wrong direction
- Letters that grow bigger or smaller across the line instead of staying a consistent size
- Slumping or turning sideways instead of sitting correctly at the table

TRY THIS AT HOME

- Practice the tripod grip with a short pencil or crayon before writing
- Say the starting point and direction out loud as your child forms each lowercase letter
- Write a few of the same letter in a row on lined paper and check they are all the same size
- Do a quick 'sitting check' \u2014 back straight, feet flat, paper in front \u2014 before every writing session

2 THE LESSON • READ TOGETHER

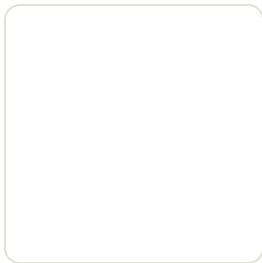
When you are learning to write, two things matter before you even make a single letter: how you SIT and how you HOLD your pencil. Getting these ready is like a runner tying their shoes before a race. If you sit correctly at a table and hold your pencil comfortably and correctly, your hand can move smoothly and your letters will look their best.

Let's start with sitting. Sit at your table with your back straight and your feet flat, so your body feels steady and comfortable. When you sit correctly at a table, your paper stays in front of you and your writing hand can reach across it easily. A wobbly, slumped body makes wobbly writing, but a calm, upright body helps your hand stay in control.

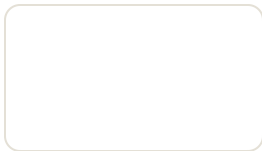
Next comes the pencil. The best way to hold it is called a tripod grip. 'Tripod' means three, and you hold the pencil with three fingers: your thumb and your first finger pinch the pencil, and it rests on your middle finger. Hold it gently and comfortably \u2014 not squeezing hard \u2014 so your fingers don't get tired. A comfortable tripod or near-tripod grip lets you guide the pencil exactly where you want it to go.

Now the fun part: making letters. Every lowercase letter has a starting point and a direction. That means there is a special spot where your pencil begins and a path it travels, so the letter finishes in the right place. For example, many letters start at the top; some start with a curve going one way. Learning the correct starting point and direction for each letter helps you write all 26 lowercase letters neatly and quickly. When you write on lined paper, try to keep your letters the same size \u2014 the little lines are like fences that help each letter stay just the right height. Same size, right start, right direction: that is beautiful handwriting!

PICTURE CARDS TO TALK ABOUT



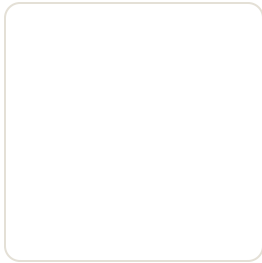
SAY I sit correctly at the table so my body is steady and my hand can move across the paper easily.



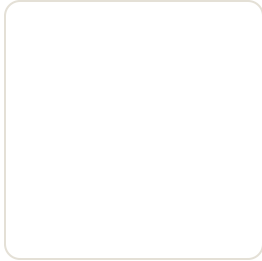
SAY This is my tripod grip \u2014 three fingers holding the pencil comfortably, not too tight.



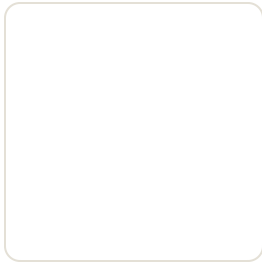
SAY I start 'c' at the top, curve around to the left, and finish in the right place.



SAY I start 'l' at the top and pull straight down \u2014 the right starting point and direction.



SAY I keep my letters the same size so they sit neatly between the lines.



SAY I can form all 26 lowercase letters, each one starting and finishing in the right spot.

3 SHOW WHAT YOU KNOW • TALK IT THROUGH

Ask each question out loud. Check the box when Hugh shows they understand.

- ☐ **Show me how you would sit at the table to write. Tell me what your body should do.**
 - Describes sitting correctly at a table
 - Mentions an upright/steady body position
- ☐ **Explain how to hold a pencil the correct way for writing.**
 - Describes a tripod or near-tripod grip using the fingers
 - Notes that the pencil should be held comfortably, not too tightly
- ☐ **Pick one lowercase letter. Tell me where your pencil starts and which direction it goes to form it correctly.**
 - Identifies a correct starting point for the letter
 - Describes the direction so the letter finishes in the right place
- ☐ **When you write letters on lined paper, what should you do to keep your writing neat?**
 - Mentions keeping letters a consistent size
 - Connects size to writing neatly on the lines

WHERE THESE FACTS COME FROM

Listening and responding

1 FOR THE GROWN-UP · TEACHING GUIDE

Why it matters. Listening is one of the four core skills of language, alongside speaking, reading, and writing, and people spend about 45 percent of their communication time listening. Strong listening helps children understand others, avoid conflict caused by misunderstandings, and build closer, more positive relationships. It is a foundation for learning and for getting along with both adults and peers.

HOW TO TEACH IT

Help your child understand the difference between hearing (which happens automatically) and listening (which is a choice to pay attention and understand). Teach the three habits of active listening: be attentive, be nonjudgmental, and don't interrupt. Point out that we also 'read' faces and gestures, and that responding — with words or with nods and eye contact — completes the loop by showing the speaker they were understood.

WATCH FOR THESE MIX-UPS

- Interrupting or jumping in before the speaker finishes
- Only hearing what they want to hear, or having a closed mind
- Being distracted and not making eye contact
- Confusing listening with obeying — a child can truly listen and understand even if they later disagree

TRY THIS AT HOME

- Practice paraphrasing: after someone speaks, have your child repeat the main idea in their own words
- Play 'sound detective' — close your eyes, listen, and guess what made each sound (this builds attention to sound cues)
- Set a family discussion rule of taking turns and not interrupting during dinner
- Read the story of Samuel together and talk about how he was ready to listen and respond

2 THE LESSON • READ TOGETHER

Have you ever noticed that your ears are working all the time? Even right now, they are picking up little sounds all around you. That is called hearing, and it happens all by itself, without you even trying. But listening is something different and very special. Listening is when you CHOOSE to pay attention to a sound and try to understand what it means. Hearing just happens, but listening is a choice you make with your brain and your heart.

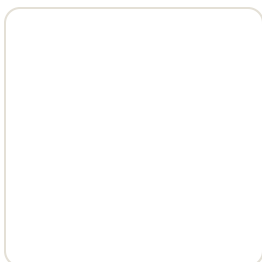
When you listen to another person, you do two big jobs at once. First, you hear the words they are saying. Second, you try to figure out what those words really mean and how the person feels. This is called interpersonal listening, which is a fancy way of saying 'listening to people.' It is one of the most important things we do, because we spend almost half of our talking-and-listening time just listening! Good listening helps us understand friends, solve problems, and feel closer to each other. When people listen poorly, by interrupting or only hearing what they want to hear, it can lead to arguments and mix-ups.

The very best kind of listening is called active listening. An active listener does three things: they pay close attention, they do not judge or make fun, and they do not interrupt. An active listener also watches the speaker's face and hands, because people say a lot without words! When you listen this way, conversations become clearer, and your friendships grow stronger. There is an old, wise saying: 'We should listen harder than we speak.'

After we listen, we respond. Responding means letting the speaker know we understood. We can respond with words, like saying 'I agree' or asking a question. We can also respond without words, by nodding, smiling, or making eye contact. In your classroom, there are agreed-upon rules that help everyone get a turn: we raise our hand, we wait for our turn, and we listen while others speak. These rules make sure every voice is heard. Remember the Bible advice: be 'quick to listen, slow to speak.' When you listen well and respond kindly, you show people that they matter.

PICTURE CARDS TO TALK ABOUT

SAY That is hearing! Certain places have familiar sounds. When a strange sound sneaks in, your ears alert you that something new is happening — but you did not choose it, it just happened.



SAY This is listening on purpose! The child is picking up sound clues that mean 'Mom is home!' Putting clues together to find meaning is a big part of listening.



SAY This is active listening. Making eye contact and staying quiet until they finish shows you truly care about what they are saying.



SAY Following the discussion rules makes sure everybody gets a fair chance to speak and be heard.



SAY Samuel showed us that even children can be wonderful listeners who are ready to pay attention and respond.



SAY Repeating back what you heard in your own words is called paraphrasing. It proves you were really listening and understood.

3 SHOW WHAT YOU KNOW • TALK IT THROUGH

Ask each question out loud. Check the box when Hugh shows they understand.

- ☐ **Explain the difference between hearing and listening, and give one example of each.**
 - States that hearing happens automatically/physically
 - States that listening is a choice to pay attention and understand
 - Gives a relevant example for each
- ☐ **Describe three things an active listener does to show they are really paying attention.**
 - Names being attentive/paying attention
 - Names not interrupting
 - Names being nonjudgmental OR watching nonverbal cues like eye contact and nodding
- ☐ **Why are classroom rules like raising your hand and waiting your turn helpful during a discussion?**
 - Explains that the rules help everyone get a turn to speak
 - Connects the rules to listening to others / being heard
 - Uses a complete, relevant idea
- ☐ **A classmate tells you a long story. Explain how you would respond to show you understood, using both words and body language.**
 - Describes a verbal response such as paraphrasing or answering relevantly
 - Describes a nonverbal response such as eye contact or nodding
 - Response would clearly show understanding to the speaker

WHERE THESE FACTS COME FROM

English Wikipedia — “Listening is the act of paying attention to sounds... When listening to another person, one hears what they are saying and tries to understand what it means. Hearing is a physiological phenomenon; listening is a psychological act.” · “behavioral processes include responding to others with verbal and nonverbal feedback; active listening requires that the listener be attentive, nonjudgmental, and non-interrupting.”

Building sentences

1 FOR THE GROWN-UP · TEACHING GUIDE

Why it matters. Understanding that words combine to make complete sentences is the foundation of both speaking clearly and writing. A sentence expresses a complete thought, and children who grasp this early can produce and expand their own sentences in speech and writing, which supports every subject that follows.

HOW TO TEACH IT

Start with the idea that a sentence tells a WHOLE thought so a listener understands. Teach the two building blocks in kid words: the subject (the who or what) and the verb (the action, the doing word). Show that even a two-word sentence like 'Bill reads' is complete because it has both parts. Then contrast it with a fragment like 'Because Bill wants to learn more,' which leaves you asking what happens. Finally, play with stretching sentences by adding details.

WATCH FOR THESE MIX-UPS

- Thinking a sentence must be long — remind them 'Dogs bark' is short but complete.
- Confusing a fragment with a sentence because it 'sounds like words' — ask 'Does it tell the whole thought, or does it leave you guessing?'
- Leaving out the verb (the action word) so the thought isn't finished.
- Struggling to find the subject — ask 'Who or what is this about?'

TRY THIS AT HOME

- Play 'Sentence or Not?': say a phrase and have your child give a thumbs up for a complete sentence or thumbs down for a fragment.
- Take turns naming a subject and a verb to build a two-word sentence together, like 'Mommy cooks' or 'Birds fly.'
- Play 'Stretch It': start with a small sentence like 'The dog runs' and add one detail at a time to make it grow.
- Point to things around the house and ask your child to make a complete sentence about each one.

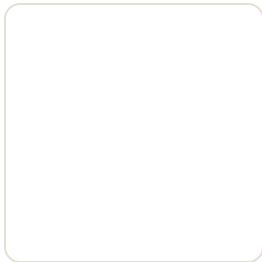
2 THE LESSON • READ TOGETHER

A sentence is a group of words that work together to tell a whole, complete thought. Think of a sentence like a little train: the words are cars that hook together, and when they are all connected the right way, the train can go somewhere. When you say a sentence, the person listening understands exactly what you mean. That is the special job of a sentence — to share a complete idea so it makes sense all by itself.

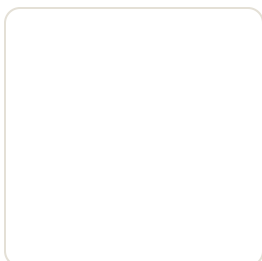
Every good sentence needs two important parts. The first part is called the subject. The subject is the who or the what the sentence is about. The second part is called the verb. The verb is the action — the doing word. When you put a subject and a verb together, you make a sentence! For example, "Bill reads." Here, "Bill" is the subject (that's who), and "reads" is the verb (that's what he does). Even though it is only two words, it is a whole sentence, because it tells a complete thought.

A sentence does not have to be long to be complete. A short sentence like "Dogs bark" is finished and makes sense. But sometimes a group of words is NOT finished. If words do not tell a complete thought, we call it a fragment. A fragment is like a train car that got left behind — it can't go anywhere by itself. For example, "Because Bill wants to learn more." When you hear that, you wonder, "What happens? What comes next?" It leaves you guessing, so it is not a complete sentence.

The fun part is that once you can build a simple sentence, you can make it grow! You start with a subject and a verb, and then you add more words to tell more about it. "The dog runs" can become "The big brown dog runs fast in the park." Every extra word paints a clearer picture in the listener's mind. Learning to build and stretch sentences helps you talk clearly and write down your own ideas so everyone can understand them.

PICTURE CARDS TO TALK ABOUT

SAY This is a complete sentence! "Bill" is the subject (who it is about) and "reads" is the verb (the action). It tells a whole thought all by itself.



SAY Short, but complete! Two words — a subject (Dogs) and a verb (bark) — can make a real sentence.



SAY This is a fragment, not a sentence. It leaves you asking, 'What happens?' It does not finish the thought, so it can't stand on its own.



SAY Here the subject is 'The cat' and the verb is 'sleeps.' A subject plus a verb equals a complete thought.



SAY This is the same sentence, but stretched! We added details to tell where the cat sleeps. Now we can picture it clearly.



SAY Watch a sentence grow! Start with a subject and verb, then add words to make a bigger, clearer picture.

3 PRACTICE WORKSHEET

Read each question aloud with Hugh if they are still learning to read.

1. Which one is a complete sentence?

- ☐ Because Bill wants
 - ☐ The frog jumps.
 - ☐ Under the big tree
-

2. In the sentence 'The cat sleeps,' which word is the verb (the action)?

- ☐ The
 - ☐ cat
 - ☐ sleeps
-

3. Does a sentence have to be long to be complete?

- ☐ Yes, always long
 - ☐ No, it can be short like 'Dogs bark.'
-

4. Build a complete sentence using the subject 'The bird.' Add a verb (an action word).**5. Which words are missing an ending — a fragment that leaves you guessing?**

- ☐ The sun shines.
- ☐ When we go outside
- ☐ Kids play.

4 SHOW WHAT YOU KNOW • TALK IT THROUGH

Ask each question out loud. Check the box when Hugh shows they understand.

- ☐ **Say or write a complete sentence. Then tell which word is the subject (the who or what) and which word is the verb (the action).**
- Produces a complete sentence that expresses a whole thought
 - Correctly identifies a subject (who or what)
 - Correctly identifies a verb (the action word)
- ☐ **Look at these two groups of words: (1) 'The puppy barks.' and (2) 'Because the puppy is happy.' Which one is a complete sentence and which one is a fragment? Explain how you know.**
- Identifies 'The puppy barks.' as the complete sentence
 - Identifies 'Because the puppy is happy.' as the fragment
 - Explains that a fragment does not tell a complete thought / leaves you guessing
- ☐ **Start with the sentence 'The dog runs.' Make it bigger by adding details. Say your new, stretched sentence.**
- Keeps a complete sentence with a subject and verb
 - Adds at least one detail (where, when, or how)
 - New sentence still makes sense and tells a clear thought

WHERE THESE FACTS COME FROM

Simple English Wikipedia — “A simple sentence... contains one subject and one verb... It must also express a complete thought.” · “‘Because Bill wants to learn more.’ ... that is not a sentence. It is a dependent clause / sentence fragment.”

Understanding print

1 FOR THE GROWN-UP · TEACHING GUIDE

Why it matters. Before children can read words, they need to understand how print works — that those marks carry meaning, that we move through them left-to-right and top-to-bottom, and that books are organized page by page. These 'concepts of print' are the foundation of the kindergarten reading standard (RF.K.1) and make later reading much smoother.

HOW TO TEACH IT

Read aloud together every day and make the mechanics visible. Run your finger under the words as you read so your child sees the left-to-right, top-to-bottom flow. Pause to ask 'Where do we start reading?' and let your child point. Hand the book to your child and let them turn the pages one at a time. Point out that the print, not the pictures, is what you are actually reading.

WATCH FOR THESE MIX-UPS

- Child 'reads' the picture instead of noticing the print carries the meaning
- Sweeping a finger right-to-left or randomly instead of left-to-right
- Skipping or flipping several pages at once instead of turning one at a time, in order
- Pointing to the middle or bottom of the page as the starting spot instead of the top-left

TRY THIS AT HOME

- Give your child a favorite book and ask them to show you where the reading starts on a page.
- Let your child be the 'page turner,' turning one page at a time in the correct order.
- Play 'finger train': have your child slide a finger under the words left to right as you read the line aloud.
- Find print out in the world — signs, labels, menus — and talk about how the words carry a message.

2 THE LESSON • READ TOGETHER

Have you ever opened a book and seen all those little marks on the page? Those marks are called print. Print is words made of letters, and sometimes pictures too, that people put on paper. Books and newspapers are made by printing — that is when words and images are put on paper, usually with ink. The most exciting secret about print is this: it is not just squiggles. Print carries meaning! Every group of letters stands for a word that someone can say out loud. When you learn to read, you are learning to unlock the meaning hidden in the print.

Print follows special rules so that everybody reads it the same way. In English, we read from the LEFT side of the page to the RIGHT side. Your eyes start at the left, slide along to the right, and when the words run out at the end of that line, you jump back to the left and drop down to the next line. That is called going left-to-right and top-to-bottom. We start at the top of the page and work our way down, line by line, all the way to the bottom.

Books are also organized page by page. That means the story does not jump around — it goes in order. You read one whole page, then you turn to the very next page, then the next, until you reach the end. If you turned the pages in the wrong order, the story would get all mixed up and stop making sense!

So when you sit down with a book, there are three helpful things to remember. First, find where the reading STARTS — usually the top line, at the left side. Second, move your finger under the words as you go, sliding from left to right. Third, turn the pages one at a time, in order. When you do these things, you are showing that you understand how print works. That makes you a real reader!

PICTURE CARDS TO TALK ABOUT

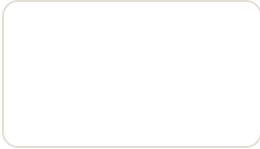
SAY These marks are print, and print carries meaning. Each word stands for something we can say!



SAY I read from left to right. My finger follows the words like a little train on a track.



SAY I read top to bottom. When one line ends, I hop down to start the next one on the left.



SAY Reading starts here — at the top, on the left side of the page.



SAY Books go page by page. I turn one page at a time, in order, so the story makes sense.



SAY Printing lets us make lots of copies of the same words, so everyone can have a book to read!

3 PRACTICE WORKSHEET

Read each question aloud with Hugh if they are still learning to read.

1. In English, which way do we move our eyes and finger when we read a line?

- ☐ From right to left
 - ☐ From left to right
 - ☐ From the bottom to the top
 - ☐ In a circle
-

2. Where do we usually START reading on a page?

- ☐ At the bottom-right corner
 - ☐ In the middle
 - ☐ At the top, on the left side
 - ☐ Anywhere we want
-

3. How should you turn the pages in a book?

- ☐ Skip a few pages each time
 - ☐ Turn one page at a time, in order
 - ☐ Start at the back and go forward
 - ☐ Flip them all at once
-

4. What does print do?

- ☐ It is just pretty squiggles with no meaning
 - ☐ It carries meaning — the letters stand for words
 - ☐ It is only for decoration
 - ☐ It tells us the color of the page
-

5. When you finish reading the top line of words, what do you do next?

6. Why do we read pages in order in a book?

4 SHOW WHAT YOU KNOW • TALK IT THROUGH

Ask each question out loud. Check the box when Hugh shows they understand.

- ☐ **Show me with your finger: which way do the words go across the page, and where do you begin?**
- Points to the top-left as the starting place
 - Moves finger from left to right across the line
 - Mentions or shows moving down to the next line (top to bottom)
- ☐ **Pretend you are reading a book to a younger child. Tell them how to turn the pages the right way.**
- Says to turn one page at a time
 - Says the pages go in order, not skipped or backward
 - Connects orderly page-turning to the story making sense
- ☐ **Look at a page of print. What are these marks, and what do they do?**
- Identifies the marks as print/words
 - Explains that print carries meaning
 - May note the letters stand for words we can say

WHERE THESE FACTS COME FROM

Simple English Wikipedia — “Printing is putting words and images on paper by mechanical means. Books and newspapers are made by printing. Usually, the images are formed by ink.” · “It is an efficient way to multiply writing, so that copies may be made available to all people.”

Knowing all letters

1 FOR THE GROWN-UP · TEACHING GUIDE

Why it matters. Recognizing and naming every uppercase and lowercase letter is the foundation of learning to read and write. Before children can sound out or spell words, they need to know the 26 letters by name in both cases and understand that letters follow a set order from A to Z.

HOW TO TEACH IT

Start by showing letters one at a time and naming them together, pointing out the big (uppercase) and small (lowercase) versions of each. Sing the alphabet song to build the A-to-Z sequence, then mix letters up so your child names them out of order. Finally, talk about letter names when spelling simple words, asking questions like 'Is it c or k in cat?'

WATCH FOR THESE MIX-UPS

- Mixing up letters that look similar, like b and d, or p and q
- Knowing the alphabet song but not being able to name a single letter shown alone or out of order
- Confusing a letter's name with the sound it makes
- Only recognizing uppercase letters and not their lowercase partners

TRY THIS AT HOME

- Play 'letter hunt' — find and name letters on signs, cereal boxes, and books, spotting both uppercase and lowercase versions
- Lay out alphabet cards in a jumble and take turns naming them in random order
- Sing the alphabet from A to Z, then pause and ask 'What comes after M?'
- When writing a word together, ask which letter to use, like 'Is it c or k?', so your child practices using letter names to explain spelling choices

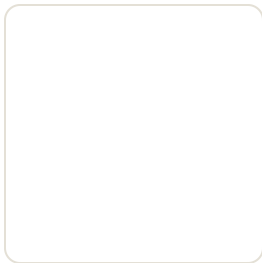
2 THE LESSON • READ TOGETHER

Every word you read and write is built out of letters — the little shapes that make up the alphabet. There are 26 letters in all, and each one has a name, like A, B, and C. Knowing all your letters means you can recognize each letter when you see it and say its name out loud, even when the letters are all mixed up and not in order. That is a big, important step in learning to read!

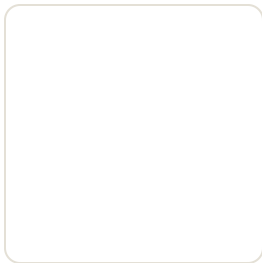
Here is a fun secret: every letter comes in two sizes. There is a BIG one called the uppercase (or capital) letter, and a little one called the lowercase letter. The letter 'A' can look like 'A' when it is big and 'a' when it is small. The letter 'B' looks like 'B' and 'b'. They have the same name and make the same sound, but they wear two different outfits. Uppercase letters like to stand at the start of a name or a sentence, while lowercase letters do most of the everyday work inside words. When you truly know your letters, you can name all 26 of them in both cases, even when someone shows them to you in a jumbled, random order.

Letters also love to line up in a special order — the alphabet — that always goes the same way: A, B, C, D, E, F, G, all the way to X, Y, Z. Singing or saying the alphabet from A to Z helps you remember which letter comes next. It is like a friendly parade where every letter marches in its own spot.

Best of all, once you know the NAMES of letters, you can use them to talk about how to spell words. Imagine you want to write the word 'cat.' You might ask, 'Does it start with a c or a k?' Both letters can sound the same, so knowing their names helps you choose the right one and explain your choice. Letter names are like tools that help you build words correctly. Long ago a man named Timothy Dexter wrote a whole book with many misspellings and letters capitalized in random places — it makes reading very tricky! That is a good reminder of why knowing your letters and using them carefully really matters.

PICTURE CARDS TO TALK ABOUT

SAY This is the letter A. When it is big it looks like 'A' (uppercase) and when it is small it looks like 'a' (lowercase). Same name, two sizes!



SAY Even when letters are all mixed up and not in order, you can still name each one: m, R, t, B, s, K. Knowing them in random order shows you really know your letters.



SAY Sing the alphabet from A all the way to Z. Every letter has its own spot in the parade, and it always goes in the same order.



SAY To spell 'cat,' I ask: is it a c or a k at the start? I choose c. Using letter names helps me explain my spelling choice.



SAY In his old book, Timothy Dexter capitalized letters in random places and misspelled many words, which makes it hard to read. That shows why we name our letters and use them carefully.

3 SHOW WHAT YOU KNOW • TALK IT THROUGH

Ask each question out loud. Check the box when Hugh shows they understand.

- ☐ **I will show you letters in a mixed-up order, like 'r, T, m, B, s.'** Name each letter you see. Tell me how you know its name.
- Names each shown letter correctly
 - Names both uppercase and lowercase letters when shown
 - Can name letters even when they are in random, non-alphabetical order
- ☐ **Recite the whole alphabet out loud, from A to Z, in order. You may sing it if you like.**
- Says all 26 letters
 - Keeps the letters in the correct A-to-Z order
 - Does not skip or repeat letters
- ☐ **Pretend you are spelling the word 'cat.' Explain which letter goes first and how you decided — is it a c or a k?**
- Uses letter names to explain the choice
 - Chooses a reasonable letter for the sound
 - Shows understanding that different letters can be options for a spelling

WHERE THESE FACTS COME FROM

English Wikipedia — “The book contains no punctuation, many misspellings and grammatical mistakes, and randomly capitalized letters.”

ENGLISH · SPEAKING & LISTENING · AGES 4–6

Exploring Ideas Through Talk

1 FOR THE GROWN-UP · TEACHING GUIDE

Why it matters. Spoken language is where thinking begins. An idea is simply the result of thought, and being able to make and understand ideas is a defining feature of being human. When children talk through their ideas, they are learning to speculate, imagine, and explore, and they build the words they will later use in reading and writing.

HOW TO TEACH IT

Model the language of wondering: say 'What if...' and 'I think it might be because...' during everyday moments. When a new word comes up in a story or lesson, use it again yourself and invite your child to try it in conversation. Before your child writes or draws, ask them to tell you their idea out loud first, so the idea is clear before it goes on paper.

WATCH FOR THESE MIX-UPS

- A child who says 'I don't know' instead of guessing — encourage any 'What if' guess, since speculating has no wrong answer here
- New words heard once but never used again — gently reuse the word so it sticks
- Rushing to write before the idea is clear — pause and talk it through first
- Confusing imagining (making things up) with explaining a real reason — both are great, just name which one they are doing

TRY THIS AT HOME

- Play 'What if...?' at dinner: take turns finishing 'What if animals could talk?'
- Pick one 'word of the day' from a story and challenge everyone to use it out loud three times
- Before drawing or writing, have your child tell you the whole idea like a mini story
- Wonder aloud together about everyday things: 'I think it might be raining because the sky is gray — what do you think?'

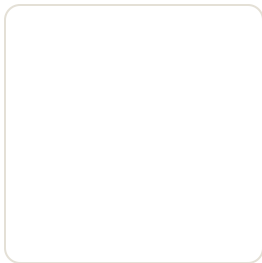
2 THE LESSON • READ TOGETHER

Have you ever had a little picture pop into your head, like a fire-breathing dragon, or a way to build the tallest block tower ever? That little picture in your mind is called an idea. An idea is what you get when you think! Grown-ups who study words say an idea is the result of thought. The word idea even comes from a very old language, and long ago it meant 'a form' or 'a pattern,' and it came from a word that means 'to see.' So when you have an idea, it is almost like you are seeing a little something inside your mind that no one else can see yet.

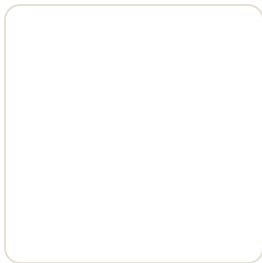
Here is something amazing: being able to make ideas and understand them is a special thing about being a person. It is one of the things that makes human beings so wonderful. And guess what? The best way to grow your ideas bigger and clearer is to TALK about them! When you say your ideas out loud, you can hear them, add to them, and share them with a friend.

There are fun ways to explore ideas with talk. One way is to WONDER. You can say, 'What if the moon was made of cheese?' or 'I think it might be because the plant needs water.' Wondering out loud helps you guess and imagine. Another way is to IMAGINE. You can use your imagination and fantasy to make up whole new pictures in your mind, like inventors do when they dream up brand new things. A fresh, brand-new idea can even lead to something no one has made before!

Talking is also how you catch new words. When you hear a new word in a story or a lesson, you can try saying it yourself in a conversation. The more you use a new word, the more it becomes YOUR word. And here is a great trick: before you write something down, talk about it first! Say your idea to a friend, a teacher, or even to yourself. Talking helps you get your idea ready. Then, when it is time to write, your idea is already clear and strong.

PICTURE CARDS TO TALK ABOUT

SAY That is WONDERING out loud! Saying 'What if...' helps you explore an idea and imagine a possibility.



SAY That is guessing an answer, called speculating. Saying 'I think it might be because...' lets you share your thinking.



SAY That is trying out a new word! Hearing 'enormous' in a story and then using it in talk helps the word become yours.



SAY That is using your imagination and fantasy, just like inventors do when they dream up brand-new ideas.



SAY That is exploring an idea through talk before writing. Talking first makes the idea clear and ready to write down.

3 PRACTICE WORKSHEET

Read each question aloud with Hugh if they are still learning to read.

1. Which sentence is a way of WONDERING out loud about an idea?

- ☐ 'Please pass the crayon.'
 - ☐ 'What if fish could fly?'
 - ☐ 'I am done now.'
-

2. You just heard the new word 'gigantic' in a story. What is a good way to make it your word?

- ☐ Forget it right away
 - ☐ Use it in a conversation, like 'That truck is gigantic!'
 - ☐ Only think it, never say it
-

3. What should you do before you write your story down?

- ☐ Talk about your idea out loud first
 - ☐ Close your eyes and stop thinking
 - ☐ Erase everything
-

4. Finish this wondering sentence out loud: 'I think the flower is growing because...'

5. An idea is what you get when you do what? (Hint: it starts with 'th')

4 SHOW WHAT YOU KNOW • TALK IT THROUGH

Ask each question out loud. Check the box when Hugh shows they understand.

☐ **Tell me a 'What if...?' idea about anything you like. Say it out loud in a full sentence.**

- Uses the 'What if...' form or clearly wonders about a possibility
- Expresses an imagined or speculated idea, not just a fact
- Speaks in a complete, understandable sentence

☐ **Here is a new word: 'curious,' which means wanting to know more. Use 'curious' in your own sentence about something in your day.**

- Correctly tries out the new word in a sentence
- Uses it in a way that fits its meaning
- Shows willingness to use vocabulary heard in the lesson

☐ **Think of a story you might write. Before writing, tell me your idea out loud: who is in it and what happens?**

- Explores the idea through talk before writing
- Includes at least a character and an event
- Speaks the idea clearly enough for a listener to picture it

WHERE THESE FACTS COME FROM

English Wikipedia — “An idea (from the Greek word: ἰδέα (idea), meaning 'a form, or a pattern') is the result of thought.” · “The word idea comes from Greek ἰδέα, romanized: idea, 'form, pattern', from the root of ἰδεῖν idein, 'to see'.”

Wikijunior — “relying on their human imagination, intuition and emotions ... Inventing novel electronic circuits is a result of human fantasy, imagination and enthusiasm.”

Single Letter Sounds

1 FOR THE GROWN-UP · TEACHING GUIDE

Why it matters. Knowing the primary sound for each consonant and both the short and long sounds for the five vowels is the foundation of reading. Being able to respond speedily to a single-letter grapheme lets a child sound out and blend words instead of guessing, which is the goal set out in early reading standards.

HOW TO TEACH IT

Show one letter at a time and ask for its most common sound, aiming for instant, automatic responses. Once single sounds are quick, move to hearing sounds inside short CVC words like 'cat' and 'bed', focusing especially on the middle vowel. Then introduce that vowels have two sounds — a short sound and a long sound that matches the letter's name.

WATCH FOR THESE MIX-UPS

- Confusing similar short vowels, such as the /a/ in 'cat' and the /e/ in 'bed'
- Slow, effortful recall — the goal is a speedy, automatic response to each letter
- Expecting one letter to always make one sound; in English, vowels especially make both a short and a long sound
- Mixing up a letter's NAME with its SOUND (the long-vowel sound happens to match the name, which can help or confuse)

TRY THIS AT HOME

- Flash single letters and have your child say the sound as fast as they can, like a game
- Say a CVC word slowly ('c-a-t') and ask what sound is in the middle
- Sort small objects or pictures by their short vowel sound
- Play 'short or long?' — say a vowel sound and have your child tell you whether it's the short sound or the letter's name (long sound)

2 THE LESSON • READ TOGETHER

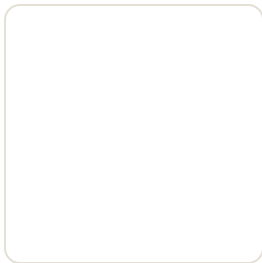
Every word you say and read is built out of tiny sounds. When you learn to read, one of the first and most important jobs is to look at a single letter and know the sound it makes. This is called a single-letter sound, and being able to say it quickly — almost as fast as you can blink — is a big step toward becoming a reader.

Letters come in two teams: consonants and vowels. Most consonants have one main sound they make most of the time. When you see the letter, your job is to say its most common sound right away. For example, when you see 'm' you make the /mmm/ sound, and when you see 's' you make the /sss/ sound. Learning the primary sound for each consonant, and saying it speedily when you see the letter, is the heart of this lesson.

Then there are the five special vowel letters: a, e, i, o, and u. The letter 'a' is the very first letter of our alphabet and the first vowel of all. Vowels are extra-special because they can make more than one sound! Each vowel has a short sound and a long sound. The short 'a' is the sound you hear in the middle of 'cat' and 'pad' — /a/. The long 'a' is the sound you hear when you say the letter's own name, like in 'ace' or 'say' — it sounds just like the name of the letter 'a'! Every vowel works this way: a short sound and a long sound that matches the letter's name.

Sounds live inside little words. A word like 'cat' has three sounds — /c/ /a/ /t/ — and a word like 'bed' has /b/ /e/ /d/. If you can hear that the middle of 'cat' is /a/ and the middle of 'bed' is /e/, you can tell those short vowel sounds apart, and that helps you read and spell lots of words.

Here is a tricky-but-fun secret: in English, one letter can stand for more than one sound, which is why we practice so carefully. In some languages, like Spanish, each letter almost always makes just one sound — but English letters, especially the vowels, like to surprise us. That is exactly why fast, sure knowledge of each letter's main sound is such a superpower for a young reader.

PICTURE CARDS TO TALK ABOUT

SAY When you see 'm', say its most common sound right away: /mmm/. Practicing until it pops out fast is the goal — quick and sure!



SAY Look at 's' and hiss like a snake: /sss/. Each consonant has one main sound you say the moment you see it.



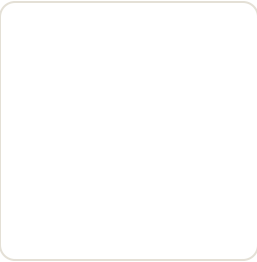
SAY The middle sound of 'cat' is the short a: /a/. You can also hear it in 'pad'. That's the short 'a' sound.



SAY The middle sound of 'bed' is the short e: /e/. Notice it is different from the /a/ in 'cat' — listen closely and you can tell them apart!



SAY In 'ace' and 'say', the 'a' says its own name: /ay/. That's the long 'a' — it sounds exactly like the letter's name.



SAY a, e, i, o, and u are the five vowels, and each one has TWO sounds — a short one and a long one. Consonants usually have just one main sound.



SAY See how 'a' can be /a/ in 'cat' but /ay/ in 'ace'? English letters can make more than one sound, so we practice the short and long sounds for every vowel.

3 SHOW WHAT YOU KNOW • TALK IT THROUGH

Ask each question out loud. Check the box when Hugh shows they understand.

☐ **I show you the letters m, s, and t. Say the most common sound for each one.**

- Gives one primary sound per consonant (e.g., /m/, /s/, /t/)
- Responds quickly and confidently rather than slowly
- Says the sound, not the letter name

☐ **Listen to 'cat' and 'bed'. Tell me the middle sound in each word and explain how they are different.**

- Identifies /a/ as the middle of 'cat'
- Identifies /e/ as the middle of 'bed'
- Explains that these are two different short vowel sounds

☐ **The letter 'a' can make two sounds. Give an example of the short 'a' sound and an example of the long 'a' sound, and tell how the long sound connects to the letter's name.**

- Gives a short 'a' example (e.g., 'cat' or 'pad')
- Gives a long 'a' example (e.g., 'ace' or 'say')
- Notes that the long sound matches the letter's own name

WHERE THESE FACTS COME FROM

English Wikipedia — “A is the first letter and the first vowel letter of the Latin alphabet; its name in English is the long A sound as in say; the letter represents the near-open front unrounded vowel /æ/ as in pad and the diphthong /eɪ/ as in ace and major”

Wikijunior — “In Spanish each letter represents a single sound and each vowel represents only one sound, while many letters and letter combinations in English represent multiple sounds”

Rhyming words

1 FOR THE GROWN-UP · TEACHING GUIDE

Why it matters. Hearing and making rhymes is one of the earliest and most important reading skills. Rhyme is a powerful memory helper — it makes patterns pleasant to hear and easy to remember — and it trains a child to notice the sounds inside words, which is the foundation for later reading and spelling.

HOW TO TEACH IT

Read rhyming books and nursery rhymes aloud often, then pause at the end of a line so your child can supply the matching word. Move from noticing rhymes ("cat and hat sound the same!") to producing them ("what else rhymes with cat?"). Sing songs, clap the beat, and make up silly rhymes together during everyday chores.

WATCH FOR THESE MIX-UPS

- Confusing rhyme (matching endings, like cat/hat) with words that just start the same (like cat/car).
- Offering a word with the right meaning but wrong sound — e.g., saying 'kitten' to rhyme with 'cat.'
- Thinking two words must be spelled alike to rhyme; rhyming is about the sound your ears hear, not how the word looks.

TRY THIS AT HOME

- Play 'Stand Up for Rhymes': recite a nursery rhyme and have your child stand up and repeat the pair whenever they hear a rhyme (like 'fox/box').
- Start a rhyme chain in the car or kitchen: you say 'day,' and take turns adding 'way, say, play, stay.'
- Pick one nursery rhyme — such as 'Hey Diddle Diddle' or 'Baa Baa Black Sheep' — and learn it by heart together to recite at bedtime.

2 THE LESSON • READ TOGETHER

Say these two words out loud: cat and hat. Do you hear how they end with the very same sound — "-at"? When words end with the same sound, we call them rhyming words. A rhyme means words that sound the same or similar in their endings. Your ears are like little sound-detectives: when two words rhyme, your ears smile because the endings match!

Rhymes are everywhere in the songs and poems you already love. Poems and popular songs use rhyme all the time, and a simple little poem can even be called a rhyme. Think of "Roses are red, violets are blue, sugar is sweet, and so are you." Listen to the endings — blue and you match! That is a rhyme hiding at the end of the lines. Nursery rhymes, folk songs, and children's songs are full of these matching sounds.

Here is a fun secret: once you know one rhyme, you can make a whole family of them. Try "day, way, say, may, bay, play, stay" — they all end in the same sound. Or "me, we, be, see, tree, knee." You can start with one word and keep the pattern going, adding word after word after word. If someone says "fox," you can rhyme back "box, ox, socks, rocks, locks, knocks"!

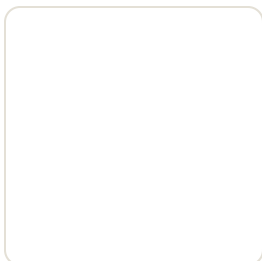
Why do we love rhymes so much? A rhyme is a pattern that is simply pleasant to hear, like a little beat for your ears. Rhymes also help us remember things — that is why it is so easy to keep a nursery rhyme in your head and say it by heart. When a grown-up reads a rhyming story and pauses at the end of a line, you can often shout out the word that comes next, because your ears already guessed the rhyme!

PICTURE CARDS TO TALK ABOUT

SAY Cat and bat rhyme because they both end with the "-at" sound. Dog is the odd one out — it does not match, so it does not belong in this rhyming family.



SAY Listen to the ends of the lines: blue and you sound the same. That is a rhyme sitting right at the end of the line, the most common place rhymes like to hide.



SAY Start with one rhyming word and you can make a whole chain! These all end in the same sound. Try it with 'me': we, be, see, tree, knee.



"Welcome Home - SS 2009
seconda prova" by the Italian
voice is licensed under CC

SAY When the grown-up pauses, you can fill in the rhyme: box! And you don't have to stop there — fox also rhymes with ox, socks, rocks, locks, and knocks.

SAY These are famous nursery rhymes you can learn by heart and recite. Once you know your favorite, you can say it all by yourself — the rhyme helps you remember the words.

3 SHOW WHAT YOU KNOW • TALK IT THROUGH

Ask each question out loud. Check the box when Hugh shows they understand.

☐ **Look at these three words: cat, bat, dog. Tell which two rhyme and explain how you know.**

- Correctly chooses cat and bat as the rhyming pair
- Explains that they rhyme because they sound the same at the end (the '-at' sound)
- Recognizes that dog does not rhyme with them

☐ **I say the word 'day.' Give me two more words that rhyme with it.**

- Produces at least two words that truly rhyme (e.g., way, say, may, play, stay)
- Words match the ending sound, not just the meaning or the first letter

☐ **Recite a nursery rhyme you know by heart, then point out one pair of words in it that rhyme.**

- Recites a nursery rhyme (such as Hey Diddle Diddle or Baa Baa Black Sheep) mostly from memory
- Correctly identifies one rhyming pair from the rhyme
- Shows enjoyment and joins in with the repeated pattern

WHERE THESE FACTS COME FROM

Simple English Wikipedia — “Rhyme means words that sound the same or similar in their endings. Poems and popular song lyrics often use rhyme. A simple poem can also be called a rhyme.” · “Roses are red, violets are blue, Sugar is sweet, and so are you.”

Core Knowledge — “We’ll catch a fox, and put him in a box... students may say: ox, docks, knocks, locks, pox, rocks, or socks.” · “Your child may come home singing and doing the motions for “Bat, Bat,” “Row, Row, Row Your Boat,” and “Once I Saw a Little Bird.””

English Wikipedia — “Rhyme partly seems to be enjoyed simply as a repeating pattern that is pleasant to hear. It also serves as a powerful mnemonic device, facilitating short-term memory.”

Onsets & Rimes (word families)

1 FOR THE GROWN-UP · TEACHING GUIDE

Why it matters. Hearing and playing with the sounds inside words is one of the first big steps to reading and spelling. Syllables are the building blocks of words, and when a child can blend, break apart, and change sounds, they are getting ready to match those sounds to letters. This skill supports the Year 1 English expectation to work with onsets, rimes, and individual phonemes in single-syllable words.

HOW TO TEACH IT

Start with blending: say an onset and rime slowly (/c/ ... /at/) and have your child smoosh them together to say 'cat.' Then practice segmenting by having them break a short word into first, middle, and last sounds. Finally, play the 'change one sound' game to make new words. Keep it playful and out loud, focusing on SOUNDS, not letters, at first.

WATCH FOR THESE MIX-UPS

- Confusing the middle vowel sound; the middle sound is the one you can sing or shout on its own.
- Adding extra 'uh' sounds to consonants (say a crisp /t/, not 'tuh').
- Mixing up first and last sounds, remind them 'first' is at the start and 'last' is at the end.
- Trouble hearing the rime as one chunk, model /c/ + /at/ a few times slowly.

TRY THIS AT HOME

- Play 'Robot Talk': you say a word in sound chunks (/s/ /u/ /n/) and your child blends it into 'sun.'
- Silly Rhymes: keep the rime and change the onset to make a family, cat, bat, hat, mat, sat.
- 'What's the last sound?' Point to objects and ask for just the first, middle, or last sound.
- Sing a word slowly and stretch the middle vowel like a eraser band.

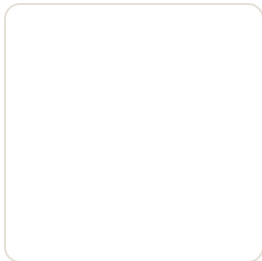
2 THE LESSON • READ TOGETHER

Every word we say is made of little sounds. When you put those sounds together, they build a whole word, kind of like blocks build a tower. Words are made of syllables, and syllables are the building blocks of words. A short word like 'dog' has just one syllable, so we call it a one-syllable word.

Inside a one-syllable word there are two special parts: the onset and the rime. The onset is the first sound, the sound at the very start of the word. The rime is the vowel sound in the middle plus any sound that comes after it. Let's use the word 'cat.' The onset is /c/, and the rime is /at/. If you say /c/ and then /at/ quickly, they smoosh together: 'cat!' That is called blending, when you put sounds together to make a word.

We can also do the opposite and pull a word apart. This is called segmenting. If someone says 'sun,' you can break it into its little sounds: /s/... /u/... /n/. Every one-syllable word like this has three parts you can listen for: the FIRST sound, the MIDDLE sound, and the LAST sound. The middle sound is almost always a vowel, like the /a/ in 'cat' or the /u/ in 'sun.' The middle vowel is the sound you can shout or sing all by itself.

Here is the most magical part. Once you can hear the sounds in a word, you can CHANGE a sound to make a brand-new word! Start with 'cat.' Take away the /c/ and put in a /b/, and now you have 'bat.' Change the /b/ to /h/ and you have 'hat.' Same rime, brand-new words. You can play with the first sound, the middle sound, or the last sound. This is how good readers and spellers train their ears, and it is a super-fun sound game you can play anywhere.

PICTURE CARDS TO TALK ABOUT

SAY This is blending an onset and a rime. The onset is the first sound /c/, and the rime is /at/. Say them fast together and you hear 'cat.'



SAY A word with just one syllable is like one building block. You can still hear a first sound /d/, a middle sound /o/, and a last sound /g/.



SAY The first sound (onset) is /c/, the middle sound (nucleus) is /a/, and the last sound (coda) is /t/. The middle and last sounds together, /at/, make the rime.



SAY The very last consonant sound after the vowel is called the coda. In 'cat' the coda is /t/. Listen for it at the end of the word.



SAY We swapped the first sound and made a new word! Keep the rime /at/ the same and just change the onset.



SAY You can change the middle vowel sound too. Trade the /a/ for an /o/ and 'cat' becomes 'cot.'

3 SHOW WHAT YOU KNOW • TALK IT THROUGH

Ask each question out loud. Check the box when Hugh shows they understand.

- ☐ **I say the sounds /c/ and /a/. Put them together and tell me the word. Then explain what you did.**
- Correctly blends to say 'cat'
 - Shows understanding that blending means putting sounds together to make a word
- ☐ **Say the word 'cat' slowly and tell me its first sound, middle sound, and last sound.**
- Identifies first sound /c/
 - Identifies middle sound /a/
 - Identifies last sound /t/
- ☐ **Start with the word 'cat.' Change one sound to make a brand-new word, and tell me which sound you changed.**
- Produces a valid new word (e.g., bat, hat, cot, cap)
 - Explains which sound (first, middle, or last) was changed
- ☐ **Why is a word like 'dog' called a one-syllable word? Explain in your own words.**
- Recognizes it has just one syllable/one chunk
 - Connects idea that syllables are the building blocks of words

WHERE THESE FACTS COME FROM

English Wikipedia — “In the one-syllable English word cat, the nucleus is a, the onset c, the coda t, and the rime at. This syllable can be abstracted as a consonant-vowel-consonant syllable, abbreviated CVC.” · “In phonology and studies of languages, syllables are often considered the 'building blocks' of words. A word that consists of a single syllable (like English dog) is called a monosyllable.”

Simple English Wikipedia — “A coda is the final consonant sound or sounds that follow the nucleus of a syllable. The nucleus and the coda together make up the rhyme.”

Blending Sounds to Read Words

1 FOR THE GROWN-UP · TEACHING GUIDE

Why it matters. Blending sounds is the engine of early reading. Reading is the process of taking meaning from written symbols through sight, and blending is how a child turns separate letter-sounds into a whole spoken word. Once children can blend fluently and read words without excessive sounding out, their working memory is freed up to focus on meaning — which research links to stronger comprehension and to reading becoming a source of pleasure rather than a chore.

HOW TO TEACH IT

Teach the letter-sound (grapheme-phoneme) links first, then model blending: point to each grapheme, say its sound crisply, then slide the sounds together into the word. Start with short, phonically regular words like 'cat', 'shop', and 'ship', including words where two letters make one sound (like 'sh'). Practice recognizing graphemes quickly so your child isn't stuck decoding letter by letter. Then move to comparing similar words such as 'big' and 'bag' so your child learns to notice the one differing sound.

WATCH FOR THESE MIX-UPS

- Saying sounds too slowly with big gaps, so the child can't hear the word inside them — model a smooth slide.
- Adding an extra 'uh' to consonants (saying 'buh' instead of a clean /b/), which makes blending harder.
- Guessing a word from the first letter or the picture instead of reading all the sounds left to right.
- Mixing up similar words like 'pin' and 'pan' or 'big' and 'bag' — a sign to slow down and listen to the middle sound.
- Treating letter pairs like 'sh' as two separate sounds instead of one.

TRY THIS AT HOME

- Play 'robot talk': you say a word in slow sounds (/f/-/i/-/sh/) and your child blends it into 'fish', then swap roles.
- Write three-sound words on cards and have your child touch each letter as they sound it out, then say it fast.
- Make minimal pairs ('big/bag', 'cat/cot', 'pin/pan') and ask your child which sound changed.
- Keep sessions short and playful, and read a favorite book together afterward so reading feels fun, not like a chore.
- Practice speedy sound flashcards so your child can name letter-sounds quickly without long pauses.

2 THE LESSON • READ TOGETHER

When you read, your eyes look at little marks called letters, and your brain turns those marks into meaning. Reading is the way we take in the sense of symbols of a written language by using our sight (and, for some people, by touch). But how do the letters turn into words you can actually say? The secret is a skill called blending!

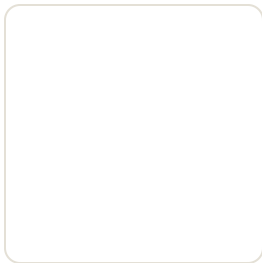
Every word is made of small sounds. The tiny sounds in words are called phonemes, and the letters (or groups of letters) that stand for those sounds are called graphemes. When you learn which sound each grapheme makes, that is called phonics. In English there are more than 40 different sounds, and a good reader learns to see a letter and say its sound quickly, without stopping to think for a long time.

Blending means saying each sound in a word one at a time, and then pushing them together to make the whole word. Watch: the word 'shop' is made of three sounds. First /sh/, then /o/, then /p/. Say them slowly: /sh/... /o/... /p/. Now say them faster and faster until they slide together — 'shop'! That sliding-together is blending. You can sound out words you have never seen before this way, as long as you know the letter sounds inside them.

At first you sound out every single letter out loud, and that is perfectly fine. But with practice, your brain remembers words and you can read them smoothly, without needing to sound out every sound. Reading depends on memory: you keep each sound fresh in your mind just long enough to blend them into a word. When you can read words quickly, you have more brain space left to enjoy the story and understand what it means.

Blending also helps you tell apart words that look almost the same. Look carefully at the letters! 'Big' and 'bag' look alike, but the middle letter is different: /b/-/i/-/g/ makes 'big', and /b/-/a/-/g/ makes 'bag'. That one little sound in the middle changes everything. Good readers listen closely to each sound, so they never mix up a cat and a cot, or a pin and a pan.

PICTURE CARDS TO TALK ABOUT



SAY Three sounds. Say them slowly, then slide them together: /c../a../t/ → 'cat'! That's blending three single-letter sounds.



SAY Here two letters, s and h, make one sound: /sh/. Blend /sh/-/o/-/p/ and you read the word 'shop'. Not every sound is just one letter!



SAY These words look almost the same. Read each sound: /b/-/i/-/g/ is 'big', but /b/-/a/-/g/ is 'bag'. The middle letter sound is different, so listen carefully.



SAY Point at each letter and blend: /p/-/i/-/n/ makes 'pin', and /p/-/a/-/n/ makes 'pan'. Changing one sound makes a brand new word.



SAY Even if you've never seen this word, you can sound it out: /sh/-/i/-/p/ → 'ship'. Phonics lets you read words you don't know yet!



SAY The more you practice, the faster you read. Soon you won't sound out each letter — your brain just knows the word, so you can enjoy the whole sentence.

3 SHOW WHAT YOU KNOW • TALK IT THROUGH

Ask each question out loud. Check the box when Hugh shows they understand.

☐ **Read this word by sounding it out: 'shop'. Tell an adult each sound you hear and then say the whole word.**

- Identifies the three sounds /sh/, /o/, /p/ in order
- Recognizes 'sh' as one sound, not two
- Blends the sounds smoothly into the correct word 'shop'

☐ **Look at the words 'big' and 'bag'. Explain how they are different and how you can tell them apart when you read.**

- Notices the words are spelled almost the same
- Identifies that the middle sound differs (/i/ vs /a/)
- Shows they read each sound to tell the words apart

☐ **Here is a word you may not have seen before: 'ship'. Sound it out and read it aloud, then read it a second time as smoothly as you can.**

- Uses letter-sounds to decode the unfamiliar word
- Blends /sh/-/i/-/p/ into 'ship' correctly
- Reads it more fluently the second time, with less sounding out

WHERE THESE FACTS COME FROM

English Wikipedia — “Reading is the process of taking in the sense or meaning of symbols, often specifically those of a written language, by means of sight or touch.”

Wikijunior — “Learning to read is a complex, multi-year process of learning to recognize the sounds and meanings of symbols; children must decode words at a reasonable speed so they don't have to hold meaning in memory too long.”

Group discussions

1 FOR THE GROWN-UP · TEACHING GUIDE

Why it matters. Group discussion is a core speaking-and-listening skill. Being able to share an interest with others, stay on topic, build on what someone else said, and reach a friendly agreement helps children solve problems together and strengthens their thinking and reasoning skills.

HOW TO TEACH IT

Model the three habits: staying on topic, building on others ('I agree because...'), and keeping focus while others speak. Keep groups small—three to twenty people works best so everyone can be heard. Introduce the idea of reaching agreement through discussion and compromise rather than simply voting to win.

WATCH FOR THESE MIX-UPS

- A child who changes the subject instead of staying on topic across several turns
- A child who only shares their own idea and never connects to what someone else said
- A child who loses focus when it is not their turn
- Insisting on 'winning' instead of finding a compromise everyone can accept

TRY THIS AT HOME

- Hold a two-minute family discussion on one topic (like dinner) and challenge everyone to stay on that topic the whole time.
- Practice sentence starters like 'I agree because...' or 'That makes me think of...' to build on each other's ideas.
- Make a small decision by consensus—talk, compromise, and agree together instead of voting.
- Praise your child when they politely 'stand aside' and let the group move forward.

2 THE LESSON • READ TOGETHER

Have you ever sat in a circle with friends and talked about something together—like what game to play, or a story you all read? That is a group discussion! A discussion group is a bunch of people who usually share the same interest, and who come together to talk about ideas, solve problems, or share what they think. Sometimes people talk in person, sitting close together, and sometimes people even talk using messages or websites, but the heart of it is always the same: taking turns sharing and listening.

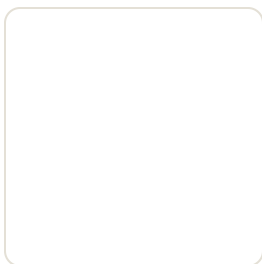
Most of the best discussions happen in a small group—somewhere from about three people up to around twenty. When the group is just the right size, everybody gets a chance to speak and everybody can hear each other. In a small group it is easier to pay attention, to solve a problem, and to make sure no one is left out.

So how do you become a great discussor? There are three big skills. First, **STAY ON TOPIC**. If the group is talking about which book to read next, you keep your ideas about books—not about your new puppy. Staying on topic for many turns helps the group actually get somewhere. Second, **BUILD ON WHAT OTHERS SAY**. Instead of only sharing your own idea, listen to your friend and connect to it. You might say, 'I agree because...' or 'That makes me think of...' This shows you were really listening, and it grows the conversation like adding a new block to a tower. Third, **KEEP YOUR FOCUS AND CONTRIBUTE**. That means paying attention even when it is not your turn, and then adding something helpful when it is your turn.

Groups also have to make choices together, and there is a friendly way to do this called consensus. Consensus means the group talks, finds a compromise, and comes to a general agreement—not just 'most people win,' but almost everybody agreeing to move forward. Sometimes you might not get exactly what you wanted, and you can 'stand aside,' which means politely saying, 'This isn't my first choice, but I won't stop the group.' Being respectful and courteous like this is one of the most appreciated things you can do. When people discuss ideas this way, they grow their thinking, their reasoning, and their ability to look at problems carefully—skills that help you your whole life.

PICTURE CARDS TO TALK ABOUT

SAY A discussion group is people who share an interest coming together to discuss ideas, solve a problem, or make comments—like friends choosing a book together.



SAY Small-group discussions of about three to twenty people work really well, because everyone gets a turn and can hear each other.



SAY Building on another speaker's idea—like saying 'I agree because...'—shows you listened and helps the conversation grow.



SAY Maintaining focus means paying attention even when it isn't your turn, so you can add something helpful when it is.



SAY Consensus is when a group discusses, finds a compromise, and comes to a general agreement everyone can accept.



SAY Standing aside politely—being respectful and courteous—lets the group move forward together.



SAY Discussing topics together helps grow your thinking, critical-thinking, and problem-solving skills.

3 SHOW WHAT YOU KNOW • TALK IT THROUGH

Ask each question out loud. Check the box when Hugh shows they understand.

☐ **Explain what it means to 'stay on topic' during a group discussion, and why it helps the group.**

- Describes keeping ideas connected to the same subject across multiple turns
- Explains that staying on topic helps the group discuss or solve the problem together
- Uses a clear example

☐ **Pretend a friend says, 'I think we should read the pirate story.' Write a reply that BUILDS on their idea instead of ignoring it.**

- Connects to the friend's idea rather than starting a new one
- Uses a building phrase such as 'I agree because...' or adds a reason
- Stays polite and on topic

☐ **Your group cannot all agree on the same choice. Describe how the group could reach an agreement in a friendly way.**

- Mentions discussing and finding a compromise (consensus)
- Notes coming to a general agreement rather than just the majority winning
- Includes being respectful, such as politely standing aside

WHERE THESE FACTS COME FROM

English Wikipedia — “A discussion group is a group of individuals, typically who share a similar interest, who gathers formally or informally to discuss ideas, solve problems, or make comments.” · “Small-group discussions, consisting of a minimum of three and maximum of about 20 people, have been found to be more effective in medical teaching.”

Wikijunior — “Consensus is when the community discusses a situation, finds compromise, and comes to a general agreement on a proper course of action... people who are in the minority tend to 'stand aside'... by being respectful and courteous, we can always improve and make forward progress.”

Describing Aloud

1 FOR THE GROWN-UP · TEACHING GUIDE

Why it matters. Describing aloud builds the foundation for clear speaking, confident presenting, and later, strong writing. Standards for young children ask them to describe familiar people, places, things, and events with relevant detail, and to speak audibly and express thoughts, feelings, and ideas clearly. These are core communication skills children use every day at school and home.

HOW TO TEACH IT

Model it first: describe something out loud yourself using several details in a sensible order, then invite your child to try. Give them a familiar topic they love (a pet, a favorite meal, their bedroom). Prompt for more with gentle questions like 'What color? What does it feel like? What happened next?' Praise both the details and the clear, loud voice.

WATCH FOR THESE MIX-UPS

- One-word or very short answers with no details " ask a follow-up question to draw out more
- A voice that is too quiet or mumbled for a group to hear
- Jumbled order, such as telling the end of an event before the beginning " help them use 'first, next, last'
- Wandering off topic instead of sticking to the person, place, thing, or event being described

TRY THIS AT HOME

- Play 'Guess What I'm Describing': describe an object in the room with three details and let your child guess, then switch roles
- Have a nightly 'best part of my day' share where your child tells the event in order: first, next, last
- Practice the 'back-of-the-room voice' by having your child describe something so a family member across the room can hear clearly
- Describe a family photo together, taking turns adding one new detail each

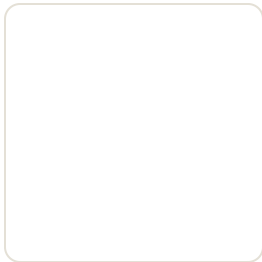
2 THE LESSON • READ TOGETHER

When you describe something aloud, you use your voice to paint a picture with words. You tell about a person, a place, a thing, or something that happened, and you add lots of little details so the people listening can see it in their minds, almost like they are right there with you. Describing well means two things: choosing good details, and saying them clearly and loudly enough for everyone to hear.

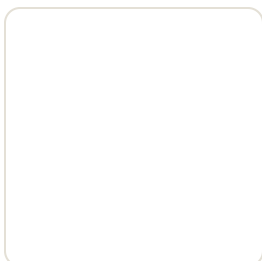
The first part of describing is **DETAIL**. A detail is a small true fact that helps your picture come alive. Instead of just saying "I saw a dog," you can say "I saw a fluffy brown dog with floppy ears that wagged its whole body when it ran." Now your listener can really see the dog! You can describe familiar people (like your grandma or your best friend), familiar places (like your kitchen or the park), familiar things (like your favorite toy), and events (like a birthday party or a trip). Good describers use **SEVERAL** details, not just one.

The second part is **SPEAKING CLEARLY AND LOUDLY**. It does not matter how wonderful your details are if nobody can hear them! When you share with a group, you sit or stand tall, look at your listeners, open your mouth, and speak so the person farthest away can hear every word. You also say your words clearly, not too fast and not mumbled, so your thoughts, feelings, and ideas come across.

The third part is **ORDER**. A good description is organized, which means your ideas come in a sensible order that is easy to follow. When you describe a place, you might go from top to bottom or front to back. When you tell about an event, you tell what happened first, next, and last. When you describe a thing, you might tell what it looks like, then what it feels like, then what it smells like. Putting your ideas in order helps your listeners understand and enjoy what you are sharing.

PICTURE CARDS TO TALK ABOUT

SAY "My baby brother has round cheeks, tiny fingers, and a big giggly laugh. When he is happy, he claps his hands." That uses several details so your listener can picture him.



SAY "Our kitchen has a red table by the window, a shelf full of cups, and it smells like warm toast in the morning." Notice how the details move around the room in order.



SAY "My teddy bear is soft and brown. It has one button eye and a little bow. It feels warm when I hug it." You told what it looks like, then what it feels like.



SAY "First we blew up balloons. Next we sang the birthday song. Last we ate cake." Telling it first, next, last keeps your story organized and easy to follow.



SAY A good describer can even tell how something looks and smells. In an old story, a monster called the Wendigo is described as ugly and smelly here, without seeing it, you can imagine it because of the details.



SAY Before you share, take a breath, stand tall, and imagine your voice reaching the very back of the room: "THE PARK HAS A TALL SILVER SLIDE!" Speaking audibly means everyone in the group can hear you.

3 SHOW WHAT YOU KNOW • TALK IT THROUGH

Ask each question out loud. Check the box when Hugh shows they understand.

- ☐ **Describe a person you know well, like a family member or friend. Use at least three details so a listener can picture them.**
 - Names or clearly identifies a familiar person
 - Includes at least three relevant details
 - Details help create a clear picture (appearance, actions, or personality)
- ☐ **Describe your favorite room or place. Try to put your details in an order that is easy to follow.**
 - Identifies a familiar place
 - Uses several details about the place
 - Organizes details in a logical order (such as around the room or top to bottom)
- ☐ **Tell about something that happened to you recently. Use the words 'first,' 'next,' and 'last' to keep your story in order.**
 - Tells about a real event
 - Uses first/next/last or another clear order
 - Ideas are organized logically and easy to follow
- ☐ **Practice being heard: describe an object in the room in one clear sentence, spoken loudly enough for everyone to hear. Write down the sentence you would say.**
 - Sentence describes a real object with detail
 - Sentence is complete and clear
 - Shows understanding of speaking audibly for a group

WHERE THESE FACTS COME FROM

Simple English Wikipedia — “The wendigo is said to be ugly and smelly.”

Expressing & Justifying Opinions

1 FOR THE GROWN-UP · TEACHING GUIDE

Why it matters. Being able to say what you think and add a reason is the beginning of clear thinking and confident speaking. At this age children learn that an opinion is a viewpoint that is not the same as a fact, and that a reason turns a simple opinion into a real argument. These spoken-language skills — discussing, presenting, and explaining thinking — are core to the English curriculum and help children in every subject later on.

HOW TO TEACH IT

Model the sentence frame 'I think ____ because ____' many times a day in real situations: choosing snacks, books, or games. Ask 'why do you think that?' whenever your child gives a one-word answer, gently coaching them to add a 'because.' Play fact-versus-opinion games and give lots of low-pressure chances to speak up — at dinner, during story time, and in pretend play.

WATCH FOR THESE MIX-UPS

- Giving only one-word answers ('yes,' 'good') without any reason — prompt with 'because...'
- Mixing up facts and opinions (thinking 'I like it' is the same as 'it is true for everyone')
- Being too shy to speak in front of others — build confidence with small, friendly audiences first
- Getting upset when someone disagrees — remind them different people can have different opinions and both can be okay

TRY THIS AT HOME

- At dinner, everyone shares 'My favorite part of today was ____ because ____.'
- Play 'Fact or Opinion?': say a sentence and have your child sort it — 'The grass is green' (fact) vs. 'Green is the best' (opinion).
- Set up a mini pretend show or role play where your child presents a toy and tells the family why they like it.
- Have a friendly two-person debate: 'Are cats or dogs better?' Each person gives one reason, then hug it out.

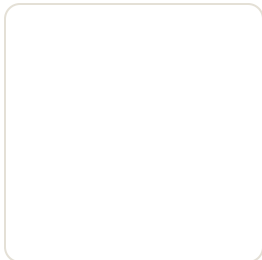
2 THE LESSON • READ TOGETHER

An opinion is something YOU think or feel. It is a viewpoint or a statement that is not the same as a plain fact. A fact is something true that everybody can check and agree on, like "the sky is up high." An opinion is what you believe or like, like "blue is the best color." Two friends can both look at the very same thing and still have different opinions about it — and that is okay!

Here is the special trick that makes an opinion strong: give a REASON. When you tell why you think something, your opinion turns into an argument, which just means a good, careful reason for what you believe. You can use these magic words: "I think ____ because ____." The "because" part is the reason. "I think dogs are the best pets BECAUSE they play with me." Now people understand your thinking, not just your answer.

Your opinions come from YOU — from the way you see things, the way you feel, and the things you like and want. That is why your opinion is special and worth sharing. But remember, since it comes from your feelings, other people might feel differently, and that is fine.

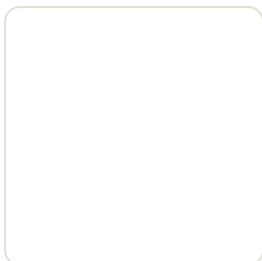
We share our opinions by talking out loud. You can share in a class discussion, when you answer a teacher's question, when you stand up to give a presentation or a performance, when you play pretend (role play), or in a friendly debate where two people share different opinions and their reasons. When it is your turn, try not to give just a one-word answer. Explain your thinking! Say the whole idea, add your "because," and speak with confidence so everyone can hear your good ideas.

PICTURE CARDS TO TALK ABOUT

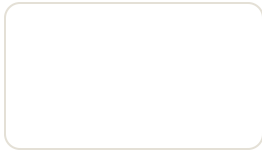
SAY This is a strong opinion! It says what you think AND gives a reason with the word 'because.'



SAY A fact is true for everybody. An opinion is what YOU think or feel.



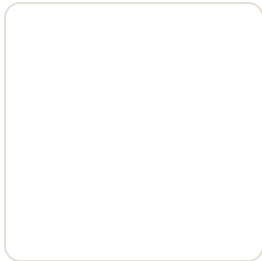
SAY Different people can have opposite opinions about the very same thing — and both can give reasons!



SAY This is giving a presentation with confidence — speaking so everyone can hear your idea.



SAY Instead of just saying 'yes,' you explained your thinking. That's a full, strong answer!



SAY Even in pretend play you can practice giving an opinion and a reason.

3 SHOW WHAT YOU KNOW • TALK IT THROUGH

Ask each question out loud. Check the box when Hugh shows they understand.

☐ **Tell me one thing you like and WHY. Use the words 'I think ____ because ____.'**

- States a clear opinion (what they like or think)
- Includes a reason using 'because'
- Speaks in a full sentence rather than one word

☐ **I say a sentence and you tell me if it is a FACT (true for everyone) or an OPINION (what someone thinks). Then explain how you knew.**

- Correctly sorts fact from opinion
- Explains that a fact can be checked/true for everyone and an opinion is what someone thinks or feels
- Gives more than a one-word answer

☐ **Stand up and give a short 'presentation' to your family: name your favorite animal, then give one reason you like it.**

- Speaks in front of others with confidence and a clear voice
- States an opinion (favorite animal)
- Supports it with at least one reason

WHERE THESE FACTS COME FROM

English Wikipedia — “An opinion is a judgement, viewpoint, or statement that is not conclusive, as opposed to facts, which are true statements.” · “Distinguishing fact from opinion is that facts are verifiable, i.e. can be agreed to by the consensus of experts.”

Vowel Digraphs

1 FOR THE GROWN-UP · TEACHING GUIDE

Why it matters. Vowel digraphs (vowel teams) are a big step up from sounding out single letters. Once your child knows that two letters can join to make one sound — and that a team like 'oo' can even make more than one sound — they can read and spell hundreds of common words like 'tree', 'boat', 'rain', and 'coin'. This is a core early-reading skill in both the U.S. and U.K. curricula.

HOW TO TEACH IT

Teach one sound family at a time. Show that the two letters make ONE sound, not two. Use word building: say a word, ask which vowel team is hiding inside, and have your child write it. For the two-sound teams like 'oo', pair the words side by side (moon vs. book) so your child hears and feels the difference. Encourage the 'try both sounds' detective strategy when reading a new word.

WATCH FOR THESE MIX-UPS

- Sounding out each vowel separately (saying b-o-a-t as two vowel sounds instead of one /oa/).
- Getting stuck when a team like 'oo' makes an unexpected sound — remind them to try the other sound and pick the word that makes sense.
- Mixing up which team goes where, like using 'ai' at the end of a word where 'ay' belongs (day, not dai).

TRY THIS AT HOME

- Go on a 'vowel team hunt' in a picture book and circle every ee, oa, ai, oi, and oo you find.
- Play the moon-or-book game: say 'oo' words and have your child point up (long /oo/, moon) or curl up small (short /oo/, book).
- Say a word aloud (rain, boat, coin) and have your child spell it by choosing the right vowel team.
- Watch a short spelling show together, like an Alphablocks vowel-team episode (Train, Beep, Toad, Book), then find those words around the house.

2 THE LESSON • READ TOGETHER

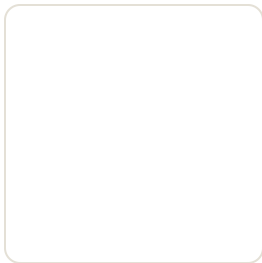
Have you ever noticed that some words have two letters standing right next to each other, holding hands, that make just ONE sound? When two letters team up to make one sound, we call them a digraph. The word digraph comes from old words that mean "two" and "to write" — so it really means "two letters written together" that make a single sound. When those two teammates are vowels, we call them a vowel digraph, or a vowel team.

Why do we need vowel teams? Because some sounds in English are hard to spell with just one letter. So the letters work together like partners on the playground. In the word 'tree', the two e's (ee) hold hands and say one long /ee/ sound. In the word 'boat', the letters o and a (oa) team up to say a long /oa/ sound. Long ago, people started doubling letters like 'ee' and 'oo' to show a long, stretched-out vowel sound, and we still spell words that way today.

There are lots of vowel teams to meet. Some make the long /ai/ sound you hear in 'rain': the teams ai and ay. Some make the long /ee/ sound: the teams ee and ea. Some make the long /oa/ sound: the teams oa and oe. And some make bouncy sounds like the /oy/ you hear at the end of 'boy': the teams oi and oy.

Here is the tricky, fun part: the same vowel team can sometimes make more than one sound! Think about the team 'oo'. In the word 'moon' it says a long, hooting /oo/. But in the word 'book' it says a short, gentle /oo/. Same two letters — two different sounds! English is sneaky like that. The letters 'ou' sound different in 'noun' and in 'cough', too. That is why reading is like being a detective: you look at the letters, try the sounds you know, and listen for the word that makes sense.

So when you read, hunt for the vowel team, remember it makes ONE sound, and try it out loud. When you spell, listen for the vowel sound in the middle of the word and choose the right team of letters to write it. With a little practice, you will spot these letter partners everywhere!

PICTURE CARDS TO TALK ABOUT

SAY The team 'ai' says the long /ai/ sound. In 'train' the a and i hold hands to make one sound.



SAY The team 'ay' also says the long /ai/ sound, and it likes to sit at the end of words like 'day'.



SAY The double letters 'ee' say the long /ee/ sound. Two e's stretch the vowel out long.



SAY The team 'ea' can also say the long /ee/ sound, just like in 'ee' words.



SAY The team 'oa' says the long /oa/ sound. The o and a work together in 'boat'.



SAY The team 'oe' can say the long /oa/ sound too, like at the end of 'toe'.

PICTURE

moon (oo) vs. book
(oo)

SAY The team 'oo' can make TWO sounds: a long hooting /oo/ in 'moon' and a short gentle /oo/ in 'book'.



SAY The team 'oi' makes a bouncy /oy/ sound in the middle of words like 'coin'.



SAY The team 'oy' makes the same bouncy /oy/ sound and likes to sit at the end of words like 'boy'.

3 SHOW WHAT YOU KNOW • TALK IT THROUGH

Ask each question out loud. Check the box when Hugh shows they understand.

☐ **Read these three words aloud: 'tree', 'boat', and 'coin'. For each one, tell which two letters are the vowel team.**

- Reads all three words correctly
- Names 'ee' in tree, 'oa' in boat, and 'oi' in coin
- Understands the two letters make one sound together

☐ **The team 'oo' can make two different sounds. Say the words 'moon' and 'book' and explain how the 'oo' sound is different in each one.**

- Pronounces both words correctly
- Explains that 'moon' has a long/hooting /oo/ and 'book' has a short/gentle /oo/
- Shows understanding that the same digraph can make more than one sound

☐ **I will say three words: 'day', 'sea', and 'boy'. Spell each one, using the correct vowel team.**

- Spells 'day' with 'ay'
- Spells 'sea' with 'ea'
- Spells 'boy' with 'oy'

WHERE THESE FACTS COME FROM

English Wikipedia — “A digraph ... is a pair of characters used in the orthography of a language to write ... a single phoneme (distinct sound)” · “In Middle English, the sequences ⟨ee⟩ and ⟨oo⟩ were used ... to represent lengthened "e" and "o" sounds respectively; both spellings have been retained in modern English orthography”

Simple English Wikipedia — “Alphablocks episodes: Train (vowel team AI), Beep (vowel team EE), Toad (vowel team OA), Book (vowel team OO), Cowboy (digraphs OY and OU)”

Wikijunior — “the vowel digraph "ou" in "noun" and "cough" ... the written and spoken language can differ a great deal”

Consonant Digraphs

1 FOR THE GROWN-UP · TEACHING GUIDE

Why it matters. Learning that two letters can make one sound is a huge step in early reading and spelling. Without knowing digraphs like ch, sh, th, ck, tch, ng, ph, and wh, a child would try to sound out each letter separately and get stuck. Mastering these letter teams lets your child read and spell hundreds of everyday words confidently.

HOW TO TEACH IT

Start by explaining that a digraph is two letters that make one single sound. Point to the digraph in a word and cover the rest, having your child make just that one sound, then blend the whole word together. Teach the tricky ones directly: 'ph' says /f/ and 'wh' usually says /w/. For spelling, show the pattern that short words often end in -ck for the /k/ sound (duck) and -tch for the /ch/ sound (match).

WATCH FOR THESE MIX-UPS

- Sounding out both letters separately (saying 's-h' instead of one /sh/ sound).
- Reading 'ph' as 'puh' instead of the /f/ sound.
- Forgetting the silent letters and writing 'duk' instead of 'duck' or 'mach' instead of 'match'.
- Confusing similar-looking digraphs like ch and sh at the start of words.

TRY THIS AT HOME

- Go on a 'digraph hunt' around the house or in a storybook—find words with sh, ch, th, or ck.
- Play a sound game: say 'phone' and 'photo' and ask what sound ph makes.
- Use letter cards to build and spell words like 'duck', 'sock', 'match', and 'catch'.
- Take turns whispering 'sh' words and clapping the /ch/ 'chug' sound in ch words.

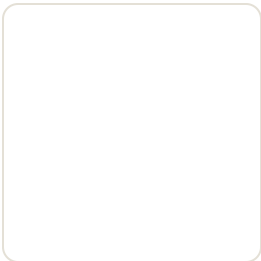
2 THE LESSON • READ TOGETHER

Have you ever noticed that some words have two letters standing side by side that team up to make just ONE sound? When two letters work together like that, we call them a digraph. The word "digraph" means "two letters," and even though your eyes see two letters, your mouth only makes one sound! This is a special trick that shows up in many, many words you will read and write.

Let's meet the most common consonant digraphs. First there is 'sh', which makes a quiet "shhh" sound, like when you tell someone to be quiet. You hear it in 'shop'. Next is 'ch', a short punchy sound like a train chugging, which you hear in 'chip'. Then comes 'th', a soft buzzy sound made with your tongue peeking between your teeth, like in 'that'. There is also 'wh', which usually just makes the /w/ sound, like the word 'when'. And 'ng' hums at the back of your mouth, like the end of 'ring' or 'song'.

Some digraphs are trickier because the sound does not match the letters at all! The digraph 'ph' does not say "puh-huh." Instead, 'ph' makes the /f/ sound, just like the letter f. So when you see 'phone', you say it like it starts with an f. And the letter 'k' can help spell the /k/ sound too, the same sound that 'c' can make.

We also use digraphs when we spell certain endings. When a short little word ends with the /k/ sound right after one short vowel, we usually spell it with '-ck', like in 'duck', 'sock', and 'kick'. And when a short word ends with the /ch/ sound right after one short vowel, we often spell it with '-tch', where the silent t joins ch, like in 'match', 'catch', and 'fetch'. Once you learn to spot these letter teams, reading and spelling become much easier, because you know exactly what sound to make!

PICTURE CARDS TO TALK ABOUT

SAY The letters s and h team up to make one "shhh" sound. Blend it: sh-o-p, shop!



SAY c and h together make the chugging /ch/ sound. Read it: ch-i-p, chip!



SAY t and h make the buzzy /th/ sound with your tongue near your teeth: th-a-t, that!



SAY w and h together make the /w/ sound. Say it: wh-e-n, when!



SAY p and h do not say 'puh'—they make the /f/ sound, so 'phone' begins like 'fun'.



SAY n and g hum together at the end to make one /ng/ sound: r-i-ng, ring!



SAY After the short vowel, we spell the /k/ sound with -ck: d-u-ck, duck!



SAY After the short vowel, we spell the /ch/ sound with -tch: m-a-tch, match!

3 SHOW WHAT YOU KNOW • TALK IT THROUGH

Ask each question out loud. Check the box when Hugh shows they understand.

☐ **Read these three words out loud: 'shop', 'chip', and 'that'. Tell an adult what the underlined letter team sounds like in each one.**

- Correctly reads all three words by blending sounds
- Identifies sh as /sh/, ch as /ch/, and th as the /th/ sound
- Treats each digraph as one sound, not two separate letter sounds

☐ **Look at the words 'phone' and 'when'. Say each word and explain what sound 'ph' makes and what sound 'wh' makes.**

- States that ph makes the /f/ sound
- States that wh makes the /w/ sound
- Uses the words correctly when reading them aloud

☐ **Spell two words: one that ends in -ck (like a bird that quacks) and one that ends in -tch (something you do with a ball). Write both words.**

- Spells a -ck word correctly, such as 'duck'
- Spells a -tch word correctly, such as 'match' or 'catch'
- Uses the full digraph endings with the silent letters included

WHERE THESE FACTS COME FROM

English Wikipedia — “This is a list of digraphs used in various Latin alphabets... For instance, ⟨ch⟩ becomes ⟨Ch⟩.”

Simple English Wikipedia — “A few consonant digraphs (tr, dr, ts) are used... Consonant Digraphs reference.”

Segmenting words into sounds

1 FOR THE GROWN-UP · TEACHING GUIDE

Why it matters. Segmenting spoken words into their individual sounds is the foundation of spelling. Before a child can write a word, they need to hear the separate sounds (phonemes) inside it and then match a letter to each one. This skill lets young writers spell CVC words like 'cat,' 'bed,' and 'hot' from dictation and move on to other simple, phonetically regular words.

HOW TO TEACH IT

Teach in three clear steps: SAY the word, SEGMENT it into each sound out loud, then CHOOSE a letter for each sound. Start with CVC words that spell exactly the way they sound. Have your child stretch the word slowly and push a counter, clap, or tap a finger for each sound before writing. Always do the listening (segmenting) first and the letter-writing second.

WATCH FOR THESE MIX-UPS

- Writing only the first and last sound and skipping the vowel in the middle — remind them every small word needs a vowel sound
- Rushing to write letters before saying and breaking the word into sounds
- Confusing the number of letters with the number of sounds — start with words where they match one-to-one
- Blurring sounds together; slow the word down so each segment is separate and in order

TRY THIS AT HOME

- Play 'Robot Talk': say a word in slow robot sounds (/d/ /o/ /g/) and have your child guess and then write it
- Use three coins or buttons and push one forward for each sound you hear in a word
- Dictate a short list of CVC words (cat, bed, hot, sun, map) for your child to spell by sounding out
- Say a word and ask 'What sound is in the middle?' to build attention to the vowel

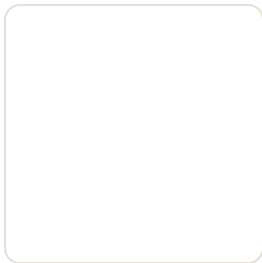
2 THE LESSON • READ TOGETHER

When you say a word out loud, it may sound like one smooth sound. But if you listen very closely, most words are really made of tiny sounds joined together. These tiny sounds are called segments, and a segment is one of the smallest units you can hear in the stream of speech. Grown-ups who study language call the smallest sound you can hear a phone, and the sounds that matter in a word phonemes. When we segment a word, we break it apart, sound by sound, in the same order we say it.

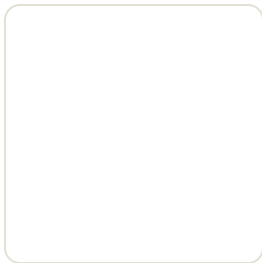
Here is the exciting part: the sounds come one after another, in order, from the beginning of the word to the end. Because they are separate little units, we can point to each one, stretch it out, and count it. Try saying the word "cat" slowly, like this: /c/ ... /a/ ... /t/. Did you hear three sounds? That word has three segments. Say "hot" the same way: /h/ ... /o/ ... /t/. Three sounds again!

When we speak, sounds come in two big families. Some are consonants and some are vowels. A vowel is a very special open sound. The letter A is a vowel, and so are E, I, O, and U. Every little word needs a vowel sound in the middle to hold it together. Words like "cat," "bed," and "hot" are called CVC words, which stands for consonant-vowel-consonant, because they go: a first sound, then a vowel sound, then a last sound.

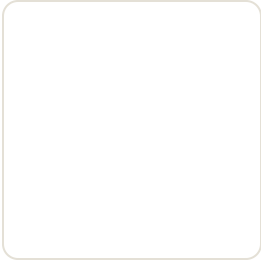
Segmenting is the secret trick that helps you spell. Here is how it works: first you SAY the word, then you SEGMENT it into each sound, and THEN you choose a letter (or letters) to write for each sound. Sound first, letter second! If you say "bed" and hear /b/ /e/ /d/, you write b, then e, then d. When you match one letter to each sound you hear, you are spelling phonetically, which just means spelling by listening. Push a counter or tap your fingers for each sound, and soon you can write whole words all by yourself.

PICTURE CARDS TO TALK ABOUT

SAY Stretch it out: /c/ /a/ /t/. Three sounds! Write a letter for each one: c-a-t. This is a CVC word with the vowel /a/ in the middle.



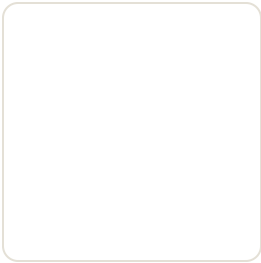
SAY Say it slowly: /b/ /e/ /d/. Three sounds. Choose a letter for each: b-e-d. The vowel sound /e/ holds the middle together.



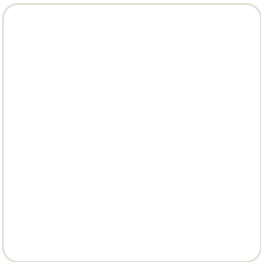
SAY Break it apart: /h/ /o/ /t/. Three sounds, three letters: h-o-t. Notice the vowel /o/ is right in the middle.



SAY A, E, I, O, and U are vowels. The letter A is a vowel sound. Every small word needs a vowel sound in the middle, like the /a/ in cat or the /u/ in sun.



SAY Listen: /m/ /a/ /p/. Say each sound, then write it: m-a-p. Because you can hear every sound clearly, you can spell it by sound.



SAY Sounds come one after another in order. Tap a finger for each: /s/ /i/ /t/ = three taps = s-i-t. Tapping helps you keep the sounds separate and in the right order.

3 SHOW WHAT YOU KNOW • TALK IT THROUGH

Ask each question out loud. Check the box when Hugh shows they understand.

☐ **I say the word 'bed.' Break it into each sound out loud, then write the word.**

- Segments the word into three sounds: /b/ /e/ /d/
- Writes the letters b-e-d in the correct order
- Includes the middle vowel sound

☐ **Explain the steps you use to spell a word you hear, from start to finish.**

- States that you say the word first
- States that you break (segment) it into separate sounds
- States that you choose a letter for each sound and write them in order

☐ **Spell these three words that I say aloud: 'cat,' 'hot,' 'sun.' Write a letter for each sound you hear.**

- Correctly segments each word into its sounds
- Spells each word phonetically with a letter for each sound
- Includes the vowel in the middle of each word

WHERE THESE FACTS COME FROM

English Wikipedia — “a segment is 'any discrete unit that can be identified, either physically or auditorily, in the stream of speech’” · “segments will typically be grouped into consonants and vowels”

Simple English Wikipedia — “A is the first letter of the English alphabet. The small letter, a, is used as a lowercase vowel.”

Wikijunior — “research-based activities... Phonemic Awareness, Phonics, Vocabulary, Fluency, and Comprehension”

Phonics Vocabulary

1 FOR THE GROWN-UP · TEACHING GUIDE

Why it matters. Phonics is one of the five big ideas in beginning reading (along with phonemic awareness, vocabulary, fluency, and comprehension). When children know the vocabulary of phonics — words like phoneme, grapheme, blend, and segment — they can talk about exactly what they are doing when they read and spell, which makes learning faster and clearer.

HOW TO TEACH IT

Keep it playful and hands-on. Say a sound and point to the letter that makes it so your child sees GPC in action. Practice blending by saying sounds slowly and sliding them together to read a word, and practice segmenting by 'stretching' a word out to spell it. Use everyday CVC words like cat, dog, and sun, and point out digraphs (sh, ch, th) when you spot them in books.

WATCH FOR THESE MIX-UPS

- Confusing phoneme (the sound) with grapheme (the letter) — remind them: sound comes from the mouth, letter is what we write
- Trying to read a digraph like 'sh' as two separate sounds instead of one
- Mixing up blending (for reading) and segmenting (for spelling) — blend to read, segment to spell
- Forgetting that some sounds are made by more than one letter

TRY THIS AT HOME

- Play 'Robot Talk': you say a word in separate sounds (/c/ /a/ /t/) and your child blends it back together
- Clap the syllables in family members' names
- Hunt for digraphs (sh, ch, th, ph, wh) on cereal boxes and signs
- Say a short word and have your child segment it into sounds using their fingers, one finger per sound

2 THE LESSON • READ TOGETHER

Have you ever wondered how a bunch of little marks on a page turn into words you can read? Reading means taking in the meaning of symbols, using your eyes to see them (or even your fingers to touch them, the way some people read with bumpy dots). To become a strong reader, you learn something called phonics — one of the five big ideas that help kids learn to read, along with phonemic awareness, vocabulary, fluency, and comprehension. Phonics is all about matching the sounds you say with the letters you see.

Just like builders have special words for the parts of a house — bricks, doors, windows — readers have special words for the building blocks of words. Learning these words helps you talk about what your mouth and eyes are doing. Let's meet the building blocks!

A phoneme is the tiniest little sound in a word. When you say 'cat' slowly, you hear three sounds: /c/ /a/ /t/. Each one is a phoneme. A grapheme is the letter or letters we write down to show that sound. So the sound /c/ can be written with the grapheme 'c'. When you match a grapheme to its phoneme — letter to sound — that matching game is called GPC, which stands for grapheme-phoneme correspondence.

Sometimes two letters team up to make just one sound. That team is called a digraph. English uses digraphs like 'ch', 'sh', 'th', 'ph', and 'wh' — two letters, but only one sound. Isn't that clever?

The English alphabet has 26 letters. Five of them are vowels — a, e, i, o, and u — and the other 21 are consonants. Lots of first words you read follow the pattern CVC, which means consonant-vowel-consonant, like 'cat', 'dog', and 'sun'.

Two super-important jobs use special words too. When you read, you blend: you push the sounds together to make a word (/d/ /o/ /g/ ... dog!). When you spell, you segment: you break a word apart into its separate sounds so you know which letters to write. Blend to read, segment to spell!

Longer words have beats called syllables — clap 'ba-na-na' and you hear three. Words can also have parts with jobs: the root word is the main word, a prefix is a piece added to the front, and a suffix is a piece added to the end. And watch out for homophones — words that sound the same but are spelled differently and mean different things, like 'sun' and 'son'. These building-block words are the tools that phonics is built on.

PICTURE CARDS TO TALK ABOUT

SAY Each tiny sound is a PHONEME. 'Cat' has three phonemes.

SAY A letter (or letters) we write to show a sound is a **GRAPHEME**.

SAY Matching a grapheme to its phoneme is GPC — grapheme-phoneme correspondence.

SAY These are **DIGRAPHS** — two letters that make just one sound.

SAY These five letters are the **VOWELS** of the alphabet.

SAY The other 21 letters are **CONSONANTS**.

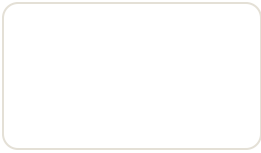
SAY These are CVC words: consonant-vowel-consonant.

SAY Pushing sounds together to read is called **BLENDING**.

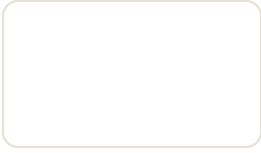
SAY Breaking a word into its sounds to spell is called **SEGMENTING**.

SAY Each beat you clap is a **SYLLABLE**.

SAY 'happy' is the **ROOT WORD** and 'un-' is a **PREFIX** added to the front.



SAY '-ing' is a SUFFIX added to the end of the root word.



SAY These are HOMOPHONES — they sound the same but are spelled differently and mean different things.

3 PRACTICE WORKSHEET

Read each question aloud with Hugh if they are still learning to read.

1. What do we call the tiniest sound in a word, like the /c/ in 'cat'?

- ☐ A grapheme
 - ☐ A phoneme
 - ☐ A syllable
-

2. 'sh' is two letters that make just ONE sound. What is that called?

- ☐ A digraph
 - ☐ A vowel
 - ☐ A prefix
-

3. Which set shows the five vowels?

- ☐ b, c, d, f, g
 - ☐ a, e, i, o, u
 - ☐ s, h, c, t, n
-

4. You push the sounds /d/ /o/ /g/ together to read 'dog'. What are you doing?

- ☐ Segmenting
 - ☐ Blending
 - ☐ Clapping
-

5. Which word is a CVC word (consonant-vowel-consonant)?

- ☐ sun
 - ☐ happy
 - ☐ banana
-

6. Break the word 'map' into its separate sounds. What is this called?

7. 'sun' and 'son' sound the same but are spelled differently. What are they?

- ☐ Digraphs
- ☐ Homophones
- ☐ Syllables

4 SHOW WHAT YOU KNOW • TALK IT THROUGH

Ask each question out loud. Check the box when Hugh shows they understand.

- ☐ **Say the sounds for these graphemes: 'sh', 'ch', and 'th'. Then explain the difference between a phoneme and a grapheme in your own words.**
- Produces the correct single sound for each digraph
 - Uses the word 'grapheme' to mean the letters we write
 - Uses the word 'phoneme' to mean the sound we say
 - Shows understanding that a grapheme can be more than one letter
- ☐ **Read the word 'sit' out loud by blending. Tell me the sounds you used and explain what 'blend' means.**
- Says the three phonemes /s/ /i/ /t/
 - Pushes the sounds together to read the whole word
 - Correctly explains that blending means pushing sounds together to read
- ☐ **You want to spell the word 'hat'. Segment it into its sounds and explain what 'segment' means.**
- Breaks the word into /h/ /a/ /t/
 - Names one letter/grapheme for each sound
 - Correctly explains that segmenting means breaking a word into its sounds for spelling
- ☐ **Name one vowel and one consonant, and tell why 'cat' is called a CVC word.**
- Names a correct vowel (a, e, i, o, or u)
 - Names a correct consonant
 - Explains CVC as consonant-vowel-consonant matching the letters in 'cat'

WHERE THESE FACTS COME FROM

Simple English Wikipedia — “The modern English alphabet is a Latin alphabet of 26 letters, of which five are vowels (a, e, i, o, and u) and 21 are consonants” · “Written English uses 18 digraphs (strings of two letters to represent just one sound), such as ch, sh, th, ph, wh”

English Wikipedia — “Reading is the process of taking in the sense or meaning of symbols, often specifically those of a written language, by means of sight or touch.”

Wikijunior — “the five Big Ideas in Beginning Reading (Phonemic Awareness, Phonics, Vocabulary, Fluency, and Comprehension)”

Asking Questions

1 FOR THE GROWN-UP · TEACHING GUIDE

Why it matters. Asking and answering questions is a core speaking-and-listening skill in school curricula from kindergarten onward, and it is how children extend their understanding, get help, gather information, and clear up confusion. A child who can ask a good question and give a specific answer becomes an active learner rather than a passive one.

HOW TO TEACH IT

Model the three reasons to ask a question — when something is unclear, when more information is needed, and when help is needed — and think out loud so your child hears real questions in action. Practice the question words (What, Where, When, Why, Who, Which, Whose, How) and show how each one leads to a different kind of answer. Then practice answering back with relevant, specific details rather than vague ones.

WATCH FOR THESE MIX-UPS

- A child who stays quiet and confused instead of asking when something is unclear.
- Answers that are off-topic or too vague, like 'I don't know' or 'stuff,' instead of relevant, specific information.
- Interrupting or demanding rather than requesting clarification politely with words like 'Can you explain that again?'

TRY THIS AT HOME

- During story time, pause and invite your child to ask one question about the story to extend their understanding.
- Play a question-word game: point to something and ask your child to make up a 'What?', 'Where?', or 'Why?' question about it.
- Practice polite clarifying phrases together, such as 'Can you say that again, please?' so they come naturally when your child is confused.

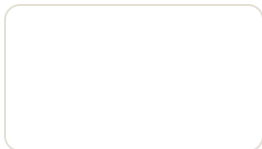
2 THE LESSON • READ TOGETHER

A question is a special kind of sentence you use when you want to find something out. When you ask a question, you are inviting another person to share what they know so you can understand better. Good learners ask questions all the time! Asking a question is one of the most useful things you can do in a classroom, at home, or with your friends, because it helps you learn more, get the help you need, or clear up something that felt confusing.

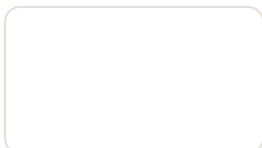
There are three big reasons to ask a question. The first is when something is unclear — maybe a word or an idea just doesn't make sense to you yet. The second is when you need more information — you understand part of something, but you want to know more. The third is when you need help — you are stuck and someone else could show you the way. Whenever you notice that little foggy feeling in your brain, that is a great signal that it's time to raise your hand and ask.

Many questions begin with special question words. In English these are words like 'What?', 'Where?', 'When?', 'Why?', 'Who?', 'Which?', 'Whose?', and 'How?'. Each one helps you dig up a different kind of answer. 'What?' asks about a thing, 'Where?' asks about a place, 'When?' asks about a time, 'Why?' asks about a reason, 'Who?' asks about a person, and 'How?' asks about the way something is done. If you pick the right question word, you'll usually get exactly the kind of answer you were hoping for.

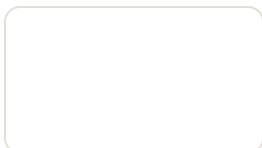
Asking is only half of the job — answering matters too. When someone asks YOU a question, try to answer with relevant and specific information. That means your answer should actually match the question, and it should give real details instead of just 'I don't know' or 'stuff.' And if you didn't understand something the first time, it is always okay to ask politely for it again. Kind, clear words like 'Can you explain that again, please?' help everyone learn together.

PICTURE CARDS TO TALK ABOUT

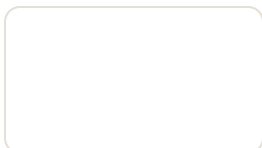
SAY Ask when something is unclear: 'What does that word mean?' Asking clears up the fog so you can keep learning.



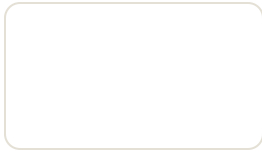
SAY Ask for more information: 'What should I do after I finish this part?' A good question helps you extend what you already understand.



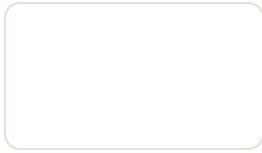
SAY Ask for help: 'Can you help me with this, please?' Asking for help is smart, not silly.



SAY Request clarification politely: 'Can you explain that again?' Polite words make it easy for people to help you.



SAY Use the question word 'When?' — for example, 'When is recess?' Different question words fetch different answers.



SAY Answer with relevant, specific information: 'I put it in my blue backpack by the door,' instead of just 'somewhere.'

3 SHOW WHAT YOU KNOW • TALK IT THROUGH

Ask each question out loud. Check the box when Hugh shows they understand.

☐ **Think of a time something was unclear to you. Write the question you would ask to understand it better.**

- Response is an actual question (asks, not just states)
- Question fits a moment when something is unclear or more information is needed
- Wording is clear and on-topic

☐ **A classmate asks you, 'How do you get to the library from our classroom?' Write a helpful answer that gives relevant and specific information.**

- Answer directly matches the question asked
- Includes specific details rather than vague words
- Communicates clearly enough to be helpful

☐ **Write a polite sentence you could use to ask someone to explain something again.**

- Requests clarification
- Uses polite, respectful wording (e.g., 'please')
- Is a complete, understandable sentence

WHERE THESE FACTS COME FROM

Wikijunior — “Ne? - What? Nerede? - Where? Ne zaman? - When? Neden?/Niye? - Why? Hangi? - Which? Kim? - Who? Kimin? - Whose? Nasıl? - How?”

Predicting what happens next

1 FOR THE GROWN-UP · TEACHING GUIDE

Why it matters. Predicting keeps children actively thinking while they read instead of just saying words. It builds comprehension, because to predict well a child must understand the story so far, notice clues, and connect them to reasons. Curriculum guidance for early readers specifically asks children to predict what might happen next on the basis of what has been read so far, and to discuss the significance of titles and events.

HOW TO TEACH IT

Read together and pause at natural stopping points. Before turning the page, ask, "What do you think happens next? Why?" Insist gently on the 'why,' because the reasoning is the real skill. Start with the title: ask what the story might be about and whether the title fits once you finish. As you read on, point out new information and ask whether it changes the earlier guess—this teaches that revising a prediction is a strength, not a mistake.

WATCH FOR THESE MIX-UPS

- A child guessing with no reason—prompt them with "What in the story made you think that?"
- Treating a changed prediction as being 'wrong' rather than as smart updating with new clues
- Ignoring the title—remind them the title is a big clue chosen on purpose
- Only predicting the very end instead of the next small step

TRY THIS AT HOME

- Cover a book and read only the title; make a prediction together, then check it after reading.
- Use sticky notes to mark 'prediction spots' and write a quick guess before turning the page.
- Pause a movie or cartoon and ask everyone to predict what happens next, then keep watching to check.
- After reading, talk about why the author chose that title and whether it fit the events.

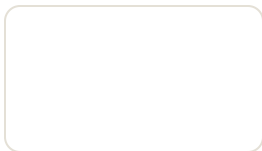
2 THE LESSON • READ TOGETHER

When you read a story, your brain likes to run ahead of the words. It asks, "What might happen next?" Making a prediction means using the clues you already have—the pictures, the characters, and the things that have happened so far—to make a smart guess about what could come on the next page. A prediction is not the same as a wild guess. A good prediction has a reason behind it, and you can point to the part of the story that gave you the idea.

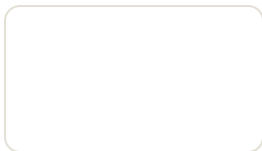
One of the very first clues in any book is the title. A title is the name of the story printed on the cover. Titles are chosen carefully, and they often hint at what the story will be about. Before you even open the book, you can look at the title and ask, "What do I think this story will be about, and why?" As you read, you can check: does the title fit what is happening? Sometimes a title tells you exactly what to expect, and sometimes it hides a little surprise that only makes sense at the end.

Here is the most important part: good readers change their minds. As you turn each page, you meet new information—a new character, a problem, a change in the weather. When you learn something new, you go back and update your prediction. This is called revising your prediction. It does not mean you were wrong before. It means you are a careful reader who keeps paying attention. So the reading rhythm goes like this: look at the clues, make a prediction and explain your reasoning, turn the page, and then keep or change your guess based on what you just learned.

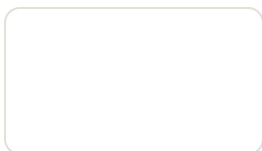
Predicting is not only for stories. People all over the world try to predict what will happen next. Scientists build simulations—models that show what might happen without it really taking place—to guess what could occur in the future. There is even a television game where players are shown a video that stops, and they must choose what happens next. And some things are very hard to predict at all: scientists still have not found a reliable way to predict when an earthquake will happen. Predicting takes clues, reasons, and the willingness to change your mind—whether you are reading a picture book or studying the planet.

PICTURE CARDS TO TALK ABOUT

SAY The title says "The Lost Puppy," so I predict this story is about a puppy that gets lost and someone tries to find it. The title gave me that clue.



SAY Before I turn the page, I predict she is going to the beach, because a bucket, shovel, and sunscreen are things you take to play in the sand.



SAY I need to revise my prediction! New information changed my mind—she's in the snow. Now I predict she will build something with the bucket and shovel in the snow instead.

SAY The title hints that the crayons in this story will stop working or complain. It makes me want to find out WHY they quit, so the title fits the surprise inside.

SAY This is real-life predicting! A simulation shows what might happen without it actually taking place, so people can guess and plan before the trip.

SAY Just like a reader, the players use the clues in the video to predict which of the three things will happen next before the video plays on.

3 PRACTICE WORKSHEET

Read each question aloud with Hugh if they are still learning to read.

1. What is the best way to make a prediction about a story?

- ☐ Guess anything with no reason
 - ☐ Use clues you have already read and give a reason
 - ☐ Skip to the last page
 - ☐ Only look at the back cover
-

2. A book is called "The Runaway Balloon." What does the title hint the story is about?

- ☐ A balloon that floats away or escapes
 - ☐ A boy eating breakfast
 - ☐ A quiet library
 - ☐ A sleepy cat
-

3. You predicted a girl was going to the beach, but the next page shows snow. What should a good reader do?

- ☐ Stop reading
 - ☐ Ignore the new page
 - ☐ Revise the prediction using the new information
 - ☐ Say the book is wrong
-

4. In your own words, what does it mean to 'revise a prediction'? Give one reason a reader might do it.

5. Which real-life example shows people predicting what might happen without it really taking place?

- ☐ Reading a menu
 - ☐ A computer simulation of a spaceship's route to the moon
 - ☐ Coloring a picture
 - ☐ Tying shoes
-

6. Name one clue in a book, besides the title, that can help you predict what happens next.

4 SHOW WHAT YOU KNOW • TALK IT THROUGH

Ask each question out loud. Check the box when Hugh shows they understand.

- ☐ **Pick a story you are reading. Stop before the last page and write a prediction about what happens next. Explain the reason for your prediction using something that has already happened in the story.**
- States a clear prediction about what happens next
 - Gives a reason drawn from what has been read so far
 - Reasoning connects logically to the prediction
- ☐ **Look at the title of a book. Explain whether you think the title fits the story and what it hints the story will be about.**
- Identifies what the title hints about the content
 - Explains whether the title fits the events of the story
 - Uses at least one specific detail as support
- ☐ **Describe a time while reading when you changed your prediction. What new information made you revise it, and what was your new prediction?**
- Describes an original prediction
 - Identifies the new information that appeared in the text
 - Explains the revised prediction based on that new information

WHERE THESE FACTS COME FROM

Simple English Wikipedia — “A simulation can be used to predict what might happen without doing it... Computers are used to make simulations of what a spaceship's route might be on the way to the moon, to help plan the mission.”

English Wikipedia — “Part Two – What Happens Next? The celebrities are shown a video that is stopped at a specified point. Both teams are then given three choices to predict what happens next in the video.”

Wikijunior — “At this time, seismologists have not found a reliable method for predicting earthquakes.”

Spaces Between Words

1 FOR THE GROWN-UP · TEACHING GUIDE

Why it matters. Spaces between words are one of the first big ideas that turns a scribble into readable writing. Spaces mark where each word begins and ends, and clear, consistent spacing is what makes reading smoother and faster for the eye. Learning this early helps a child both write more clearly and read more easily.

HOW TO TEACH IT

Start by showing your child that print is full of little gaps, and that each gap separates two words. Explain that a long time ago people wrote with no spaces at all (*scriptio continua*) and it was very hard to read—this vivid contrast helps children understand what spaces are FOR. Then teach the classic 'finger-space' habit: write a word, lay a finger on the paper, and begin the next word on the far side of the finger.

WATCH FOR THESE MIX-UPS

- Words that run together with no gaps, like *scriptio continua*, which makes writing hard to read.
- Uneven or huge gaps—spaces should be clear but consistent, roughly a finger-width.
- Confusing the space between letters with the space between words; the word gap should be bigger.
- Forgetting that you can count spaces to count words in a sentence.

TRY THIS AT HOME

- Write a short sentence with no spaces (like 'Iloveyou') and ask your child to add lines where the spaces should go.
- Practice the finger-space trick together on a simple sentence.
- Point to a sentence in a storybook and count the words by counting the gaps.
- Play 'monk and modern writer': write a phrase squished together, then rewrite it with nice spaces and talk about which is easier to read.

2 THE LESSON • READ TOGETHER

When you read a book, look closely at the words. Do you see the little empty gaps between them? Those gaps are called spaces! A space is a small empty spot that tells us where one word ends and the next word begins. Spaces mark the edges, or boundaries, of every word in print.

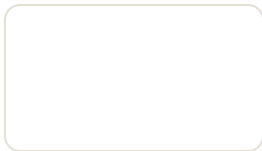
Here is something surprising. A very long time ago, people did not use spaces at all! In Ancient Greek and old Latin writing, people wrote all the letters in one long string with no gaps, no big letters, and no dots. This way of writing is called *scriptio continua*, which means "continuous script." They squished everything together to save space and ink. Can you imagine reading a whole page with no spaces? It would look like this: **FORGODSOLOVEDTHEWORLD**. That is very hard to read!

Then something wonderful happened. About 1,300 years ago, Irish monks (people who spent their days copying books by hand) began putting spaces between the words. Slowly, more and more writers started leaving gaps too. Today, when we write in English, we always put a space between each word, just like they learned to do.

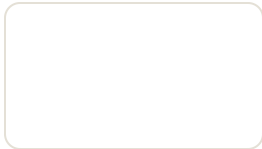
Why do we bother with spaces? Because spaces make reading easier and faster! When we talk out loud, we do not really pause between every word, but our eyes need those little visual gaps to see where each word is. When the spaces are clear and even, your eyes glide smoothly along the line and you can read the words as neat little bundles.

When you write, the best trick is to leave a finger-space between each word. After you write one word, put your finger down on the paper, then start the next word on the other side of your finger. That gives every word its own comfy gap. You can even count the words in a sentence by counting the spaces between them! Not every language uses spaces, though. In Chinese and Japanese, writers often do not put spaces between words at all, because their symbols work in a different way. But in English, spaces are one of our most important helpers.

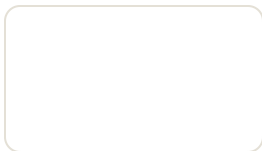
PICTURE CARDS TO TALK ABOUT



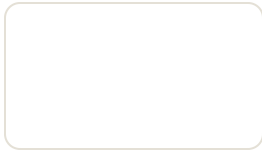
SAY Look at these three words. See the gaps? Each space tells your eyes: one word is done, here comes the next. That's a space marking a word boundary.



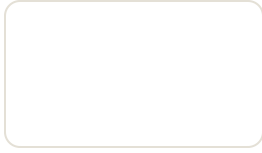
SAY Long ago people wrote with no spaces at all, like this. It's called *scriptio continua*, and it is really tricky to read because you can't tell where one word ends!



SAY Use a finger-space! Write a word, put your finger down, then start the next word after your finger. Now every word has a nice clear gap.



SAY Count the spaces to count the words! When you spot the gaps between words, you can count how many words are in a sentence.



SAY Not every language uses spaces. Chinese writing often has no spaces between words because each symbol works differently. But in English we always use them.

3 SHOW WHAT YOU KNOW • TALK IT THROUGH

Ask each question out loud. Check the box when Hugh shows they understand.

- ☐ **Look at this writing: 'Ilikecake'. Rewrite it with spaces in the right places. How many words did you make?**
- Correctly separates into 'I like cake' with clear spaces
 - States there are 3 words
 - Shows understanding that spaces mark word boundaries
- ☐ **Explain the 'finger-space' trick to a friend who is just learning to write. What should they do?**
- Describes writing a word, then placing a finger, then starting the next word after it
 - Explains this leaves a clear space between words
 - Uses own words in a friendly way
- ☐ **Long ago people sometimes wrote with no spaces at all. Why is that hard to read, and why are spaces helpful?**
- Notes that without spaces you can't tell where words begin and end
 - States that spaces make reading easier or faster
 - May mention scriptio continua or that monks began adding spaces

WHERE THESE FACTS COME FROM

English Wikipedia — “Word spacing in typography is space between words, as contrasted with letter-spacing. Irish monks first started to add word spacing to texts in the late 7th century. While spoken language has no audible gaps, the visual spaces introduced between words can make reading more efficient.”

Simple English Wikipedia — “Scriptio continua is a style of writing that does not use spaces to tell apart words. FORGODSOLOVEDTHEWORLD... written using scriptio continua. Chinese and Japanese sometimes use scriptio continua and do not use spacing.”

MATHEMATICS

21 topics

MATHEMATICS · COUNTING & CARDINALITY · AGES 4–6

One-to-one counting

1 FOR THE GROWN-UP · TEACHING GUIDE

Why it matters. One-to-one correspondence is the foundation of all counting: each object is paired with exactly one number name. Before a child can add, subtract, or understand 'how many,' they must reliably match one spoken number to one object without skipping or double-counting. This is a core kindergarten math skill.

HOW TO TEACH IT

Model slow, deliberate counting where you physically point to or touch each object exactly once while saying the number name out loud. Start with small sets (2–4 objects) like John's three hats, then grow to larger sets. Line objects up in a row at first so they are easy to track, then try scattered arrangements. Emphasize that the LAST number you say tells how many there are in all.

WATCH FOR THESE MIX-UPS

- Saying number names faster than they touch objects (words race ahead of the pointing)
- Touching the same object twice and giving it two numbers — double-counting
- Skipping an object so it never gets a number
- Losing track in a scattered pile; encourage moving each counted item aside

TRY THIS AT HOME

- Count everyday sets together: forks at the table, stairs as you climb, crackers on a plate — touch each one once.
- Deliberately make a mistake (count one block twice) and ask your child to catch the error and fix it.
- Have your child move each object into a 'counted' pile as they say its number, so nothing gets counted twice.
- Watch a favorite show like 101 Dalmatians and count a small group of dogs on screen together, one number per dog.

2 THE LESSON • READ TOGETHER

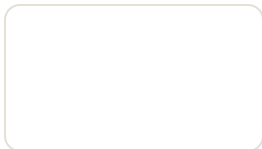
When you count things, you are finding out how many there are. But there is a special rule that makes counting work every single time. The rule is this: each thing gets exactly one number, and each number gets said only one time. This is called one-to-one counting, because you match ONE object to ONE number name.

Here is how it goes. You start with the number one and give it to the first thing you touch. Then you give the number two to the next thing. Then three to the next thing, then four, then five, and you keep going, giving each thing its very own number. When you have counted the last thing, the last number you said tells you how many there are in all!

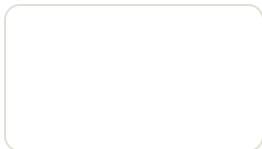
The secret to doing it right is to point to or touch each object exactly once as you say each number. If you touch a toy and say "one," you cannot touch that same toy again and say "two" that number would be wrong. And you must not skip any objects either, because every single thing needs to get a number. When each object is matched with exactly one number name, your count comes out just right.

Let's try it the way John did with his hats. John has a black hat, a brown hat, and a blue hat. He gives the number 1 to the black hat, the number 2 to the brown hat, and the number 3 to the blue hat. Now he has no more hats to count. The biggest number he said was three, so John has three hats! You can do the very same thing with blocks, crackers, buttons, or even puppies.

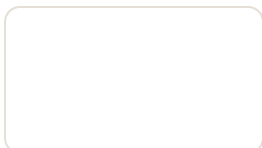
PICTURE CARDS TO TALK ABOUT



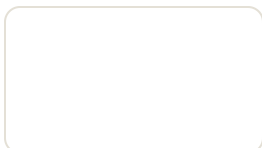
SAY Touch the black hat and say 'one.' Touch the brown hat and say 'two.' Touch the blue hat and say 'three.' There are three hats! Each hat got just one number.



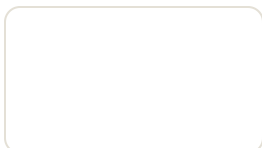
SAY Point to A and say 'one,' point to Q and say 'two,' point to L and say 'three,' point to S and say 'four.' There are four letters. No letter was skipped and none was counted twice.



SAY Counting can be done on fingers. Hold up one finger and say 'one,' the next and say 'two,' the next and say 'three.' Each finger gets its own number.



SAY Imagine touching each spotted puppy one at a time as you count: one, two, three... all the way to fifteen! Because Pongo and Perdita had a litter of fifteen puppies, if you give each puppy exactly one number, your last number will be fifteen.



SAY When Roger counts the whole family of dogs, he touches or points to each dog one time and gives it one number. The very last number he reaches is 101 — that is how the movie got its name, 101 Dalmatians!

3 PRACTICE WORKSHEET

Read each question aloud with Hugh if they are still learning to read.

1. John gives the number 1 to his black hat and the number 2 to his brown hat. What number should he give the blue hat?

- ☐ 2
 - ☐ 3
 - ☐ 1
-

2. Which is the RIGHT way to count?

- ☐ Touch each toy one time and give it one number
 - ☐ Touch one toy two times and give it two numbers
 - ☐ Skip some toys so they don't get a number
-

3. There are 4 letters: A, Q, L, S. If you give each letter exactly one number, what is the last number you say?

- ☐ 3
 - ☐ 4
 - ☐ 5
-

4. Your friend touches the same apple two times and says 'one, two.' What did your friend do wrong, and how can they fix it?

5. When you finish counting, which number tells you how many things there are?

- ☐ The first number you said
- ☐ The last number you said
- ☐ The middle number

4 SHOW WHAT YOU KNOW • TALK IT THROUGH

Ask each question out loud. Check the box when Hugh shows they understand.

☐ **Put 5 blocks in front of you and count them out loud. Explain what you do with each block as you say a number.**

- Points to or touches each block exactly once
- Says one number name per block in order
- Does not skip any block or count one twice
- States that the last number (five) tells how many

☐ **Someone counts a row of toys but touches one toy twice and says two numbers for it. Tell what mistake they made and how to count correctly.**

- Recognizes the error of counting one object twice (double-counting)
- Explains each object should be paired with exactly one number name
- Describes touching each object only once to fix it

☐ **John has three hats. Show how he counts them so each hat gets its own number, and tell how many hats he has in all.**

- Gives number 1, then 2, then 3 to the three hats in turn
- Gives each hat exactly one number
- Concludes there are three hats total

WHERE THESE FACTS COME FROM

Simple English Wikipedia — “John gives the number 1 to the black hat. John gives the number 2 to the brown hat. John gives the number 3 to the blue hat... John tells Mary, 'I have three hats'” · “Each thing gets its own number in this way. The last number given to the last thing counted shows how many things there are.”

English Wikipedia — “Pongo and Perdita... give birth to a litter of fifteen puppies... bringing the total to 101 Dalmatians.”

MATHEMATICS · COUNTING & CARDINALITY · AGES 4–6

How Many in Total?

1 FOR THE GROWN-UP · TEACHING GUIDE

Why it matters. The cardinality principle is the bridge between reciting number words and truly understanding quantity. A child can chant "1, 2, 3, 4, 5" and still not know that the final word answers "how many?" Mastering this means your child understands that the last number said names the size of the whole group — the foundation for all later math like adding and comparing.

HOW TO TEACH IT

Count a small set of real objects together, touching each one. Then pause and ask, "So how many in total?" If your child recounts instead of answering with the last number, gently model: "We said one, two, three — so there are three." Repeat with the same objects rearranged and counted in a new order so they discover the total stays the same.

WATCH FOR THESE MIX-UPS

- Recounting the whole set every time you ask "how many?" instead of giving the last number — a sign the cardinality idea hasn't clicked yet.
- Thinking the total changes when objects are moved, spread out, or piled up.
- Believing you must always start counting from a certain end, or that a different starting point gives a different answer.
- Skipping objects or counting one twice, which leads to a wrong last number.

TRY THIS AT HOME

- Count a handful of snacks, then ask "how many?" and celebrate the last-number answer before eating them.
- Count toys in a line, then scramble them into a pile and ask if the total changed — let your child check.
- Take turns: you count left-to-right, they count right-to-left, and compare the totals.
- Play "How many in total?" around the house — shoes by the door, spoons on the table, buttons on a shirt.

2 THE LESSON • READ TOGETHER

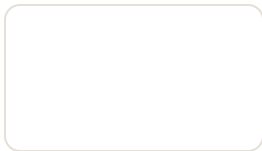
When you count a group of things, each thing gets its own number: "one, two, three, four." But here is the big, magical idea of counting — the LAST number you say is the answer to "How many?" If you count four teddy bears and the last number you say is "four," then there are four bears. That last number doesn't just belong to the last bear. It stands for the WHOLE group! This special rule is called the cardinality principle: the last number said when you count tells how many objects are in the set.

Here is why that is so wonderful. Once you know the total is four, you don't have to count again and again. "Four" tells you everything about how big the group is. If a friend asks, "How many?" you don't say the whole counting song. You just say the last number: "Four!"

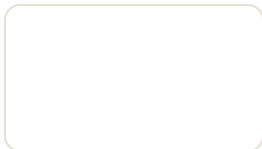
There's an even trickier and more amazing part. It doesn't matter how the objects are arranged. You can put your four bears in a straight line, in a circle, in a big pile, or spread far apart across the room. The total is still four. Moving things around does not change how many there are. Counting only tells you how many, not where they sit.

And one more secret: it doesn't matter which one you count first. You could start counting from the left, or from the right, or start in the middle. As long as you count each object exactly once and don't skip any, you will always land on the same last number. Different order, same total. When you understand all of this, you truly understand what counting is FOR — it answers the question, "How many in total?"

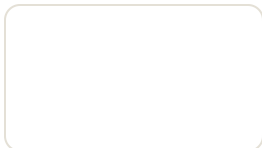
PICTURE CARDS TO TALK ABOUT



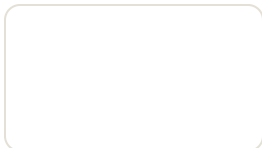
SAY The last number I said was three, so there are three apples. Three is how many in total!



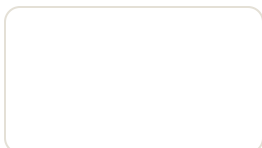
SAY I moved them into a pile, but I don't need to recount from scratch — it's still three. Rearranging doesn't change how many there are.



SAY Left to right I got five. Right to left I still got five. Counting in a different order gives the same total.



SAY I answer with the last number I said: "Four!" I don't have to count all over again.



SAY Far apart or close together, it's still six. Where things sit doesn't change the total.

3 PRACTICE WORKSHEET

Read each question aloud with Hugh if they are still learning to read.

1. You count blocks: "one, two, three, four, five." How many blocks are there in total?

- ☐ one
- ☐ five
- ☐ three
-

2. You counted 6 cars in a row. Then you push them into a pile. How many cars now?

- ☐ six
- ☐ more than six
- ☐ you can't tell
-

3. You count 4 shells from the left and get four. If you count them again starting from the right, what will you get?

- ☐ three
- ☐ four
- ☐ five
-

4. When a friend asks "how many?" after you finish counting, which number should you tell them?

5. If you spread your 5 crayons far apart, does the total change? Why or why not?

4 SHOW WHAT YOU KNOW • TALK IT THROUGH

Ask each question out loud. Check the box when Hugh shows they understand.

- ☐ **Count a set of toys out loud for a grown-up. When you finish, they ask, "How many in total?" Explain how you know your answer.**
 - Gives the last number stated as the total
 - Explains that the last number counted tells how many
- ☐ **Line up some blocks and count them. Then move them into a pile without adding or taking any away. Tell whether the total changed and explain why.**
 - States the total is the same
 - Explains that rearranging objects does not change the count
- ☐ **Count a group of objects starting from one end, then count the same group starting from the other end. Compare the two totals and explain what you notice.**
 - Reports the same total both times
 - Explains that counting in a different order gives the same total

WHERE THESE FACTS COME FROM

MATHEMATICS · ADDITION & SUBTRACTION · AGES 4–6

Subtraction as taking away or separating

1 FOR THE GROWN-UP · TEACHING GUIDE

Why it matters. Subtraction is one of the four basic operations at the very core of arithmetic, and it is the opposite (inverse) of addition. Understanding that subtraction means taking away from a group to find how many remain is a foundation your child will build on for years of math. Starting with real objects makes this abstract idea concrete and lasting.

HOW TO TEACH IT

Teach subtraction as a physical action, not a written symbol. Give your child a small group of objects, have them count how many there are, physically remove some, and then count what is left. Narrate it out loud: 'We had 5, we took away 2, now how many are left?' Repeat with different objects and small numbers until the pattern feels natural. Only later connect it to the word 'minus' and the number sentence.

WATCH FOR THESE MIX-UPS

- Recounting the whole group instead of only counting what remains after taking some away
- Confusing subtraction with addition — remind them subtraction makes the group smaller
- Forgetting the answer to subtraction is 'how many are left,' not 'how many we took away'
- Struggling when everything is taken away — reassure them that leaving zero, or none, is a fine answer

TRY THIS AT HOME

- At snack time, count the pieces, eat some, and ask 'How many are left?'
- Line up toys, take a few away, and have your child count what remains
- Play a finger game: hold up fingers, fold some down, and count the ones still up
- Sing about animals or objects going away, one or two at a time, and count who is left

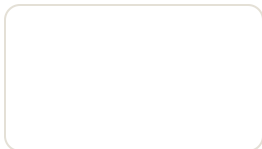
2 THE LESSON • READ TOGETHER

Subtraction is a math word that means taking away. When you have a group of things and you take some away, subtraction helps you find out how many are left. It is one of the four basic building blocks of arithmetic, along with addition, multiplication, and division. Addition puts things together, but subtraction does the opposite — it takes things away or separates them from a group.

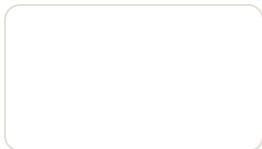
Let's picture it. Imagine you have 5 crackers on your plate. You feel hungry, so you eat 2 of them. Now you look at your plate — how many crackers are still there? You would count and find 3 crackers left. You just did subtraction! You started with a group of 5, you took away 2, and 3 remained. We can say it like this: five take away two leaves three.

Subtraction always answers the question 'How many are left?' You can find the answer by taking the objects away one at a time and then counting what stays behind. Some grown-ups write this with numbers, like 10 minus 3 equals 7, which means if you start with 10 and take away 3, then 7 are left. But you do not need numbers on paper to do it — you can do it with your toys, your snacks, your crayons, or your fingers.

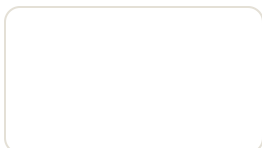
Subtraction is the opposite, or 'inverse,' of addition. Addition means putting groups together to make more. Subtraction means taking away to have fewer. Whenever you eat some food, give a toy to a friend, or watch some birds fly away, you can use subtraction to figure out how many are left behind.

PICTURE CARDS TO TALK ABOUT

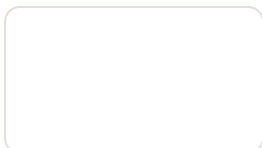
SAY I take away the 2 cookies I ate and count the rest: 1, 2, 3. There are 3 cookies left!



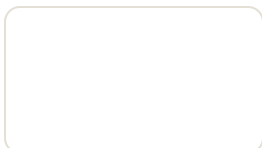
SAY I take 1 apple out of the bowl. Now I count what stays: 1, 2, 3. Three apples are left.



SAY The 4 ducks that flew away are gone, so I count the ones still swimming: 1, 2. Two ducks are left.



SAY I put down 2 fingers and count the ones still standing up: 1, 2, 3. Three fingers are left.



SAY I took away every crayon, so the box is empty. Zero crayons are left inside.

3 PRACTICE WORKSHEET

Read each question aloud with Hugh if they are still learning to read.

1. You have 5 cookies and you eat 2. How many cookies are left?

- ☐ 2
- ☐ 3
- ☐ 5
-

2. What does subtraction mean?

- ☐ Putting groups together to make more
- ☐ Taking away to find how many are left
- ☐ Counting to ten
-

3. There are 4 apples in a bowl. You take 1 out. How many are left in the bowl?

- ☐ 3
- ☐ 4
- ☐ 1
-

4. You hold up 5 fingers and fold down 3. Count and tell how many fingers are still standing up.

5. 6 ducks are swimming and 4 fly away. How many ducks are left in the water?

- ☐ 4
- ☐ 2
- ☐ 6
-

6. There are 3 crayons in a box and you take out all 3. How many crayons are left in the box?

4 SHOW WHAT YOU KNOW • TALK IT THROUGH

Ask each question out loud. Check the box when Hugh shows they understand.

☐ **Use real objects like blocks or toys. Make a group, take some away, and tell me how many are left. Explain what you did.**

- Child models 'taking away' with physical objects
- Child counts and correctly states how many remain
- Child explains they removed some from the group

☐ **Act out this story: There are 5 cookies and someone eats 2. Show it with objects and tell how many cookies are left.**

- Child acts out the 'take from' situation with objects or gestures
- Child removes the correct number (2)
- Child says the correct number left (3)

☐ **In your own words, tell me what subtraction means.**

- Child explains subtraction means taking away or separating
- Child connects it to finding how many are left
- Response is in the child's own words

WHERE THESE FACTS COME FROM

Wikijunior — “Subtraction, the inverse operation of addition, involves finding the difference between numbers, such as determining how much remains when one quantity is taken away from another. For example, subtracting 3 from 10 yields 7.” · “It encompasses the basic operations of addition, subtraction, multiplication and division.”

Addition as combining or putting together two

1 FOR THE GROWN-UP · TEACHING GUIDE

Why it matters. Addition is one of the four basic building blocks of arithmetic, and combining sets is the most basic and easy-to-picture way to understand it. Learning that addition means 'putting groups together to find the total' gives your child the foundation for all the math that comes later.

HOW TO TEACH IT

Start with real objects your child can touch and move. Make two small groups, physically slide them together into one group, and count the whole thing out loud to find the total. Name what you're doing: 'We put 3 and 2 together — that's 5 altogether.' Then act out 'add to' stories where an amount grows because more items or people arrive. Introduce the words plus, total, sum, and altogether naturally as you play.

WATCH FOR THESE MIX-UPS

- Counting only one group instead of counting everything after combining
- Forgetting to recount the whole combined group and just guessing the total
- Thinking the order matters — remind them $3 + 2$ and $2 + 3$ both make 5
- Confusing 'adding zero' — nothing new comes, so the number stays the same

TRY THIS AT HOME

- Snack math: put some crackers in two piles, combine them, and count how many altogether before eating.
- Act out an 'add to' story with toys: 'Three cars are parked, then two more drive in — how many now?'
- Play the swap game: count 2 spoons plus 3 forks, then 3 forks plus 2 spoons, and notice the total is the same.
- Use the words 'plus,' 'total,' and 'altogether' during everyday moments like setting the table.

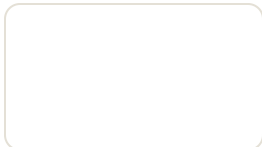
2 THE LESSON • READ TOGETHER

Addition means putting two groups together to find out how many you have altogether. When you bring things together into one big group and count them all, you are adding! Grown-ups write addition with a special mark called the plus sign, which looks like this: $+$. It tells you to combine, or put together.

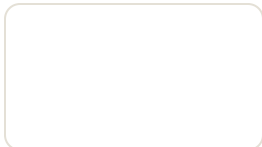
Here is a yummy way to picture it. Imagine one plate with 3 apples on it and another plate with 2 apples on it. If you scoot all the apples together onto one plate and count them, you get 5 apples. We can write that as $3 + 2 = 5$, and we read it out loud as "three plus two equals five." The little groups you start with (the 3 apples and the 2 apples) get combined into one big group, and the big number you end with is called the total, or the sum.

Addition works with anything you can count — apples, blocks, teddy bears, even people! If some children are playing and then more children run over to join them, you can add to find out how many children are playing now. That is called an "add to" story. You start with one amount, then more come, and you count everybody together.

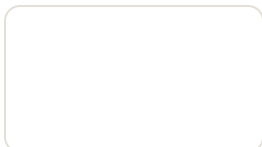
Here is a fun secret about addition: it does not matter which group you count first! $3 + 2$ gives 5, and $2 + 3$ also gives 5. You get the same total either way. And here is another secret — if you add zero (which means nothing new comes), the number stays exactly the same. Adding is one of the very first math jobs people learn, and even tiny babies can notice when 1 and 1 make 2! The best way to learn addition is to put real things together with your hands and count them, so let's practice combining groups to find how many altogether.

PICTURE CARDS TO TALK ABOUT

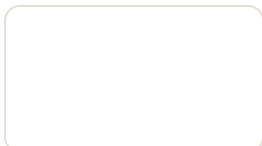
SAY Three apples put together with two apples makes five apples altogether. We write it as $3 + 2 = 5$.



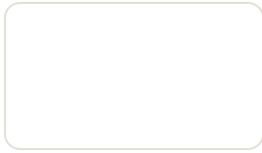
SAY This is an 'add to' story! Three children, then two more come, so we count everyone: five children are playing altogether.



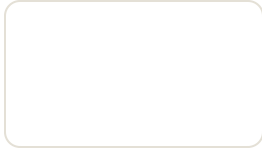
SAY Four blocks and one block combine to make five blocks in all. Putting groups together is adding!



SAY Two bears plus three bears equals five bears. We found how many altogether by combining them.



SAY $2 + 3 = 5$ and $3 + 2 = 5$. It does not matter which group you count first — the total is still five!



SAY Five crayons plus zero crayons is still five crayons. Adding nothing keeps the number the same.

3 PRACTICE WORKSHEET

Read each question aloud with Hugh if they are still learning to read.

1. You have 3 apples and put them together with 2 more apples. How many apples altogether?

- ☐ 4
- ☐ 5
- ☐ 6
-

2. What does the plus sign (+) tell you to do?

- ☐ Take some away
- ☐ Put groups together
- ☐ Count only one group
-

3. 2 children are playing, then 3 more join them. How many children are playing now?

- ☐ 5
- ☐ 3
- ☐ 2
-

4. Grab some blocks. Make a group of 4 and a group of 1, push them together, and tell me the total.

5. Which one is the same as $3 + 2$?

- ☐ $2 + 3$
- ☐ $5 + 5$
- ☐ $3 + 3$
-

6. You have 5 crayons and nobody adds any more. How many crayons do you have?

4 SHOW WHAT YOU KNOW • TALK IT THROUGH

Ask each question out loud. Check the box when Hugh shows they understand.

- ☐ **Use real objects (like toys or blocks). Make one group of 3 and one group of 2, put them together, and tell me how many there are altogether.**
 - Physically combines the two groups into one
 - Counts all the objects in the combined group
 - States the correct total (5)
- ☐ **Act out this story with people or toys: 3 children are playing, then 2 more children arrive to join them. Show what happens and tell how many children are playing now.**
 - Starts with the first group of 3
 - Adds the second group of 2 by having them 'join'
 - Counts everyone together and says 5 altogether
- ☐ **In your own words, tell me what addition means and what we are trying to find out when we add.**
 - Explains that addition means putting or combining groups together
 - Says that we are finding how many there are altogether (the total)
 - Uses a clear example if helpful

WHERE THESE FACTS COME FROM

English Wikipedia — “two columns of apples, one with three apples and the other with two apples, totaling to five apples... expressed as “ $3 + 2 = 5$ ”, read as “three plus two equals five”” · “Addition, usually denoted with the plus sign +, is one of the four basic operations of arithmetic... The addition of two whole numbers results in the total or sum of those values combined.”

Division as equal sharing

1 FOR THE GROWN-UP · TEACHING GUIDE

Why it matters. Division as equal sharing and grouping is the very first step toward understanding division, part of early Year 1 multiplication and division learning. It builds fairness thinking and prepares children for later math by letting them experience 'equal groups' with their hands before they ever see a division symbol.

HOW TO TEACH IT

Start with real objects and the word 'fair.' Show that division means sharing equally into groups, or grouping to see how many equal groups you can make. Let your child physically deal out counters one at a time between plates (sharing), and also scoop objects into fixed-size sets and count the groups (grouping). Always end by counting each group to check it is equal.

WATCH FOR THESE MIX-UPS

- Groups that are not equal — remind them division means everyone gets the same amount.
- Confusing the two questions: 'how many on each plate?' (sharing) versus 'how many groups can I make?' (grouping).
- Guessing in their head instead of moving the objects; encourage hands-on dealing and grouping.
- Forgetting to count the final groups to check the answer.

TRY THIS AT HOME

- Share 10 counters (or dried beans) equally between 2 plates and count what's on each.
- Give your child 12 small objects and ask them to make sets of 3, then count the groups.
- Ask 'How many groups of 2 are in 8?' using grapes, buttons, or toys.
- At snack time, practice sharing food equally between family members so everyone gets a fair amount.

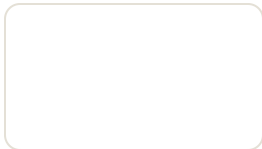
2 THE LESSON • READ TOGETHER

Have you ever had a bag of yummy crackers and a friend right next to you? What do you do? You share! Division is a special math word that means sharing equally. When we share equally, everyone gets the same amount — nobody gets more, and nobody gets less. That is the big idea of division: putting things into fair, equal groups.

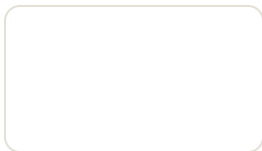
There are two ways to think about division. The first way is called **SHARING**. Imagine you have 10 counters and 2 plates. You give one counter to this plate, one to that plate, one to this plate, one to that plate, back and forth, until they are all gone. When you count, each plate has 5. You shared 10 equally between 2 plates! Sharing equally means each group ends up the same size.

The second way is called **GROUPING**. Instead of asking 'how many go on each plate,' you ask 'how many groups can I make?' Say you have 12 blocks and you want to make little sets of 3. You scoop 3 blocks — that's one group. Scoop 3 more — that's two groups. Keep going until the blocks run out. When you count your sets, there are 4 groups of 3! Grouping tells you how many equal groups of a certain size you can make.

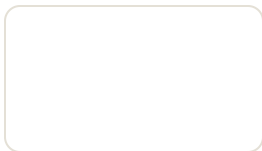
Both ways are division, and both ways keep things fair and equal, so every group is the same. The best part is you don't have to do it in your head. You can use real things you can touch — counters, buttons, grapes, toy cars — and move them into groups with your hands. Sharing one at a time, then counting each group, helps you see the answer right in front of you.

PICTURE CARDS TO TALK ABOUT

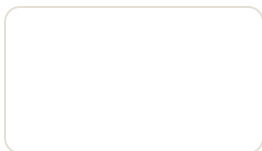
SAY Sharing 10 equally between 2 plates gives 5 counters on each plate. That is division as equal sharing!



SAY When I group 12 into sets of 3, I can make 4 equal groups. That is division as grouping!



SAY How many groups of 2 are in 8? I count my groups: 1, 2, 3, 4. There are 4 groups of 2!



SAY Sharing equally means we each get the same. We each get 3 stickers — that's fair!

3 PRACTICE WORKSHEET

Read each question aloud with Hugh if they are still learning to read.

1. You share 10 counters equally between 2 plates. How many are on each plate?

- ☐ 4
 - ☐ 5
 - ☐ 10
-

2. Division means putting things into groups that are...

- ☐ Equal (the same size)
 - ☐ Different sizes
 - ☐ All in one pile
-

3. You have 12 blocks. You make sets of 3. How many groups can you make?

- ☐ 3 groups
 - ☐ 4 groups
 - ☐ 6 groups
-

4. How many groups of 2 are in 8? Use counters to help, then tell the number.

5. Which question is about GROUPING?

- ☐ How many go on each plate?
- ☐ How many groups of 2 can I make?
- ☐ What color are the counters?

4 SHOW WHAT YOU KNOW • TALK IT THROUGH

Ask each question out loud. Check the box when Hugh shows they understand.

- ☐ **Using 10 counters and 2 plates, show how you would share them equally. Tell how many end up on each plate and why that is fair.**
- Deals out or describes sharing the counters one at a time between the 2 plates
 - States each plate has 5
 - Explains that equal sharing means each group is the same
- ☐ **You have 12 objects. Group them into sets of 3. How many groups did you make? Show your groups.**
- Makes equal sets of 3
 - Counts and reports 4 groups
 - Groups are all the same size
- ☐ **How many groups of 2 are in 8? Use objects you can touch to solve it, then explain how you found your answer.**
- Uses concrete objects to make groups of 2
 - Arrives at 4 groups
 - Explains counting the equal groups

WHERE THESE FACTS COME FROM

English Wikipedia — “the couple's property is to be divided equally between them ... each ... is entitled to share equally”

simplewith — “The main idea of it is equal division ... the angle is shared equally between the two smaller angles”

MATHEMATICS · COUNTING & CARDINALITY · AGES 4–6

Comparing groups: more or fewer

1 FOR THE GROWN-UP · TEACHING GUIDE

Why it matters. Comparing groups is one of the first true math-reasoning skills. Before children can add or subtract, they need to understand that amounts can be bigger, smaller, or the same, and to compare two groups to decide which has more, fewer, or whether they are equal.

HOW TO TEACH IT

Teach both strategies from the grounding: one-to-one matching (lining objects up so each has a partner) and counting both groups and comparing the numbers. Use real objects, then connect the action to the words 'more than,' 'less than,' 'fewer,' 'equal to,' 'most,' and 'least.'

WATCH FOR THESE MIX-UPS

- Thinking the longer line always has more, even when objects are just spread out — encourage matching or counting to check.
- Counting one group but forgetting to count the other before deciding.
- Mixing up 'more' and 'fewer' — pause and ask which pile is bigger.

TRY THIS AT HOME

- Line up two rows of spoons or socks and match them one-to-one to see which group has more or fewer.
- Put snacks on two plates and ask your child to count both and say which has more.
- Set out three cups with different numbers of coins and ask which has the most and which has the least.

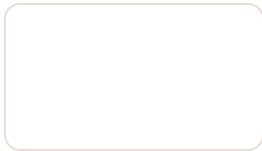
2 THE LESSON • READ TOGETHER

When you have two groups of things, you can ask a fun question: which group has MORE, and which group has FEWER? Comparing groups means looking at two piles or lines of objects and figuring out which one has a bigger amount, which one has a smaller amount, or whether they are the SAME.

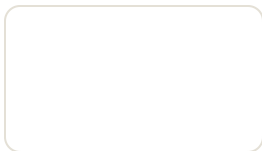
There are two smart ways to find out. The first way is called matching. You line up your two groups next to each other and draw an imaginary line from one object in the top group to one object in the bottom group. This is called one-to-one matching, because each thing gets exactly one partner. If every object has a partner and none are left over, the two groups are EQUAL — they have the same amount. But if one group has objects left over with no partner, that group has MORE. The group that ran out first has FEWER.

The second way is called counting. You count how many are in the first group, then count how many are in the second group, and compare the two numbers. If one group has more when you count, it is 'more than' the other. The smaller group is 'less than' or has 'fewer.' When the two numbers are the same, the groups are 'equal to' each other.

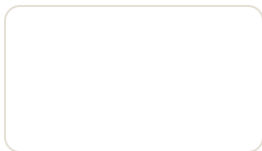
We use special comparing words to talk about groups. We say one group is 'more than' another, or 'less than' another. We say a group has 'fewer' things when it has a smaller amount. And when we compare three or more groups, we can point to the one with the 'most' (the biggest amount) and the one with the 'least' (the smallest amount). Learning these words helps you talk like a math detective!

PICTURE CARDS TO TALK ABOUT

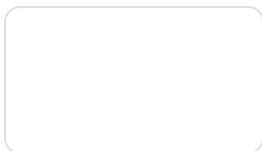
SAY Match them up one-to-one. The top has one apple with no partner, so the top group has MORE and the bottom group has FEWER.



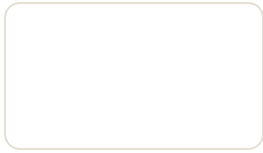
SAY Each cat has exactly one fish partner and none are left over. The groups are EQUAL — they have the same amount.



SAY Count both: 4 and 6. Six is bigger, so 6 buttons is MORE THAN 4 buttons, and 4 buttons is LESS THAN 6 buttons.



SAY The plate with 5 cookies has the MOST. The plate with 2 cookies has the LEAST.



SAY Every red block has a blue partner and nothing is left over, so the groups are EQUAL to each other.

3 PRACTICE WORKSHEET

Read each question aloud with Hugh if they are still learning to read.

1. You match 4 forks with 4 spoons, one-to-one, and nothing is left over. The two groups are...

- ☐ Equal to each other
 - ☐ More forks than spoons
 - ☐ Fewer spoons than forks
-

2. One group has 7 stickers and another group has 3 stickers. Which is true?

- ☐ 3 is more than 7
 - ☐ 7 is more than 3
 - ☐ They are equal
-

3. Three baskets hold 2, 8, and 5 balls. Which basket has the MOST balls?

- ☐ The basket with 2
 - ☐ The basket with 5
 - ☐ The basket with 8
-

4. You have 5 crayons and your friend has 5 crayons. Use a comparing word to describe the two groups.

5. When you match two groups and one group has objects left over with no partner, what does that tell you about that group?

4 SHOW WHAT YOU KNOW • TALK IT THROUGH

Ask each question out loud. Check the box when Hugh shows they understand.

☐ **Show or tell how you could use one-to-one matching to find out which of two groups of toys has more.**

- Describes pairing each object in one group with one object in the other group
- Explains that leftover objects with no partner mean that group has more (or none left over means equal)

☐ **You count 6 red blocks and 4 blue blocks. Tell which group has more and which has fewer, and use comparing words to explain how you know.**

- Correctly states 6 red blocks is more and 4 blue blocks is fewer
- Uses comparing language such as 'more than,' 'less than,' or 'fewer'
- Refers to counting both groups to decide

☐ **Three jars have 3, 9, and 6 marbles. Tell which jar has the most and which has the least, and explain why.**

- Identifies the jar with 9 as the most and the jar with 3 as the least
- Uses the words 'most' and 'least' correctly

WHERE THESE FACTS COME FROM

MATHEMATICS · MEASUREMENT · AGES 4–6

Measurable Attributes of Objects

1 FOR THE GROWN-UP · TEACHING GUIDE

Why it matters. Learning about measurable attributes is part of the measurement strand of early math, where children learn about the measurable attributes of objects. These skills connect directly to the world your child lives in and are also used in science, and they build the foundation for more advanced math later on.

HOW TO TEACH IT

Start by naming one attribute at a time: length, height, weight, and capacity. Give your child real objects to hold and compare. Measurement means giving a description of an object's size based on a particular attribute, so help your child pick which attribute they are looking at, then use comparing words like longer, shorter, heavier, lighter, more, and less.

WATCH FOR THESE MIX-UPS

- Mixing up length and height — remind them that height is how tall something stands from bottom to top.
- Thinking a bigger object is always heavier — let them lift things to feel that size and weight are different attributes.
- Only naming one attribute for an object — challenge them to find more than one, like a bottle having height, weight, AND capacity.

TRY THIS AT HOME

- Line up three pencils or spoons and ask which is longer and which is shorter.
- Let your child hold two objects, one in each hand, and say which is heavier and which is lighter.
- Pour water into cups of different sizes and ask which holds more and which holds less, and whether a cup is full or empty.
- Pick one object, like a water bottle, and name all the attributes you can measure about it together.

2 THE LESSON • READ TOGETHER

Everything around you can be measured in some way! When we measure something, we look closely at it and describe how big it is in a special way. Grown-up mathematicians say that measuring means giving a description of the size of an object based on one thing about it, called an attribute. An attribute is just a special word for something you can measure about an object.

Let's meet the four attributes we will learn about. The first is **LENGTH**. Length tells how long something is from one end to the other end. A pencil has length! You can hold a pencil and see it goes from the pointy tip all the way to the eraser. We can say a pencil is long or short, and we can measure it.

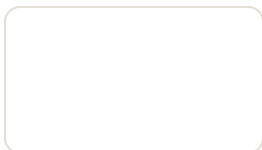
The second attribute is **HEIGHT**. Height tells how tall something is from the bottom to the top. You have height when you stand up straight. A tree is tall, and a little flower is short. We use the words tall and short to talk about height.

The third attribute is **WEIGHT**. Weight tells how heavy or how light something is. When you pick something up, a heavy thing is hard to lift and a light thing is easy to lift. A big rock is heavy. A feather is light.

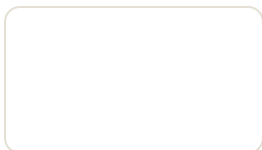
The fourth attribute is **CAPACITY**. Capacity tells how much a container can hold inside it. A cup can hold water. When a cup is all the way filled up, we say it is full. When there is nothing inside, we say it is empty. A big bucket can hold more water, and a tiny cup can hold less.

Here is the exciting part: one object can have **MANY** attributes at the same time! Think about a water bottle. It has **HEIGHT** because it stands tall. It has **WEIGHT** because you can feel how heavy it is when you carry it. And it has **CAPACITY** because it can hold water inside. So the very same bottle has height, weight, and capacity all at once!

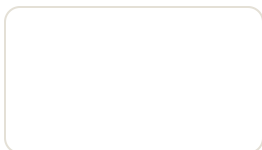
When we compare two objects, we use special comparing words. We say longer or shorter for length, taller or shorter for height, heavier or lighter for weight, and more or less for capacity. These words help us tell how two things are different when we put them side by side.

PICTURE CARDS TO TALK ABOUT

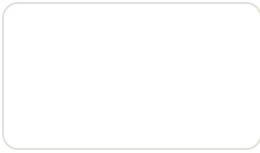
SAY A pencil has length! It goes from the sharp tip to the eraser. We can say the pencil is long, and we can measure how long it is.



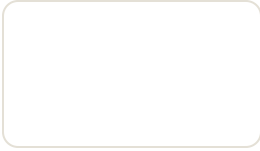
SAY The tree is tall and the flower is short. That is height! Height tells us how tall something is from the bottom to the top.



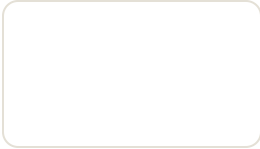
SAY The rock is heavy and the feather is light. That is weight! Weight tells us how heavy or light something is when we lift it.



SAY The cup is full! Capacity tells us how much a cup can hold inside. When it is filled up, we say it is full; when nothing is inside, we say it is empty.



SAY This one bottle has three attributes! It has height because it stands tall, weight because it feels heavy to carry, and capacity because it holds water inside.



SAY When we put two crayons side by side, we can compare them. This crayon is longer and that crayon is shorter. Comparing words help us describe the difference!

3 PRACTICE WORKSHEET

Read each question aloud with Hugh if they are still learning to read.

1. Which word describes how LONG a pencil is?

- ☐ heavy
 - ☐ long
 - ☐ full
-

2. A big rock is hard to lift. What is it?

- ☐ heavy
 - ☐ light
 - ☐ empty
-

3. A cup is filled all the way up with water. We say it is...

- ☐ empty
 - ☐ short
 - ☐ full
-

4. Which attribute tells how TALL a tree is?

- ☐ capacity
 - ☐ height
 - ☐ weight
-

5. Look at a water bottle. Name one thing about it that we can measure.

6. You put a long crayon next to a short crayon. The short crayon is...

- ☐ longer
- ☐ shorter
- ☐ heavier

4 SHOW WHAT YOU KNOW • TALK IT THROUGH

Ask each question out loud. Check the box when Hugh shows they understand.

- ☐ **Look at a pencil. Tell me one thing we can measure about it, and use a describing word.**
 - Names an attribute like length (or says how long it is)
 - Uses a describing word such as 'long' or 'short'
- ☐ **Think about a water bottle. Name as many things about it that we can measure as you can, and tell what each one means.**
 - Names at least two attributes (height, weight, capacity)
 - Correctly connects an attribute to the bottle, e.g. it holds water (capacity) or it is tall (height)
 - Bonus: names all three attributes
- ☐ **Pick any object in the room. Describe it using a measuring word like 'long', 'heavy', or 'full'.**
 - Chooses a real object
 - Uses an appropriate describing word (long, heavy, full, tall, light, etc.) that matches the object

WHERE THESE FACTS COME FROM

English Wikipedia — “In the measurement strand students learn about the measurable attributes of objects... Measurement is the process of assigning a qualitative or quantitative description of size to an object based on a particular attribute.” · “Many of the concepts that students are taught in this strand are also used in other subjects such as science, social studies, and physical education.”

Number bonds to 9

1 FOR THE GROWN-UP · TEACHING GUIDE

Why it matters. Number bonds to 10 are one of the most important early math skills. Knowing which numbers pair up to make 10 helps children add and subtract quickly and lays the foundation for working with larger numbers later.

HOW TO TEACH IT

Start with something your child can touch and count, like 10 fingers, buttons, or a drawn ten-frame. Say a number from 1 to 9 and have your child find how many more are needed to reach 10. Then help them write the pair as an equation such as $6 + 4 = 10$.

WATCH FOR THESE MIX-UPS

- Counting all the objects again instead of finding just the missing part
- Forgetting that the two numbers always add up to exactly 10
- Mixing up the partner (saying 7 goes with 4 instead of 3)
- Struggling to write the equation with the plus and equals signs

TRY THIS AT HOME

- Play 'Make 10': you show some fingers, your child shows the rest to reach 10
- Draw a ten-frame and fill in dots, then count the empty spaces together
- Say a number and race to name its partner to 10
- Write the pairs you find as equations on paper or a whiteboard

2 THE LESSON • READ TOGETHER

When two numbers join together to make a bigger number, we call them a number bond. Today we are learning about number bonds to 10. That means we want to find the number that makes 10 when we add it to a number from 1 to 9. Ten is a special, friendly number, and every number from 1 to 9 has a partner that helps it reach 10.

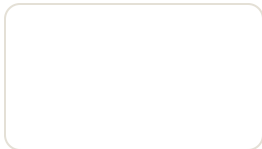
Imagine you have 10 fingers. If you hold up 7 fingers, how many are still folded down? You would see 3 folded fingers, because 7 and 3 are partners that make 10. When someone says 7, you can say 3! When someone says 6, you can say 4! Each number has just one special partner that fills the gap up to 10.

A great tool for finding these partners is a ten-frame. A ten-frame is a little box with 10 spaces, made of 2 rows of 5. When you put some counters in the box, the empty spaces show you exactly how many more you need to fill all 10. If you drop in 6 counters, you can count the empty spaces and find 4. That is how a ten-frame helps you find the complement to 10 — the number that finishes the set.

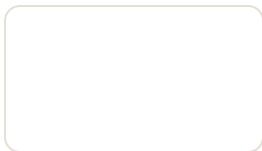
Once you find a pair, you can write it down as an equation. An equation is a math sentence with an equals sign. If 6 and 4 make 10, we write it like this: $6 + 4 = 10$. Writing the pairs down helps you remember them, and soon you will know all the partners to 10 by heart!

PICTURE CARDS TO TALK ABOUT

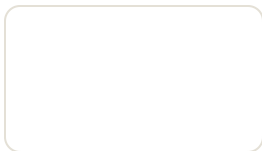
SAY 7 needs 3 more to make 10, so $7 + 3 = 10$.



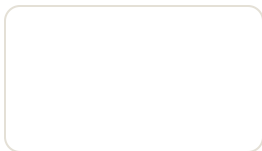
SAY 6 needs 4 more to make 10, so $6 + 4 = 10$.



SAY 9 needs just 1 more to make 10, so $9 + 1 = 10$.



SAY 8 and 2 are partners, so $8 + 2 = 10$.



SAY 5 needs 5 more to make 10, so $5 + 5 = 10$ — a matching pair!

3 PRACTICE WORKSHEET

Read each question aloud with Hugh if they are still learning to read.

1. $48 - 10 =$ _____

2. $86 - 36 =$ _____

3. $65 - 23 =$ _____

4. $65 - 23 =$ _____

5. $39 - 22 =$ _____

6. $45 - 13 =$ _____

7. $31 - 22 =$ _____

8. $84 - 66 =$ _____

9. $41 - 5 =$ _____

10. $73 - 12 =$ _____

11. $31 - 17 =$ _____

12. $70 - 25 =$ _____

4 SHOW WHAT YOU KNOW • TALK IT THROUGH

Ask each question out loud. Check the box when Hugh shows they understand.

- ☐ **Someone says the number 7. What number do you say to make 10, and how do you know?**
- Answers 3
 - Explains that 7 and 3 add together to make 10, using fingers, counting, or a ten-frame idea
- ☐ **Draw or describe a ten-frame with 6 dots. Use it to find how many more you need to make 10.**
- Identifies 4 empty spaces
 - Connects the empty spaces to the missing number that makes 10
- ☐ **Write an equation for a pair of numbers that makes 10, and explain your pair.**
- Writes a correct equation such as $6 + 4 = 10$
 - Both numbers correctly add to 10 with proper use of + and =

WHERE THESE FACTS COME FROM

MATHEMATICS · ADDITION & SUBTRACTION · AGES 4–6

Numbers up to 10 into pairs

1 FOR THE GROWN-UP · TEACHING GUIDE

Why it matters. Breaking numbers into pairs (part-part-whole) is the foundation of addition and subtraction. When a child sees that 5 can be $1+4$, $2+3$, or $0+5$, they understand that numbers are flexible and made of parts. This makes later mental math, fact fluency, and problem solving far easier.

HOW TO TEACH IT

Start with real objects your child can touch and move — buttons, beans, or blocks. Give them a small whole like 5, and have them physically slide the objects into two groups. Then help them say and write the equation. Repeat with the SAME whole to show more than one pair, so they see the whole stays fixed while the parts change.

WATCH FOR THESE MIX-UPS

- Thinking the whole changes when the parts move — remind them to recount the total each time.
- Missing pairs like $0 + 5$ or $5 + 0$; help them see that zero is a valid part.
- Forgetting that swapping the parts ($1+4$ and $4+1$) is another correct way.
- Writing the equation without understanding — make sure the objects match the numbers.

TRY THIS AT HOME

- Snack sort: give 6 crackers and ask how many different ways to split them onto two plates.
- Hidden hands: hold a number of coins, put some in each hand behind your back, then reveal and name the pair.
- Staircase counting: line up all the pairs for a number in order ($0+5$, $1+4$, $2+3$...) and count how many ways there are.
- Draw and label: have your child draw dots, circle two groups, and write the matching equation.

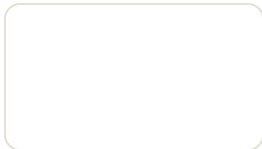
2 THE LESSON • READ TOGETHER

Every number is like a little box that can be filled in more than one way. When we take a number and break it into two smaller groups, we call those two groups a pair, and we call this breaking apart "decomposing." The whole number stays the same, but the two parts can change. This is a big math idea called part-part-whole: there is a whole (the number you start with), and it is made of two parts that fit together to make it.

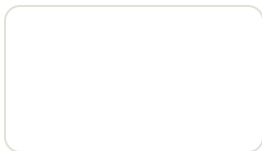
Let's start with the number 5. Imagine you have 5 red buttons. You can push some to one side and some to the other. If you push 1 button to the left and 4 to the right, you have made the pair 1 and 4: that is $5 = 1 + 4$. But you could slide them differently! Push 2 to the left and 3 to the right, and now you have $5 = 2 + 3$. You could even push 0 to one side and keep all 5 on the other side: $5 = 0 + 5$. Every single time, you still have exactly 5 buttons. Only the two parts changed. That is the magic of decomposing a number in more than one way.

We can do this with any number up to 10. The number 4 can be $0 + 4$, $1 + 3$, or $2 + 2$. The number 10 can be $5 + 5$, or $6 + 4$, or $8 + 2$, and lots more. A great way to find every pair is to keep the whole the same and count up one part while the other part goes down. Start with 0 and the whole, then 1 and one less, then 2 and one less, like climbing a little staircase.

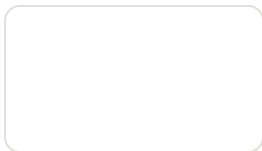
When you find a pair, you can write it down as an equation. An equation uses the plus sign (+) to join the two parts and the equal sign (=) to show they make the whole. So "three and two make five" becomes $3 + 2 = 5$. Objects you can move, drawings you can make, and equations you can write all tell the same story: two parts joining to make one whole.

PICTURE CARDS TO TALK ABOUT

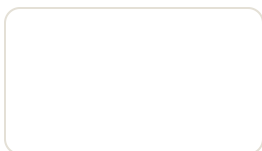
SAY $5 = 1 + 4$. One part is 1, the other part is 4, and together they make the whole, 5.



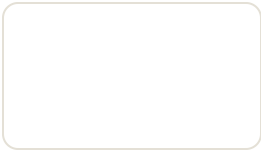
SAY $5 = 2 + 3$. We didn't add or take away any counters — we just made two different parts of the same whole.



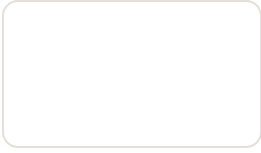
SAY $5 = 0 + 5$. Zero is allowed! One part can be empty, and the whole is still 5.



SAY $4 = 2 + 2$. Sometimes the two parts are equal to each other, and that still makes a real pair.



SAY $10 = 6 + 4$. Even bigger numbers up to 10 can be decomposed into pairs the same way.



SAY $3 = 2 + 1$. We can also flip it: $3 = 1 + 2$. Swapping the parts still makes the same whole.

3 PRACTICE WORKSHEET

Read each question aloud with Hugh if they are still learning to read.

1. Which pair makes 5?

- ☐ $2 + 3$
 - ☐ $2 + 4$
 - ☐ $1 + 1$
 - ☐ $3 + 3$
-

2. You have 4 blocks. Which is another way to make 4?

- ☐ $2 + 2$
 - ☐ $2 + 3$
 - ☐ $1 + 4$
 - ☐ $0 + 3$
-

3. Is $0 + 5$ a real way to make 5?

- ☐ Yes, one part can be zero
 - ☐ No, zero is not allowed
 - ☐ Only for the number 10
 - ☐ Only on Mondays
-

4. Use counters or drawings. Show one way to break 6 into two parts, then write the equation.

5. Fill in the missing part: $3 + \underline{\quad} = 5$

6. If the whole is 5 and one part is 4, what is the other part?

- ☐ 1
- ☐ 2
- ☐ 3
- ☐ 5

4 SHOW WHAT YOU KNOW • TALK IT THROUGH

Ask each question out loud. Check the box when Hugh shows they understand.

☐ **Use objects or drawings to find as many pairs as you can that make 5. Write each one as an equation.**

- Finds more than one correct pair for 5 (e.g., $1+4$, $2+3$, $0+5$).
- Uses objects or drawings that match the numbers.
- Records at least one decomposition as an equation with $+$ and $=$.
- Keeps the whole (5) the same across different pairs.

☐ **Show that the number 4 can be broken into two parts in more than one way. Explain how the parts can change but the whole stays 4.**

- Gives at least two different pairs for 4 (e.g., $1+3$ and $2+2$).
- Explains that the total stays 4 even though the parts change.
- Optionally records the pairs as equations.

☐ **Pick any number up to 10. Draw it, split it into two groups, and write the matching equation. Then find a second, different way and write that equation too.**

- Chooses a number up to 10 and represents it accurately.
- Splits it into two parts and writes a correct equation.
- Finds a second, different valid pair for the same number.
- Both equations use the plus and equal signs correctly.

WHERE THESE FACTS COME FROM

3-D shapes

1 FOR THE GROWN-UP · TEACHING GUIDE

Why it matters. Recognizing and naming solid shapes is one of a child's first steps into geometry. It builds the vocabulary and visual thinking they will later use for measurement, spatial reasoning, and math problem-solving. At this age it also connects math to the real world, because 3-D shapes are the actual objects children hold and play with every day.

HOW TO TEACH IT

Start with real objects your child can hold and turn, not just pictures. Name the shape clearly, then point out what makes it that shape — flat faces versus smooth curved surfaces, whether it can roll, whether it comes to a point. Let your child feel the difference: a cube has flat faces and stacks, while a sphere rolls. Repeat the six names often during everyday moments so they become familiar.

WATCH FOR THESE MIX-UPS

- Confusing a 2-D drawing with the 3-D solid — remind them a solid takes up space and can be turned around to see other sides.
- Mixing up a cube and a cuboid — the cube is the same all around, while the cuboid is stretched longer.
- Calling every round shape a 'circle' — a ball is a sphere (a solid), while a circle is flat.
- Mixing up a cylinder and a cone — the cylinder has two round ends, the cone has one round end and one point.

TRY THIS AT HOME

- Go on a shape hunt in the kitchen: find a cylinder (tin can), a cuboid (cereal box), and a sphere (an orange).
- Sort a pile of household objects into 'shapes that roll' and 'shapes that stack' and talk about why.
- Build a tower and notice which shapes stack well (cubes and cuboids) and which roll away (spheres).
- Play 'I spy a 3-D shape' — 'I spy a cone' and let your child point to the party hat or traffic cone.

2 THE LESSON • READ TOGETHER

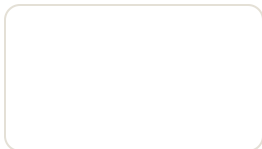
Look around your room. Almost everything you can pick up and hold is a solid — a shape you can turn around and see from many sides. A box is a good example: it is solid, not thin like a piece of paper. It has a top and a bottom, a left side and a right side, and a front and a back. Each of those flat sides is called a face. Because a solid has three ways to measure it — how long it is, how wide it is, and how tall it is — we call it a 3-D shape. "3-D" is short for three-dimensional, which just means it takes up space and has room inside it (grown-ups call that room "volume").

A flat shape, like a drawing of a square on paper, only has two ways to measure it. But a 3-D shape you can hold in your hand and spin around! Some 3-D shapes have flat faces you could stack, like a stack of books. Shapes with flat faces are one big family. Cubes belong to this family, and so do pyramids. Other 3-D shapes have smooth, curved surfaces that roll, like a ball. A sphere belongs to this rolling family.

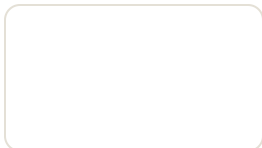
Let's meet the shapes. A cube is like a dice or a building block — all of its flat faces are square, and it looks the same no matter which way you turn it. A cuboid is like a cereal box or a brick — it also has flat faces, but it is longer in one direction, like a box stretched out. A pyramid comes to a point at the top and has a flat bottom, like the great pyramids in the desert. A sphere is a perfectly round ball, smooth all over, with no flat faces at all — it rolls in every direction. A cylinder is shaped like a tin can or a paper-towel tube: it has a round flat face on the top and bottom and a smooth curved side. A cone has a round flat bottom and rises up to a single point, like an ice-cream cone or a party hat.

Once you know these names, you can play a shape hunt anywhere you go. A tin of beans is a cylinder. A ball is a sphere. A traffic cone is a cone. When you can name a shape when you see it, and spot it hiding in real objects around you, you are thinking like a mathematician!

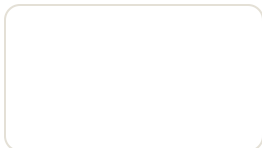
PICTURE CARDS TO TALK ABOUT



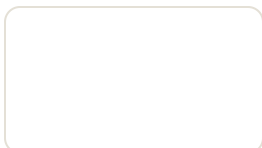
SAY This is a cube. Every one of its flat faces is a square, so it looks the same whichever way you turn it.



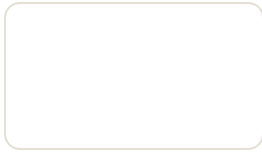
SAY This is a cuboid. It has flat faces like a cube, but it is stretched longer, like a box.



SAY This is a cylinder. It has a round flat top and bottom and a smooth curved side that can roll.



SAY This is a cone. It has a round flat bottom and comes up to a single point at the top.



SAY This is a sphere. It is round and smooth all over with no flat faces, so it rolls in any direction.

PICTURE

A model of an
Egyptian pyramid

SAY This is a pyramid. It has a flat bottom and its faces meet at a point at the top.

3 PRACTICE WORKSHEET

Read each question aloud with Hugh if they are still learning to read.

1. Which shape is perfectly round all over and rolls in every direction?

- ☐ Cube
 - ☐ Sphere
 - ☐ Cuboid
-

2. A tin of beans is which 3-D shape?

- ☐ Cylinder
 - ☐ Cone
 - ☐ Pyramid
-

3. Which shape has a round flat bottom and comes up to a single point at the top?

- ☐ Cube
 - ☐ Sphere
 - ☐ Cone
-

4. A dice with all square faces that looks the same whichever way you turn it is a:

- ☐ Cube
 - ☐ Cylinder
 - ☐ Sphere
-

5. Name one thing in your kitchen that is shaped like a cylinder.

6. Which of these is stretched longer like a cereal box?

- ☐ Cube
- ☐ Cuboid
- ☐ Sphere

4 SHOW WHAT YOU KNOW • TALK IT THROUGH

Ask each question out loud. Check the box when Hugh shows they understand.

- ☐ **I show you a ball, a tin can, and an ice-cream cone. Name each shape.**
- Names the ball a sphere
 - Names the tin can a cylinder
 - Names the cone a cone
- ☐ **Look around the room and find a real object that is a 3-D shape. Tell me the object and which shape it is.**
- Identifies a real object from the environment
 - Correctly names its 3-D shape (e.g. a tin is a cylinder, a box is a cuboid)
- ☐ **From a set of solid shapes, point to the cuboid and the pyramid, and tell me one way they are different.**
- Correctly picks out the cuboid
 - Correctly picks out the pyramid
 - Gives a reasonable difference, such as the pyramid comes to a point
- ☐ **Tell me how a cube is different from a sphere.**
- Notes the cube has flat faces or stacks
 - Notes the sphere is round/smooth or rolls

WHERE THESE FACTS COME FROM

Simple English Wikipedia — “a box is three-dimensional; it is solid... It has volume, a top and bottom, left and right (sides), as well as a front and back... viewed from another side (called a face, surface or simply side)”

English Wikipedia — “Among the most common three-dimensional shapes are polyhedra, which are shapes with flat faces... cylinders; and cones... include cubes as well as pyramids” · “Other three-dimensional shapes may be bounded by curved surfaces, such as the ellipsoid and the sphere”

MATHEMATICS · MEASUREMENT · AGES 4–6

Comparing Lengths & Heights

1 FOR THE GROWN-UP · TEACHING GUIDE

Why it matters. Comparing length and height is one of a child's very first measurement skills. Before children can use rulers, they learn to compare two objects directly and describe the difference. This builds the language and thinking they will need later for real measuring, and it gives them words to describe the world around them.

HOW TO TEACH IT

Use real objects your child can hold and line up. Always model lining up the starting ends fairly, then ask which is longer, shorter, taller, or shorter. Say the comparing words out loud together and have your child repeat them in a full sentence, such as "The blue one is longer." When two objects have a clear two-to-one relationship, introduce 'double' and 'half.'

WATCH FOR THESE MIX-UPS

- Not lining up the starting ends, which makes the comparison unfair
- Mixing up 'tall/short' (up and down) with 'long/short' (end to end)
- Saying 'more bigger' instead of the correct comparing word like 'taller' or 'longer'
- Guessing by looks instead of putting the two objects right next to each other

TRY THIS AT HOME

- Stand back-to-back with your child and decide together who is taller and who is shorter
- Line up two spoons, two socks, or two toy cars and ask which is longer
- Build two block towers and describe one as taller and one as shorter
- Make one tower with 6 blocks and one with 3, then talk about double and half

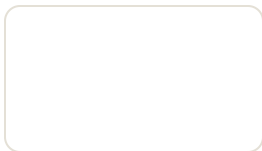
2 THE LESSON • READ TOGETHER

Have you ever stood next to a friend and wondered, "Who is taller?" Or picked up two ribbons and asked, "Which one is longer?" When you do that, you are doing math! Comparing means looking at two things and deciding how they are different. In this lesson we will compare things by their **LENGTH** (how long something is from one end to the other) and by their **HEIGHT** (how tall something is from the bottom to the top).

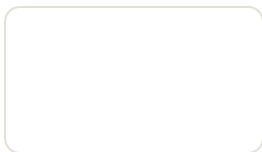
Here are some special math words we will use. Some things are **LONG**, like a jump rope. Some things are **SHORT**, like a crayon. Some things are **TALL**, like a tree. When we compare **TWO** things, we add a little sound to the end of the word. If one ribbon is longer than the other, we say it is **LONGER**. If it is not as long, we say it is **SHORTER**. If one tower is taller, we say it is **TALLER**. If one person is not as tall, we say they are **SHORTER**. These words always compare two things to each other.

The best way to compare is to put the two objects right next to each other. This is called comparing directly. To see who is taller, two children stand back-to-back, with their heels touching the ground and the tops of their heads close together. Then you can see whose head is higher up. To compare two ribbons, lay them side by side and line up one end of each ribbon at the same starting spot. Now your eyes can tell which ribbon reaches farther. When we line things up fairly like this, our comparison is true.

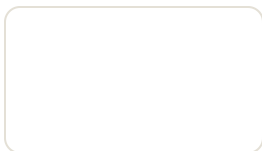
Sometimes one thing is exactly two times as big as another. Then we say it is **DOUBLE**. If your block tower is double the height of your friend's tower, yours is twice as tall. The little tower is **HALF** of yours, because two of the little towers would make one of your big ones. Double means twice as much, and half means one of two equal parts. These words help us describe not just that something is bigger, but **HOW MUCH** bigger.

PICTURE CARDS TO TALK ABOUT

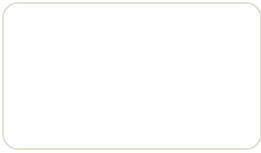
SAY Maya's head is higher than Sam's head, so Maya is **TALLER** and Sam is **SHORTER**.



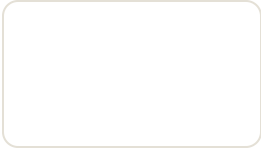
SAY This tower reaches higher, so it is **TALLER** than the other one.



SAY The red ribbon reaches farther, so it is **LONGER**, and the blue ribbon is **SHORTER**.



SAY The jump rope is LONG and the pencil is SHORT.



SAY The big tower is DOUBLE the little one, and the little tower is HALF of the big one.

3 SHOW WHAT YOU KNOW • TALK IT THROUGH

Ask each question out loud. Check the box when Hugh shows they understand.

☐ **Two children stand back-to-back. Explain how you can tell which child is taller and which is shorter.**

- Mentions standing back-to-back or lining up feet/heads
- Uses the word 'taller' for the child whose head is higher
- Uses the word 'shorter' for the other child

☐ **You build two block towers. Describe how they are different using height words.**

- Compares the two towers directly (one next to the other)
- Correctly uses 'taller' for the higher tower
- May also use 'shorter' for the lower tower

☐ **You have two ribbons. Tell how you would compare them and use the words 'longer' and 'shorter' in your answer.**

- Says to place the ribbons side by side and line up the ends
- Uses 'longer' for the ribbon that reaches farther
- Uses 'shorter' for the ribbon that does not reach as far

WHERE THESE FACTS COME FROM

MATHEMATICS · MEASUREMENT · AGES 4–6

Comparing Capacity

1 FOR THE GROWN-UP · TEACHING GUIDE

Why it matters. Comparing capacity helps children describe and compare how much containers hold using everyday words like full, empty, more than, less than, and half full. This is an early measurement skill that builds real mathematical thinking children use at the sink, at meals, and in play.

HOW TO TEACH IT

Use real containers and water, rice, or sand. Let your child pour from one container to another and talk out loud using the describing words: 'Is this cup full or empty? Which one holds more?' Comparing two containers directly by pouring is the clearest way for a young child to see which holds more or less.

WATCH FOR THESE MIX-UPS

- Thinking a tall, thin container always holds more than a short, wide one \u2014 pour to check instead of just looking
- Confusing 'full' with 'a lot' \u2014 remind them full means holding all it can, right to the top
- Mixing up 'more than' and 'less than' \u2014 practice pointing and saying both phrases aloud

TRY THIS AT HOME

- At bath time, ask your child to make one cup full and one cup empty, then say which is which
- Fill a clear glass to the middle and ask if it is half full; then fill another to the bottom and ask if it is quarter full
- Line up two different cups at snack time and ask 'Which one holds more juice?' \u2014 then pour to find out

2 THE LESSON • READ TOGETHER

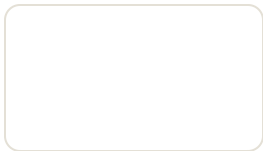
Have you ever poured juice into a cup, filled a bucket at the beach, or watched a bathtub fill up with water? Every time you do, you are learning about capacity! Capacity means how much a container can hold inside it. Some containers can hold a lot, and some can hold only a little.

When we talk about capacity, we use special describing words. A container is full when it is holding all it can, like a cup filled right to the top. A container is empty when there is nothing inside it at all, like a cup after you drink every drop. In between full and empty, we can say a container is half full when it holds about the same amount as the empty space above it, or quarter full when it holds just a little bit at the bottom.

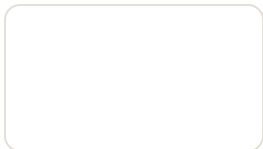
The most fun part is comparing two containers. We can pick up two cups and ask, 'Which one holds more, and which one holds less?' When one container has more water inside than another, we say it holds more than the other. When it has less, we say it holds less than the other. Sometimes a small cup that is completely full still holds less than a big cup that is only half full \u2014 so it always helps to look carefully and pour to check!

Learning these words \u2014 full, empty, more than, less than, half full, and quarter full \u2014 helps you describe and compare how much things hold. You can use them at the sink, at snack time, or in the bathtub. You are doing real math when you compare capacity!

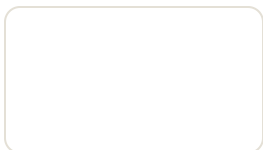
PICTURE CARDS TO TALK ABOUT



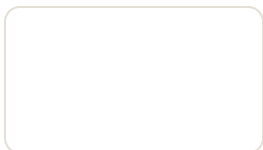
SAY This cup is full \u2014 it is holding all the water it can hold.



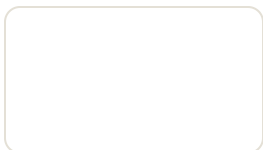
SAY Now the cup is empty \u2014 there is nothing inside it.



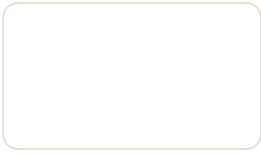
SAY This glass is half full \u2014 the milk fills half and empty space fills the other half.



SAY This jar is only a quarter full \u2014 it has just a little bit inside.



SAY The pitcher holds more than the teacup, and the teacup holds less than the pitcher.



SAY The first cup has more water and the second cup has less water \u2014 we can compare which holds more!

3 SHOW WHAT YOU KNOW • TALK IT THROUGH

Ask each question out loud. Check the box when Hugh shows they understand.

☐ **Point to a full container and an empty container in your house. Tell someone which is which and how you know.**

- Correctly identifies a full container as holding all it can
- Correctly identifies an empty container as having nothing inside
- Uses the words 'full' and 'empty' to describe them

☐ **Fill a clear cup partway with water. Describe how full it is using 'half full' or 'quarter full', and explain your choice.**

- Uses 'half full' or 'quarter full' appropriately
- Connects the word to how much is inside the container
- Explains reasoning in simple terms

☐ **You have two cups of water. Show how you can find out which cup has more water, and say which words you would use to tell about them.**

- Describes comparing the two containers (for example, by looking or pouring)
- Correctly says which holds more and which holds less
- Uses 'more than' and 'less than' language

WHERE THESE FACTS COME FROM

MATHEMATICS · MEASUREMENT · AGES 4–6

Measuring mass and weight (age 4+)

1 FOR THE GROWN-UP · TEACHING GUIDE

Why it matters. Comparing objects by weight is one of a child's first steps into measurement math. Before children ever use rulers or scales with numbers, they learn to compare things directly and describe the difference with words like heavier and lighter. This builds the foundation for measurement and comparison later in school.

HOW TO TEACH IT

Start with your child's own hands. Give them two very different objects — say a shoe and a sock — and let them 'heft' one in each hand, feeling which pulls down more. Then introduce a simple balance (even a coat hanger with two cups can work) so they can see the heavier object make its side sink. Always model the full sentence: 'The shoe is heavier than the sock.'

WATCH FOR THESE MIX-UPS

- Guessing that the bigger object is always heavier — a big beach ball is lighter than a small rock, so let them test surprising pairs.
- Saying only 'heavy' or 'light' without comparing; gently prompt the full 'heavier than / lighter than' sentence.
- Confusing which side of the balance means heavier — remind them the side that goes DOWN holds the heavier object.

TRY THIS AT HOME

- Do a 'two-hand test': hand your child two objects and ask which hand feels heavier.
- Build a simple balance and compare kitchen items like a spoon and a potato.
- Make a game: sort a pile of toys into a 'heavy' basket and a 'light' basket.
- Ask comparing questions at snack time: 'Is the apple heavier than the cracker?'

2 THE LESSON • READ TOGETHER

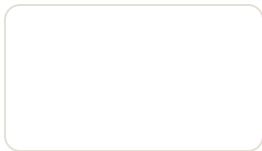
Have you ever picked up a big rock and then a soft feather? The rock feels **HARD** to lift, and the feather feels like nothing at all! When we talk about how heavy or how light something is, we are talking about its weight and its mass. Today we are going to learn how to compare two things and say which one is heavier and which one is lighter.

The word 'heavy' means something is hard to lift or pushes down a lot, like a big bag of potatoes. The word 'light' means something is easy to lift and does not push down much, like a cotton ball or a leaf. Every object in the world can be heavy or light, and we can compare them to each other.

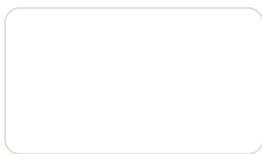
One fun way to find out which object is heavier is to use your own two hands. Hold one object in your left hand and one object in your right hand. Let your hands go up and down, like they are little scales. This is called 'hefting.' The hand that goes **DOWN** is holding the heavier object, and the hand that goes **UP** is holding the lighter object. Your hands can feel the difference!

Another way is to use a special tool called a balance. A balance has two little dishes, like a seesaw. When you put an object in each dish, the dish that goes **DOWN** holds the heavier object, and the dish that goes **UP** holds the lighter one. If both dishes stay even, the objects weigh the same!

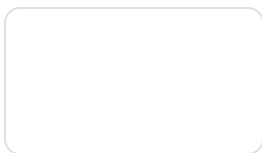
When we compare two things, we use special words. We say one thing is 'heavier than' the other, or one thing is 'lighter than' the other. For example, 'An apple is heavier than a grape,' and 'A grape is lighter than an apple.' These comparing words help us describe exactly what we found.

PICTURE CARDS TO TALK ABOUT

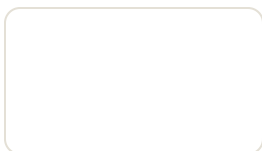
SAY The hand with the book sinks down low. The book is heavier than the crayon, and the crayon is lighter than the book!



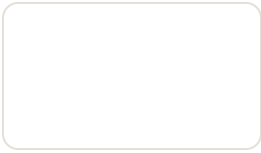
SAY The left dish went down, so the toy car is heavier than the marble.



SAY The feather is light and easy to lift. The rock is heavy and hard to lift. The rock is heavier than the feather.



SAY Neither dish goes down. The two blocks weigh the same — they are not heavier or lighter than each other.



SAY The full bottle pulls the hand down. The full bottle is heavier than the empty bottle, and the empty bottle is lighter than the full one.

3 SHOW WHAT YOU KNOW • TALK IT THROUGH

Ask each question out loud. Check the box when Hugh shows they understand.

☐ **Pick up two things near you, one in each hand. Tell me which one is heavier and which one is lighter, and how you knew.**

- Names one object as heavier and the other as lighter
- Explains that hefting/feeling which hand went down showed the difference

☐ **You put a toy car and a marble on a balance and the car's side goes down. Explain what the balance tells you about the two objects.**

- States that the toy car is heavier than the marble
- Connects the dish going down to being the heavier object

☐ **Use the words 'heavier than' in a sentence about two objects you know.**

- Uses the phrase 'heavier than' correctly
- Compares two specific objects

WHERE THESE FACTS COME FROM

MATHEMATICS · MEASUREMENT · AGES 4–6

Ordering Events in Time

1 FOR THE GROWN-UP · TEACHING GUIDE

Why it matters. Ordering events in time is an early math and thinking skill in the Year 1 measurement curriculum. It helps children understand sequence, cause and effect, and the passage of time — foundations they'll later use for telling time, reading calendars, and understanding stories.

HOW TO TEACH IT

Use your real daily routine as the teaching tool. Narrate the day out loud with time words: 'First we get dressed, next we eat, last we head out.' Anchor 'yesterday, today, tomorrow' to real events your child remembers or looks forward to, and label parts of the day (morning, afternoon, evening) as they happen so the words attach to lived experience.

WATCH FOR THESE MIX-UPS

- Mixing up yesterday and tomorrow — remind them yesterday is finished and tomorrow hasn't come yet.
- Confusing 'before' and 'after.' Act out the order slowly to make it concrete.
- Guessing morning vs. afternoon — connect each to a landmark like breakfast, lunch, or bedtime.

TRY THIS AT HOME

- At bedtime, ask your child to retell their day using 'first, next, last.'
- Each morning say, 'Yesterday we ____, today we ____, tomorrow we will ____.'
- Play 'Which part of the day?' — name an activity (eating breakfast) and have them say morning, afternoon, or evening.
- Line up three picture cards of a routine and have your child put them in order and explain.

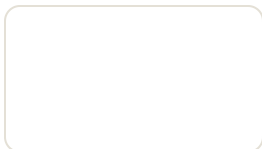
2 THE LESSON • READ TOGETHER

Every single day, things happen in an order. You wake up, you eat breakfast, you play, you eat dinner, and then you go to sleep. These are called events — the things that happen. Putting events in the order they happen is called putting them in time order, or sequencing. This lesson helps you learn special time words so you can tell everyone exactly WHEN something happens.

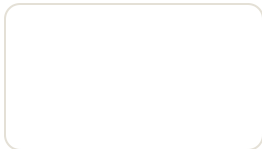
One of the most helpful sets of words is first, next, and last. Imagine getting ready for bed. FIRST you take a bath. NEXT you put on your pajamas. LAST you climb into bed and hear a story. 'First' means the very beginning. 'Next' means what comes after that. 'Last' means the ending. When you use these three words together, you can tell the whole story of what you did in the right order!

We also have three big words for whole days: yesterday, today, and tomorrow. TODAY is the day you are in right now. YESTERDAY was the day before today — it already happened and is finished. TOMORROW is the day that is coming after today — it hasn't happened yet. So you might say, 'Yesterday I went to the park, today I am at school, and tomorrow I will visit Grandma.'

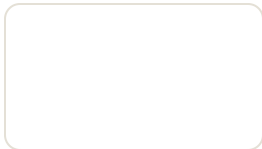
There are also words for the parts of a single day. The MORNING is the early part of the day when you wake up, the sun comes up, and you eat breakfast. The AFTERNOON is the middle part, after lunch, when the sun is high. The EVENING is the later part, when the sky gets dark and it's almost time for bed. Other useful time words are 'before' (something that happens earlier) and 'after' (something that happens later). When you learn all these words, you can put any events in the right order and tell people exactly when things happen!

PICTURE CARDS TO TALK ABOUT

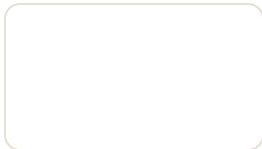
SAY Making a sandwich: FIRST get the bread, NEXT add the peanut butter, LAST take a bite. Those three words put the events in order.



SAY 'Today' is the day happening right now. If it's your birthday, you can say, 'Today I turn five!'



SAY 'Yesterday' is the day before today — it already finished. 'Yesterday it rained, so I wore my boots.'



SAY 'Tomorrow' is the day coming after today — it hasn't happened yet. 'Tomorrow we will go swimming.'

SAY The morning is the early part of the day. 'In the morning I brush my teeth and eat breakfast.'

SAY The afternoon comes after lunch, in the middle of the day. 'In the afternoon we play outside at recess.'

SAY The evening is the late part of the day when it gets dark. 'In the evening I put on my pajamas.'

SAY 'Before' means earlier and 'after' means later. 'I wash my hands BEFORE dinner and brush my teeth AFTER dinner.'

3 SHOW WHAT YOU KNOW • TALK IT THROUGH

Ask each question out loud. Check the box when Hugh shows they understand.

- ☐ **Tell me three things you did today in the right order. Use the words 'first,' 'next,' and 'last.'**
- Names three real events
 - Places them in a sensible time order
 - Uses first, next, and last correctly
- ☐ **What did you do yesterday, what are you doing today, and what will you do tomorrow?**
- Uses 'yesterday' for something already finished
 - Uses 'today' for something happening now
 - Uses 'tomorrow' for something not yet happened
- ☐ **Pick one thing you do in the morning and one thing you do in the afternoon, and tell which is which.**
- Correctly labels a morning activity
 - Correctly labels an afternoon activity
 - Shows understanding that morning comes before afternoon

WHERE THESE FACTS COME FROM

2-D shapes

1 FOR THE GROWN-UP · TEACHING GUIDE

Why it matters. Recognizing and naming flat shapes is one of the very first steps in geometry. It helps children describe the world around them, builds the vocabulary they'll need for later math, and trains them to notice details like sides and corners. Being able to name a triangle, circle, rectangle, and square is a core early-math milestone.

HOW TO TEACH IT

Start by naming shapes out loud as you see them in real life — "That window is a rectangle, that clock is a circle." Trace the edges with a finger and count the sides and corners together. Emphasize the big idea that a shape keeps its name even when it is big, small, or turned a different way, because this is exactly where young children get tripped up. Play sorting games where your child pulls all the triangles out of a mixed pile.

WATCH FOR THESE MIX-UPS

- Thinking a shape is only a triangle (or square) when it points a certain way — turn shapes to show they keep their name.
- Believing a bigger or smaller version is a different shape — line up two sizes to prove they match.
- Confusing squares and rectangles — remind them a square is a rectangle with all sides equal.
- Counting a curved circle as having sides or corners — circles have neither.

TRY THIS AT HOME

- Go on a shape hunt around the house and name each circle, triangle, rectangle, and square you find.
- Cut shapes from paper, then turn and flip them so your child names them in every position.
- Mix several paper shapes on the table and ask your child to pick out ALL the triangles.
- Make a snack picture: a square cracker, a round cracker, a triangle cheese slice — name each before eating!

2 THE LESSON • READ TOGETHER

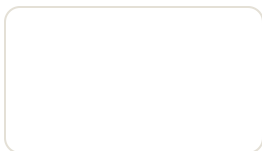
A shape is the outline of something — the edge that goes all the way around it. When we talk about a shape, we don't care what color it is or what it's made of. We only care about its form, the way its edges look. A flat shape that lies on a piece of paper is called a 2-D shape, which is short for two-dimensional shape. "Flat" is the key word: a 2-D shape has a length and a width, but it does not stick up into the air the way a ball or a box does.

Here is a wonderful secret about shapes: a shape stays the same shape no matter how big or small it is, and no matter which way it is turned. If you make a triangle tiny or giant, spin it upside down, or lay it on its side, it is **STILL** a triangle! Size and turning-direction don't change what a shape is called. That means you can be a shape detective and find a triangle even when it's hiding sideways or hiding as a big one and a little one.

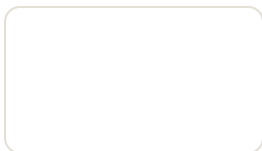
Let's meet the four shapes you will learn to name. First is the circle. A circle is perfectly round, like a wheel or the sun. It has no corners and no straight sides — just one smooth curved line that goes all the way around. Next is the triangle. "Tri" means three, and a triangle has three straight sides and three corners (we call corners "vertices"). Triangles can be tall and skinny or short and wide, but they always have exactly three sides.

The next two shapes belong to a family of four-sided shapes. A rectangle has four straight sides and four corners, and all of its corners are square corners — nice and neat, like the corner of a door or a book. The sides across from each other are the same length. A square is a very special rectangle: it also has four straight sides and four square corners, but on a square **ALL** four sides are exactly the same length, like a cracker or a window pane. So a square is really just a rectangle whose sides are all equal!

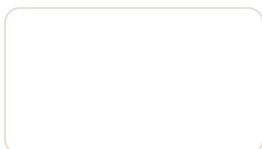
Why do we learn shapes? Because shapes are everywhere! Once you know their names, you can spot them in doors, wheels, sandwiches, signs, and windows. Naming shapes is one of the first big ideas in geometry, the part of math all about shapes and space. Ready to become a shape detective?

PICTURE CARDS TO TALK ABOUT

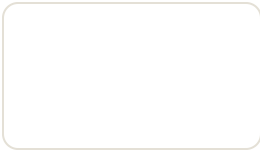
SAY These are all circles. A circle is perfectly round with one smooth curved line and no corners — a wheel is shaped like a circle too.



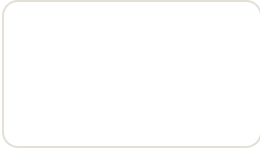
SAY These are triangles. Count the sides — one, two, three! A triangle always has three straight sides and three corners, even if it's turned upside down.



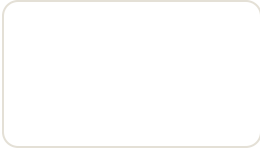
SAY These are rectangles. They have four straight sides and four square corners, and the sides across from each other match. The long sides are longer than the short sides.



SAY These are squares. A square is a special rectangle where all four sides are exactly the same length. It still has four square corners.



SAY Both are triangles! A shape doesn't change its name when it gets bigger or smaller — size doesn't matter.



SAY Still a triangle both times! Turning a shape around does not change what it is. Count the three sides and you'll know.

3 PRACTICE WORKSHEET

Read each question aloud with Hugh if they are still learning to read.

1. Which shape is perfectly round with no corners?

- ☐ Triangle
 - ☐ Circle
 - ☐ Square
-

2. How many sides does a triangle have?

- ☐ Two
 - ☐ Three
 - ☐ Four
-

3. A shape has four straight sides and four square corners, and all four sides are exactly the same length. What is it?

- ☐ Square
 - ☐ Circle
 - ☐ Triangle
-

4. A triangle is turned upside down. Is it still a triangle?

- ☐ Yes, turning it does not change the shape
 - ☐ No, it becomes a different shape
-

5. Which shape looks like a door or a book cover, with two long sides and two short sides?

- ☐ Circle
 - ☐ Rectangle
 - ☐ Triangle
-

6. Look around the room. Name one thing that is shaped like a circle.

4 SHOW WHAT YOU KNOW · TALK IT THROUGH

Ask each question out loud. Check the box when Hugh shows they understand.

☐ **I show you a triangle, a circle, a rectangle, and a square. Say the name of each shape.**

- Correctly names all four shapes
- Names the circle as round with no corners
- Distinguishes the square and rectangle

☐ **Here is a big triangle and a small triangle turned on its side. Tell me the name of each and explain why they are both the same shape.**

- Identifies both as triangles
- Explains that size does not change the shape
- Explains that turning/orientation does not change the shape

☐ **From a mixed pile of shapes (circles, squares, rectangles, and triangles), point to ALL the triangles.**

- Selects every triangle in the set
- Does not include non-triangles
- Recognizes triangles in different sizes and orientations

WHERE THESE FACTS COME FROM

English Wikipedia — “In geometry, shape excludes information about the object's position, size, orientation and chirality.” ·
“polygons are classified according to their number of edges as triangles, quadrilaterals... quadrilaterals can be rectangles, rhombi, trapezoids, squares”

Simple English Wikipedia — “The sides (faces) of a cube are squares... The corners (vertices) are at right angles.”

Positional Language

1 FOR THE GROWN-UP · TEACHING GUIDE

Why it matters. Positional language helps children describe exactly where objects are using words like above, below, beside, in front of, behind, and next to. These words are the building blocks for early math and geometry, and they help kids follow and give clear directions in everyday life.

HOW TO TEACH IT

Teach position words with real objects your child can touch and move. Name the position out loud ('The spoon is beside the bowl'), then ask your child to place objects and say where they are. Practice 'on top of,' 'behind,' and gradually add 'left' and 'right' using their own hands as a reminder.

WATCH FOR THESE MIX-UPS

- Mixing up 'in front of' and 'behind' — practice with the child facing an object.
- Confusing left and right — this is normal at this age; use a hand or a small sticker as a reminder.
- Using a position word without really checking where the object is — encourage them to look and point first.

TRY THIS AT HOME

- Play 'Where is it?' Place a toy and ask your child to tell you its position using a word like above, below, beside, or behind.
- Give simple instructions like 'Put the cube behind the box' or 'Set the cup on top of the table' and let them do it.
- During cleanup, name where each toy goes: 'The blocks go next to the books, and the ball goes below the shelf.'

2 THE LESSON • READ TOGETHER

Have you ever told a friend where to find your favorite toy? Maybe you said, "It's on top of my bed!" or "Look behind the door!" When you do that, you are using positional language. Positional language is a set of special words that tell us WHERE something is. They describe the position of an object compared to other things around it.

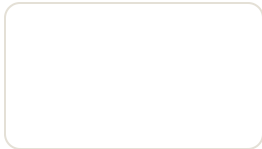
There are lots of these helpful position words. Some tell us if something is high or low, like above (higher up) and below (lower down). Some tell us if something is close by, like beside and next to, which both mean right alongside something. Some tell us which side something is facing, like in front of (the part you see first) and behind (the hidden part in back). And when we sit at a table or lay a toy down, we can say it is on top of something.

We also use two very important words: left and right. Your hands can help you! Hold them out and notice which one is your left hand and which one is your right hand. Then you can say things like "The cup is on the left" or "Turn right."

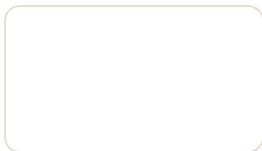
Position words are useful every single day. They help us give directions, find lost things, put toys away in the right spot, and play games. When you can say exactly where something is, other people understand you perfectly. Let's practice looking around and naming positions!

PICTURE CARDS TO TALK ABOUT

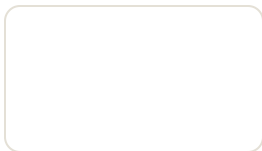
SAY The teddy bear is on top of the table. 'On top of' means it is resting up on the highest part.



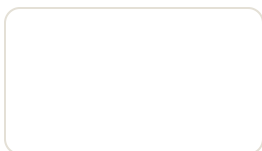
SAY The picture is above the bed (higher up), and the shoes are below the bed (lower down).



SAY First the cube is behind the box (hidden in back). Now it is in front of the box (where we can see it).



SAY The cat is beside the dog. 'Beside' and 'next to' both mean right alongside, very close.



SAY The apple is in my left hand and the banana is in my right hand.

3 SHOW WHAT YOU KNOW • TALK IT THROUGH

Ask each question out loud. Check the box when Hugh shows they understand.

☐ **Look at a toy sitting on your table. Tell me where the toy is using a position word.**

- Uses a correct position phrase such as 'on top of' the table
- Names the toy and the table clearly
- Position described matches where the toy actually is

☐ **I will say: 'Put the cube behind the box.' Show me what you do and tell me where the cube ends up.**

- Follows the instruction correctly by placing the cube behind the box
- Uses the word 'behind' when describing it
- Shows understanding that 'behind' means in back of the box

☐ **Point to something on your left and something on your right. Say which is which.**

- Correctly identifies left and right sides
- Uses the words 'left' and 'right' in a simple sentence
- Points to objects that match the direction named

WHERE THESE FACTS COME FROM

MATHEMATICS · ADDITION & SUBTRACTION · AGES 4–6

Addition and subtraction word problems

1 FOR THE GROWN-UP · TEACHING GUIDE

Why it matters. Solving simple word problems teaches your child that numbers describe real things — apples, toys, and friends. It builds the foundation for all later arithmetic by connecting counting to the two most basic operations, addition (putting together) and subtraction (taking away), within numbers up to 10.

HOW TO TEACH IT

Keep it hands-on. Read a short story aloud, then have your child act it out with counters (beans, buttons, coins) or draw quick dots. For 'add to' and 'put together' problems, physically slide groups together and count the total. For 'take from' problems, count out the starting amount and physically remove the pieces, then count what remains. Say the number sentence together, like 'six take away two equals four.'

WATCH FOR THESE MIX-UPS

- Recounting from 1 instead of trusting the total — that's normal at this age; let them count all.
- Confusing 'add to' with 'take from' — listen for clue words like 'gives more' (add) versus 'eats,' 'flies away,' or 'gives away' (take from).
- Forgetting that order doesn't matter in addition ($3 + 2$ is the same as $2 + 3$).
- Rushing to guess without acting out the story with objects.

TRY THIS AT HOME

- At snack time, ask 'You have 6 crackers and eat 2 — how many are left?' and count together.
- Combine two small groups of toys and ask how many there are in all.
- Use fingers or buttons to model 'add to' stories, like getting 3 more stickers.
- Draw dots on paper and cross some out to show 'take from' problems.

2 THE LESSON • READ TOGETHER

Have you ever put your toys together to see how many you have, or eaten some crackers and wondered how many are left? That is math! When we tell a little story with numbers and figure out the answer, we are solving a word problem. Today we will learn to solve addition and subtraction word problems using things you can touch, like counters, blocks, buttons, or beans, or by drawing simple pictures.

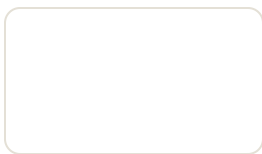
Addition means putting groups together to find the total, or sum. The plus sign, $+$, tells us to add. For example, if you have three apples in one basket and two apples in another basket, you can put them together. When you count them all, you get five apples. We write this as $3 + 2 = 5$, which we read as 'three plus two equals five.' Addition is one of the four basic math operations, and it is one of the very first math jobs children learn — even babies can notice that 1 and 1 makes more!

Here is a fun secret about adding: it does not matter which group you count first. $3 + 2$ gives the same answer as $2 + 3$. Both make 5! And if you add zero — nothing at all — your number stays the same. Adding 1 over and over is just like counting: 1, 2, 3, 4, 5.

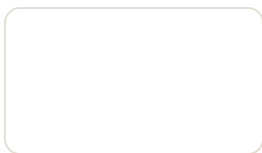
Subtraction is the opposite of addition. Instead of putting groups together, we take some away and find how many are left. If you have 6 apples and 2 get eaten, you take those 2 away and count what stays. That leaves 4 apples. Subtraction is another one of the four basic operations, and it is closely related to addition — they are like two sides of the same coin.

Word problems come in a few friendly kinds. Some are 'add to' problems, where a group grows because you add more. Some are 'take from' problems, where a group gets smaller because you take some away. And some are 'put together' problems, where you have two groups and want to know the total of both. For all of them, the same trick works: act out the story with real objects or a quick drawing, then count. Let's practice with numbers up to 10!

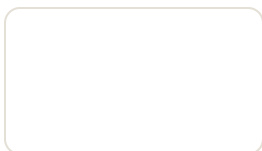
PICTURE CARDS TO TALK ABOUT



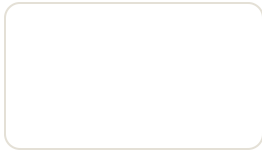
SAY This is a 'put together' problem. Slide the groups together and count them all: 1, 2, 3, 4, 5. So $3 + 2 = 5$. There are 5 apples in all.



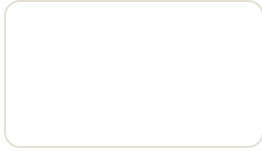
SAY This is an 'add to' problem — your group grows. Start with 4 counters, add 3 more, then count everything: $4 + 3 = 7$. You now have 7 blocks.



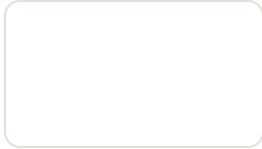
SAY This is a 'take from' problem. Put out 6 counters, take 2 away, and count what is left: $6 - 2 = 4$. There are 4 apples left.



SAY Take-from again! Put out 5 counters, remove 1, count the rest: $5 - 1 = 4$. Four birds are still on the branch.



SAY Put together to find the total. Count both groups: $2 + 5 = 7$. There are 7 marbles in the bag. Remember, $5 + 2$ makes 7 too — the order does not change the answer!



SAY Take away zero means nothing changes. $7 - 0 = 7$. All 7 dots stay. Adding or taking away zero keeps the number the same.

3 PRACTICE WORKSHEET

Read each question aloud with Hugh if they are still learning to read.

1. $88 - 14 =$ _____

2. $67 - 9 =$ _____

3. $41 - 26 =$ _____

4. $38 - 13 =$ _____

5. $90 - 59 =$ _____

6. $46 - 36 =$ _____

7. $73 - 7 =$ _____

8. $21 - 18 =$ _____

9. $94 - 29 =$ _____

10. $20 - 2 =$ _____

11. $31 - 20 =$ _____

12. $66 - 58 =$ _____

4 SHOW WHAT YOU KNOW • TALK IT THROUGH

Ask each question out loud. Check the box when Hugh shows they understand.

- ☐ **There are 6 apples and 2 are eaten. Use counters or a drawing to show how many apples are left, then tell the answer.**
- Sets out or draws 6 objects to start
 - Removes/crosses out 2 objects (shows take-from)
 - States the correct answer: 4 apples left
- ☐ **Make up your own 'add to' story using numbers up to 10, then solve it with objects and say the total.**
- Tells a story where a group grows by adding more
 - Uses objects or a drawing to model it
 - Gives the correct total (sum within 10)
- ☐ **You have 4 red blocks and 3 blue blocks. Put them together and show how many blocks there are in all.**
- Represents both groups (4 and 3) with objects or a drawing
 - Combines the groups and counts all
 - States the correct total: 7 blocks

WHERE THESE FACTS COME FROM

English Wikipedia — “Addition, usually denoted with the plus sign +, is one of the four basic operations of arithmetic... The addition of two whole numbers results in the total or sum of those values combined.” · “the adjacent image shows two columns of apples, one with three apples and the other with two apples, totaling to five apples. This observation is expressed as " $3 + 2 = 5$ ", which is read as "three plus two equals five".”

Representing Addition and Subtraction

1 FOR THE GROWN-UP · TEACHING GUIDE

Why it matters. Representing addition and subtraction with objects, drawings, and fingers is the foundation of all later math. Before a child memorizes number facts, they need to truly SEE what adding (combining groups) and subtracting (taking away) mean. This concrete stage builds real understanding, not just memory, and meets early standards like K.OA.1 and UK Year 1 addition and subtraction goals.

HOW TO TEACH IT

Start with real objects your child can touch — cubes, buttons, cereal, or toys. Model addition by physically pushing two groups together and counting the total (the sum). Model subtraction by sliding some objects away and counting what remains (the difference). Then move to drawings (dots or shapes, crossing out for subtraction), and to fingers, which children can use anywhere. Say the math out loud together: 'Three plus two equals five.' Keep numbers small (totals up to 5 or 10).

WATCH FOR THESE MIX-UPS

- Counting the objects that were taken away instead of the ones remaining during subtraction
- Recounting the whole group after adding instead of counting on from one group
- Thinking the order matters in addition — remind them $3 + 2$ equals $2 + 3$
- Forgetting that adding or subtracting zero leaves the number unchanged
- Rushing to answers without lining up and touching each object

TRY THIS AT HOME

- Use snack pieces: 'Here are 3 crackers, here are 2 more — how many all together?' Then eat some to practice subtraction.
- Play a finger game: 'Show me 4 fingers, now put down 1. How many are up?'
- Draw and cross out: draw 5 balloons and 'pop' 2 with an X to show $5 - 2$.
- Sort toys into two groups, combine them, and count the sum together.
- Point out real-life subtraction: 'You had 6 grapes, you ate 2, how many now?'

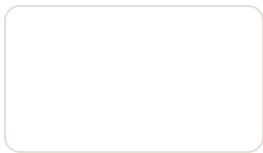
2 THE LESSON • READ TOGETHER

Have you ever put your toys together into one big pile, or taken some crackers off your plate to eat? If so, you already know a little bit about two math ideas called addition and subtraction! Today we will learn how to **SHOW** these ideas using things we can touch, pictures we can draw, and even our own fingers.

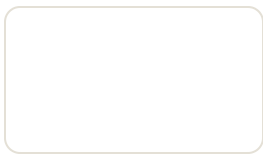
Addition is when we put two groups together to make one bigger group. When we do this, the new total is called the sum. We use a special symbol, the plus sign (+), to show addition. Imagine you have a group of 3 shapes and another group of 2 shapes. When you scoot the two groups together into one, you count all of them and get 5. So $3 + 2 = 5$! Adding is really just combining. Here is a fun secret: it doesn't matter which group you count first. $3 + 4$ gives the same answer as $4 + 3$. And if you add nothing (zero), your number stays exactly the same.

Subtraction is the opposite idea. Subtraction means taking some things away from a group. We use the minus sign (−) to show it. Picture 5 juicy peaches. If you take 2 of them away, how many are left? Count the ones still there: 3 peaches! So $5 - 2 = 3$. The answer we get when we subtract is called the difference. Just like with adding, if you take away zero, your number does not change at all.

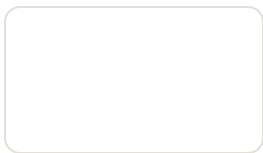
The best part is that we can show these ideas in lots of ways. We can use real objects like cubes, counters, buttons, or blocks — pushing them together to add, or sliding some away to subtract. We can draw pictures, like dots or little peaches, and cross some out to show taking away. We can even use our ten fingers, holding up some and then putting some down. People have used objects to add and subtract for a very, very long time — on a tool called an abacus, you add by gathering beads together and subtract by moving beads away. When you show a math problem in these ways, you can **SEE** the math happening right in front of you!

PICTURE CARDS TO TALK ABOUT

SAY I combined two groups to make one group. $3 + 2 = 5$. The total is called the sum.



SAY I took 2 away, so $5 - 2 = 3$. The answer left over is called the difference.



SAY $4 + 1 = 5$. My fingers are objects I can use to add anywhere I go!

SAY $2 + 4 = 6$ and $4 + 2 = 6$. Adding gives the same sum no matter which group I count first!

SAY $5 - 0 = 5$. When I take away nothing, my number stays exactly the same.

SAY I removed 3 from my group. $6 - 3 = 3$. Subtraction means taking objects away from a collection.

3 PRACTICE WORKSHEET

Read each question aloud with Hugh if they are still learning to read.

1. $61 - 7 =$ _____

2. $47 - 19 =$ _____

3. $70 - 57 =$ _____

4. $76 - 19 =$ _____

5. $84 - 38 =$ _____

6. $33 - 25 =$ _____

7. $82 - 32 =$ _____

8. $29 - 9 =$ _____

9. $81 - 59 =$ _____

10. $52 - 43 =$ _____

11. $53 - 6 =$ _____

12. $74 - 17 =$ _____

4 SHOW WHAT YOU KNOW • TALK IT THROUGH

Ask each question out loud. Check the box when Hugh shows they understand.

☐ **Use cubes or counters to show $3 + 2$. Line them up and tell how many there are all together.**

- Makes a group of 3 and a group of 2 with objects
- Combines the two groups and counts them all
- States the total (sum) is 5

☐ **Draw a picture to show this subtraction: There are 4 apples and someone eats 1. Show how many are left.**

- Draws 4 objects to start
- Shows removal by crossing out or erasing 1
- Identifies 3 remaining as the answer (difference)

☐ **Use your fingers to model $2 + 3$. Explain what you did with your fingers and what the answer is.**

- Raises 2 fingers, then 3 more fingers
- Counts all the raised fingers
- Says the total is 5

WHERE THESE FACTS COME FROM

English Wikipedia — “Subtraction (signified by the minus sign, $-$) represents removal of objects from a collection; $5 - 2$ peaches means 5 peaches with 2 taken away, resulting in 3; the result is the difference; subtraction of 0 does not change a number.” · “The result of a subtraction is called the difference; minuend $-$ subtrahend = difference.”

Simple English Wikipedia — “Addition combines two groups; one group has 3 shapes, the other 2, and when combined the sum becomes 5.” · “Addition uses the plus sign $+$; addends make a sum expressed with the equals sign.”

Wikijunior — “Addition is simulated by gathering the sets of counters representing the two addends, while subtraction is simulated by removing counters from the set.”

MATHEMATICS · MEASUREMENT · AGES 4–6

Comparing durations

1 FOR THE GROWN-UP · TEACHING GUIDE

Why it matters. Comparing durations is one of the earliest and most useful ways children begin to understand time. Before kids can read a clock, they can notice that some things take longer than others and that events happen in a certain order. This builds the language and thinking they will need later for telling time and measuring.

HOW TO TEACH IT

Teach through everyday moments and simple comparing words: quicker, slower, earlier, and later. Point them out during the normal flow of your day so the words become natural. Let your child predict and then check which of two activities takes more time, and ask them to describe it out loud using the comparing words.

WATCH FOR THESE MIX-UPS

- Mixing up 'quicker' and 'slower' — remind them quicker means less time, slower means more time.
- Confusing 'earlier' and 'later' with which happened first — anchor them to real daily routines like breakfast (earlier) and dinner (later).
- Guessing without comparing — encourage them to actually watch both events before deciding.

TRY THIS AT HOME

- Race to see who can put on their shoes first, then ask, 'Who was quicker? Who finished first?'
- At mealtimes, ask which meal comes earlier and which comes later in the day.
- Compare two chores, like tidying toys versus washing hands, and ask which one is slower.
- Talk through the morning routine in order using 'earlier' and 'later' words.

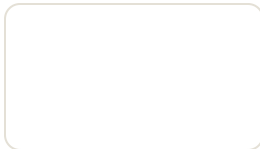
2 THE LESSON • READ TOGETHER

Time is something we cannot see or hold, but we can feel it passing all around us. Some things happen fast, and some things happen slowly. When we talk about how long something takes, we are talking about its duration. Comparing durations just means looking at two things and deciding which one takes more time and which one takes less time. This is an important math idea, and it is one of the very first ways we learn to measure and compare time.

When we compare how long two things take, we can use special comparing words: quicker and slower. If you gobble up a cookie in one big bite but your friend nibbles theirs slowly, you were quicker and your friend was slower. Quicker means something takes less time. Slower means something takes more time. You can use these words for running races, brushing teeth, tidying toys, or even how fast a snail moves compared to a puppy.

We also use comparing words to talk about the order in which things happen during the day: earlier and later. Earlier means something comes first, before another thing. Later means something comes afterward. In the morning, breakfast is earlier and lunch is later. You wake up earlier and go to sleep later. These words help you put the events of your day in the right order, from the first thing that happens to the last.

When you compare durations, you can also answer fun questions like, "Who finished first?" If two friends are cleaning up blocks and one finishes before the other, that friend was quicker and finished earlier. Learning these comparing words helps you judge time and plan your day, like knowing how much time a task will take. Once you can compare durations, you become a real time detective!

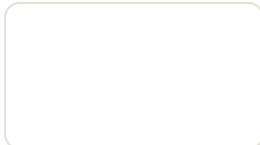
PICTURE CARDS TO TALK ABOUT

SAY The cheetah is quicker because it takes less time to move. The turtle is slower because it takes more time.

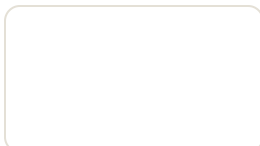
PICTURE

Blinking your eyes,
then counting to ten
out loud

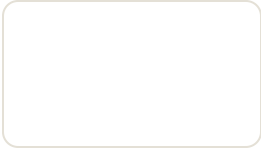
SAY Blinking is quicker and counting to ten is slower. Blinking takes less time than counting.



SAY Breakfast happens earlier in the day, and dinner happens later. Earlier comes first, later comes after.



SAY You wake up earlier and you go to sleep later. These words tell us the order of our day.



SAY The child who stops tying first finished earlier and was quicker. We can say that child finished first!

3 SHOW WHAT YOU KNOW • TALK IT THROUGH

Ask each question out loud. Check the box when Hugh shows they understand.

- ☐ **Think of two things you do. Tell which one is quicker and which one is slower, and explain how you know.**
- Names two events or activities
 - Correctly labels one as quicker and one as slower
 - Shows understanding that quicker means less time and slower means more time
- ☐ **Describe two things that happen in your day. Use the words 'earlier' and 'later' to tell which one comes first.**
- Names two daily events
 - Uses 'earlier' and 'later' correctly
 - Puts the events in the correct order
- ☐ **Two children are cleaning up toys. One finishes before the other. Who finished first, and how do you know they were quicker?**
- Identifies who finished first
 - Connects finishing first with being quicker
 - Explains that finishing first means it took less time

WHERE THESE FACTS COME FROM

Simple English Wikipedia — “quickly judging time intervals and durations; calculating how much time is needed for tasks”

MATHEMATICS · COUNTING & CARDINALITY · AGES 4–6

Representing numbers with objects

1 FOR THE GROWN-UP · TEACHING GUIDE

Why it matters. Before children can add, subtract, or compare, they need to understand that a number stands for a quantity. Representing numbers with objects, pictures, and a number line builds this foundation by connecting the spoken word, the written symbol, and the real "how many" behind it.

HOW TO TEACH IT

Move through three stages in order: first concrete (real objects your child can touch), then pictorial (dots or tally marks on paper), then the number line (a symbolic tool). Say a number, have your child show it with objects, draw it, and finally find it on a line. Keep quantities small at first — usually within 10.

WATCH FOR THESE MIX-UPS

- Counting objects but skipping or double-counting one, so the total is wrong
- Drawing a random number of dots instead of matching the exact quantity
- Pointing between marks on the number line instead of landing on the number's spot
- Thinking a bigger-looking pile of small items is a smaller number just because it takes less space

TRY THIS AT HOME

- Ask your child to "show me 6" using spoons, buttons, or fingers
- Give a number and have them draw that many dots or tally marks
- Draw a simple 0–10 number line with chalk and hop to a called-out number
- Compare two numbers by pointing to both on the number line and asking which is bigger

2 THE LESSON • READ TOGETHER

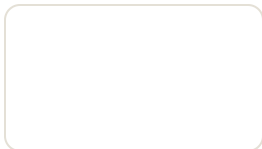
Numbers are all around you, but a number is really just an idea. The number "five" is not something you can hold in your hand — it is the idea of how many. To help our brains understand numbers, we can **SHOW** them in ways we can see and touch. This is called representing numbers, and it means using real things, pictures, or a special line to show how many a number is.

The first way is to use objects. You can grab five buttons, five blocks, five little counters, or hold up five fingers. When you can see and touch that many, the number becomes real. If a friend says "show me three," you can lay out three cubes in a row, or put up three fingers. Each object stands for one, so counting them tells you the number.

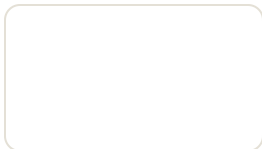
The second way is to draw a picture. You do not always have blocks nearby, but you always have a way to make marks! You can draw dots — one dot for each thing you are counting. Or you can make tally marks, which are little standing lines. If you want to show four, you draw four dots or four tally lines. The picture keeps your number safe on the paper so you can look back and count again.

The third way is to use a number line. A number line is a straight line with numbers written in order: 0, 1, 2, 3, 4, 5, and on and on. Each number has its own spot, and the numbers grow bigger as you move to the right. To show a number, you find its spot and point to it or hop to it. A number line helps you see which numbers come before and after, and which numbers are bigger.

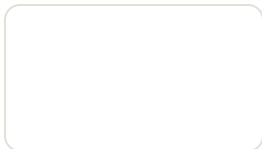
All three ways — objects, pictures, and the number line — do the same helpful job. They turn the idea of a number into something you can see, so counting and comparing numbers becomes much easier.

PICTURE CARDS TO TALK ABOUT

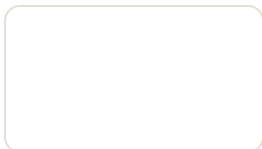
SAY I can show the number 5 by putting out 5 counters — one, two, three, four, five!



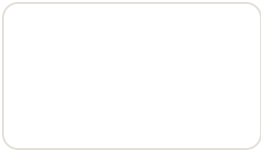
SAY Fingers are objects too! Three fingers up means the number 3.



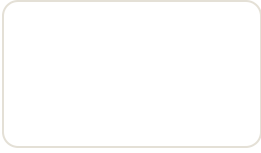
SAY I stacked 4 cubes, so my tower shows the number 4.



SAY When I have no blocks, I draw 6 dots to show 6.



SAY These tally marks show the number 4 — each line counts one.



SAY On the number line I find 3 by counting spots from 0, then I point right at it.

3 SHOW WHAT YOU KNOW • TALK IT THROUGH

Ask each question out loud. Check the box when Hugh shows they understand.

- ☐ **Use counters, cubes, or your fingers to show the number 6. Tell how you made 6.**
- Shows exactly 6 objects or fingers
 - Counts each object once without skipping or repeating
 - Connects the objects to the number 6
- ☐ **Draw a picture that shows the number 4. You may use dots or tally marks. Explain your drawing.**
- Draws exactly 4 dots or 4 tally marks
 - Uses a clear pictorial method (dots or tallies)
 - Explains that the marks stand for the quantity 4
- ☐ **Here is a number line: 0 1 2 3 4 5. Show where the number 3 is and say how you found it.**
- Locates 3 in the correct spot on the number line
 - Counts spots in order from 0 to reach 3
 - Explains that numbers grow larger to the right

WHERE THESE FACTS COME FROM

MATHEMATICS · COUNTING & CARDINALITY · AGES 4–6

One More Each Time

1 FOR THE GROWN-UP · TEACHING GUIDE

Why it matters. Understanding that each counting number is exactly one larger than the one before it is the foundation of number sense. It turns counting from a memorized chant into real meaning, and it prepares your child for addition and subtraction.

HOW TO TEACH IT

Use real objects your child can touch — blocks, snacks, coins, or fingers. Build a group, say its number, then add exactly one more object and say the new number. Then reverse it: take one away and name the number that comes before. Say the words 'one more' and 'one less' out loud every time so the language sticks.

WATCH FOR THESE MIX-UPS

- Recites number names in order but doesn't connect them to real amounts
- Adds or removes more than one object when asked for 'one more' or 'one less'
- Can find 'one more' easily but gets stuck on 'one less' (going backward is harder)
- Guesses instead of counting the objects to check

TRY THIS AT HOME

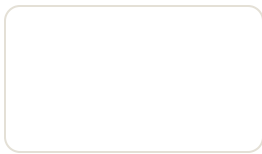
- At snack time, put out 5 crackers, then add one and ask 'How many now?'
- Play 'one more, one less': you say a number, your child says the number just after it and just before it
- Climb stairs together counting each step, pointing out that each step is one more
- Line up toys, then hide one and ask what number comes before

2 THE LESSON • READ TOGETHER

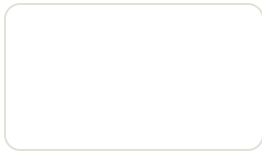
When we count, we say numbers in a special order: one, two, three, four, five, six, and on and on. Have you ever noticed that each number is just a tiny bit bigger than the one before it? That is the big idea of this lesson. Every time you move to the next counting number, you are adding exactly ONE more.

Let's think about it with your fingers or with blocks. If you have 4 blocks and you set down just one more block, now you have 5. If you have 5 and you add one more, you have 6. Each new number stands for a group that is one larger than the group before it. That is why counting works the way it does — you never skip and you never jump. You just tuck in one more object and say the next number.

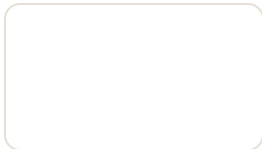
This works going up AND going down. Going up means adding one to get the next number (7, then 8). Going down means taking one away to get the number before (7, then 6). So for any number you know, you can find 'one more' by adding a single object, and 'one less' by taking a single object away. When you understand 'one more each time,' counting stops being a memorized song and starts making real sense — every number is a real amount that grows by one, step by step.

PICTURE CARDS TO TALK ABOUT

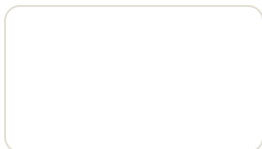
SAY 5 and one more makes 6. Six is one bigger than five!



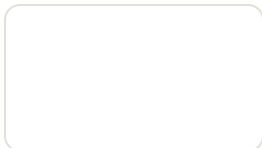
SAY 8 is one more than 7. Adding one star turned 7 into 8.



SAY Now the tower is 4. Each block up the tower is one more.



SAY One less than 9 is 8. Taking one away gives the number before.



SAY Every step you take is one more than the step before it.

3 PRACTICE WORKSHEET

Read each question aloud with Hugh if they are still learning to read.

1. You have 5 blocks. You add 1 more block. How many blocks now?

- ☐ 4
 - ☐ 6
 - ☐ 7
-

2. Which sentence is true?

- ☐ 8 is one more than 7
 - ☐ 8 is one less than 7
 - ☐ 8 is the same as 7
-

3. What is one less than 9?

- ☐ 10
 - ☐ 8
 - ☐ 7
-

4. You have 6 grapes. You eat 1. How many grapes are left, and is that one more or one less than 6?

5. When you count 3, 4, 5, how much bigger is each number than the one before?

- ☐ One more
- ☐ Two more
- ☐ Three more

4 SHOW WHAT YOU KNOW • TALK IT THROUGH

Ask each question out loud. Check the box when Hugh shows they understand.

- ☐ **Pretend you have a group of 5 toy cars. Explain what happens when you add exactly one more car. Tell how many you have now.**
 - Says the new total is 6
 - Explains that adding one object makes the next number / one more
- ☐ **Explain to a friend why 8 is one more than 7. You can talk about counting or use objects.**
 - States that 8 comes right after 7 when counting
 - Connects it to adding one more object to a group of 7
- ☐ **I am thinking of the number 6. Tell me the number that is one more than 6 and the number that is one less than 6.**
 - Correctly gives one more as 7
 - Correctly gives one less as 5

WHERE THESE FACTS COME FROM

✓ ANSWER KEY • GROWN-UP COPY

Building sentences

1. The frog jumps. · 2. sleeps · 3. No, it can be short like 'Dogs bark.' · 4. Any complete sentence with a subject and verb, such as 'The bird flies.' · 5. When we go outside

Understanding print

1. From left to right · 2. At the top, on the left side · 3. Turn one page at a time, in order · 4. It carries meaning — the letters stand for words · 5. Go back to the left and move down to the next line (top to bottom). · 6. So the story makes sense; books are organized page by page.

Exploring Ideas Through Talk

1. 'What if fish could fly?' · 2. Use it in a conversation, like 'That truck is gigantic!' · 3. Talk about your idea out loud first · 4. Any thoughtful guess, such as 'because it gets water and sunshine,' shows the child is speculating and exploring an idea through talk. · 5. Think — an idea is the result of thought.

Phonics Vocabulary

1. A phoneme · 2. A digraph · 3. a, e, i, o, u · 4. Blending · 5. sun · 6. /m/ /a/ /p/ — this is called segmenting. · 7. Homophones

Predicting what happens next

1. Use clues you have already read and give a reason · 2. A balloon that floats away or escapes · 3. Revise the prediction using the new information · 4. It means changing your guess when you learn something new in the story, such as new information on the next page that shows what you first thought was not right. · 5. A computer simulation of a spaceship's route to the moon · 6. Clues include the pictures, the characters, or the events that have already happened in the story.

One-to-one counting

1. 3 · 2. Touch each toy one time and give it one number · 3. 4 · 4. They double-counted one apple. Each apple should get only one number, so they should touch each apple just once. · 5. The last number you said

How Many in Total?

1. five · 2. six · 3. four · 4. The last number you said while counting. · 5. No. It's still five because moving things does not change how many there are.

Subtraction as taking away or separating

1. 3 · 2. Taking away to find how many are left · 3. 3 · 4. 2 fingers are left standing up. · 5. 2 · 6. Zero, or none, are left in the box.

Addition as combining or putting together two

1. 5 · 2. Put groups together · 3. 5 · 4. 5 blocks altogether ($4 + 1 = 5$). · 5. $2 + 3$ · 6. Still 5 crayons, because adding zero does not change the number.

Division as equal sharing

1. 5 · 2. Equal (the same size) · 3. 4 groups · 4. 4 groups · 5. How many groups of 2 can I make?

Comparing groups: more or fewer

1. Equal to each other · 2. 7 is more than 3 · 3. The basket with 8 · 4. They are equal to each other (the same amount). · 5. That group has more; the other group has fewer.

Measurable Attributes of Objects

1. **long** · 2. **heavy** · 3. **full** · 4. **height** · 5. **Any of these: height (how tall it is), weight (how heavy it is), or capacity (how much it holds).** · 6. **shorter**

Number bonds to 9

1. $48 - 10 = \mathbf{38}$ · 2. $86 - 36 = \mathbf{50}$ · 3. $65 - 23 = \mathbf{42}$ · 4. $65 - 23 = \mathbf{42}$ · 5. $39 - 22 = \mathbf{17}$ · 6. $45 - 13 = \mathbf{32}$ · 7. $31 - 22 = \mathbf{9}$ · 8. $84 - 66 = \mathbf{18}$ · 9. $41 - 5 = \mathbf{36}$ · 10. $73 - 12 = \mathbf{61}$ · 11. $31 - 17 = \mathbf{14}$ · 12. $70 - 25 = \mathbf{45}$

Numbers up to 10 into pairs

1. $2 + 3$ · 2. $2 + 2$ · 3. **Yes, one part can be zero** · 4. **Any correct pair, for example $6 = 4 + 2$, $6 = 3 + 3$, $6 = 5 + 1$, or $6 = 0 + 6$.** · 5. 2 · 6. 1

3-D shapes

1. **Sphere** · 2. **Cylinder** · 3. **Cone** · 4. **Cube** · 5. **Any correct example, such as a tin can, a jar, or a cardboard tube.** · 6. **Cuboid**

2-D shapes

1. **Circle** · 2. **Three** · 3. **Square** · 4. **Yes, turning it does not change the shape** · 5. **Rectangle** · 6. **Any round object works, such as a clock, a coin, a plate, or a wheel.**

Addition and subtraction word problems

1. $88 - 14 = \mathbf{74}$ · 2. $67 - 9 = \mathbf{58}$ · 3. $41 - 26 = \mathbf{15}$ · 4. $38 - 13 = \mathbf{25}$ · 5. $90 - 59 = \mathbf{31}$ · 6. $46 - 36 = \mathbf{10}$ · 7. $73 - 7 = \mathbf{66}$ · 8. $21 - 18 = \mathbf{3}$ · 9. $94 - 29 = \mathbf{65}$ · 10. $20 - 2 = \mathbf{18}$ · 11. $31 - 20 = \mathbf{11}$ · 12. $66 - 58 = \mathbf{8}$

Representing Addition and Subtraction

1. $61 - 7 = \mathbf{54}$ · 2. $47 - 19 = \mathbf{28}$ · 3. $70 - 57 = \mathbf{13}$ · 4. $76 - 19 = \mathbf{57}$ · 5. $84 - 38 = \mathbf{46}$ · 6. $33 - 25 = \mathbf{8}$ · 7. $82 - 32 = \mathbf{50}$ · 8. $29 - 9 = \mathbf{20}$ · 9. $81 - 59 = \mathbf{22}$ · 10. $52 - 43 = \mathbf{9}$ · 11. $53 - 6 = \mathbf{47}$ · 12. $74 - 17 = \mathbf{57}$

One More Each Time

1. **6** · 2. **8 is one more than 7** · 3. **8** · 4. **5 grapes, and that is one less than 6.** · 5. **One more**

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